

Microstructure Evolution in Irradiated Ferritic-Martensitic Steels: A comparison Between Cluster Dynamics and Experiments

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Abstract

The microstructural evolution of ferritic-martensitic steels under irradiation is investigated by comparing cluster dynamics simulations with experimental observations. A reaction admissibility graph is used to systematically organize the admissible reactions among point defects, helium-vacancy clusters, and dislocation loops, while a size-bin moments formulation enables efficient tracking of the evolving cluster size distributions. The model captures the nucleation, growth, and coarsening of helium-vacancy clusters and interstitial loops as functions of dose, temperature, and helium generation rate. Predicted number densities and mean sizes are shown to be in good agreement with experiments.

Keywords: cluster dynamics, radiation damage, reaction admissibility graph, ferritic-martensitic steel, helium-vacancy clusters, size-bin moments

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1. Introduction

Reduced-activation ferritic/martensitic (RAFM) steels, and in particular Eurofer97, are the leading structural material candidates for the first-wall and blanket components of future fusion reactors and advanced fission systems [1–3]. Under reactor service, however, these materials are exposed to intense fast-neutron fluences that produce high levels of atomic displacement damage, as well as transmutation-generated helium and hydrogen gasses. Together, these drive a complex, coupled microstructural evolution whose signatures are now well documented by post-irradiation examination (PIE): the accumulation of irradiation-induced dislocation loops, vacancy clusters, helium-stabilized cavities, and radiation-enhanced solute precipitates. This microstructural degradation is the root cause of macroscopic irradiation hardening, loss of ductility, and ductile-to-brittle transition temperature (DBTT) shift that ultimately set the operational limits of the material.

Post-irradiation characterization using transmission electron microscopy (TEM), weak-beam dark-field (WBDF) imaging, scanning TEM (STEM), small-angle neutron scattering (SANS), and atom probe tomography (APT) has built a quantitative database of the irradiation-induced microstructure in Eurofer97 and related F/M alloys (T91, HT9, F82H, Fe–9Cr model alloys, and pure α -iron). The principal experimental findings, compiled from neutron irradiation campaigns at BOR-60, BR2, ATR, HFR Petten, and from ion irradiation facilities, may be summarized as follows.

Dislocation loops. Loops form with two dominant Burgers vectors in bcc iron-based alloys: the glissile $\frac{1}{2}\langle 111 \rangle$ type, which dominates below $\sim 300^\circ\text{C}$, and the sessile $\langle 100 \rangle$ type, which becomes dominant above $\sim 350^\circ\text{C}$ as its lower migration energy confers higher thermal stability [4, 5]. Loop number density peaks near 300°C (typically $\sim 1.2 \times 10^{22} \text{ m}^{-3}$ in Eurofer97 at 16.3 dpa [5]) and decreases monotonically at higher temperatures. Mean loop diameter increases monotonically with temperature, from $\sim 2.5 \text{ nm}$ at 250°C to $>12 \text{ nm}$ at 415°C in Eurofer97, and to $>25 \text{ nm}$ in T91 at 460°C under BOR-60 irradiation [6]. With increasing dose at fixed temperature, loop density near 330°C approaches saturation between 15 and 32 dpa in Eurofer97—with the density increasing only from 1.4×10^{22} to $1.7 \times 10^{22} \text{ m}^{-3}$ and the mean diameter growing modestly from 3.4 to 4.8 nm [4]—whereas in T91 and HT9 at $376\text{--}377^\circ\text{C}$, the density decreases while the mean size grows. The loop size distributions evolve from a spike of sub-3 nm “black-dot” defects at low temperature toward right-skewed log-normal distributions centered at

10–16 nm at higher temperatures.

Cavities and void swelling. Cavity microstructures exhibit a qualitatively distinct evolution with dose and temperature. At doses below ~ 20 dpa and temperatures below $\sim 415^\circ\text{C}$, only a unimodal population of He-stabilized sub-critical bubbles is observed (mean diameter 1.0–2.6 nm, density $\sim 10^{21} \text{ m}^{-3}$), with swelling well below 0.02% [4, 6]. Beyond ~ 35 dpa and in the temperature window $415\text{--}510^\circ\text{C}$, a *bimodal* size distribution emerges in T91 and HT9 under BOR-60 irradiation: a high-density population of He-stabilized sub-critical bubbles ($d < 2$ nm, $\sim 10^{21} \text{ m}^{-3}$) coexists with a lower-density population of growing supercritical voids ($d > 2$ nm, $\sim 10^{20} \text{ m}^{-3}$), separated by a valley near the critical radius $r_c \approx 1$ nm [6, 7]. Swelling in this regime grows rapidly, reaching 0.07% at 35 dpa in HT9 [7] and $\sim 0.2\%$ /dpa steady-state swelling rate under self-ion irradiation of HT9 at 460°C and high dose [8]. The He/dpa ratio plays a controlling role: dual-ion irradiation experiments with simultaneous He co-injection demonstrate that high He production lowers the critical cavity radius and nucleates the bimodal distribution at substantially lower doses [9]. Under mixed-spectrum neutron irradiation at HFR Petten, where the He production rate is higher than in BOR-60, Eurofer97 develops higher void densities relative to fast-spectrum irradiation at equivalent dose and temperature [10].

Radiation-induced solute redistribution and precipitation. Beyond dislocation loops and cavities, APT and SANS studies reveal a secondary class of radiation-induced nano-features that contribute significantly to hardening. At 32 dpa and 332°C , APT measurements on Eurofer97 irradiated in BOR-60 show a high density ($\sim 10^{24} \text{ m}^{-3}$) of 3–5 nm clusters enriched in Cr, Mn, and Si [11]. These solute-rich objects—encompassing Cr-rich α' precipitates, Mn–Ni–Si G-phase-like clusters, and loop-decorating “Mn clouds” with CrMn-enriched satellite clusters—are associated with radiation-induced segregation (RIS) at dislocation loops, grain boundaries, and lath boundaries. Cr enrichment and Fe depletion have been quantified at prior-austenite grain boundaries and lath boundaries under higher-dose irradiation (≥ 72 dpa), where phase instability and enhanced segregation become significant [12]. These irradiation-enhanced precipitation phenomena, operating in parallel with dislocation and cavity evolution, contribute to dispersed-barrier hardening and DBTT elevation independently of loop and void microstructure.

Ion vs. neutron irradiation and dose-rate effects. A persistent challenge in building a comprehensive experimental database is the systematic difference between neutron-irradiated microstructures—experimentally accessible

only up to moderate doses in dedicated fast reactors—and those produced by self-ion or dual-ion irradiation, which permit rapid access to high doses at laboratory scale. Self-ion irradiation of Fe–9Cr at 2 dpa/300 °C produces loop densities 2–3 orders of magnitude higher than the matrix value under neutron irradiation, resulting in a spatially homogeneous rather than heterogeneous loop field [13]. The higher dose rate of ion irradiation suppresses defect diffusion relative to production, requiring a +60–70 °C temperature shift to reproduce the neutron-irradiated microstructure [14, 15]. Recent in-situ S/TEM investigations of neutron-irradiated Eurofer97 (15 dpa, 330 °C) further reveal that mobile line dislocations actively absorb irradiation-induced loops and form dislocation networks, a dynamic mechanism not captured by static PIE [16].

The above experimental database, while extensive, has been developed almost exclusively under fission-neutron and ion irradiation conditions that differ fundamentally in neutron energy spectrum and transmutation production ratios from the projected service environment of a fusion demonstration reactor (DEMO) or a Tokamak first wall. In fusion blankets, the 14 MeV neutron spectrum produces helium at rates of 10–15 appm/dpa and hydrogen at rates of 40–50 appm/dpa—roughly one to two orders of magnitude above the fission case—conditions that have been shown to dramatically alter swelling kinetics, bubble nucleation, and loop–void coupling [2, 17]. Experimental data at these conditions do not yet exist and will be prohibitively expensive to generate: IFMIF-DONES, the most advanced planned materials irradiation facility for fusion, will not reach full operation until the mid-2030s and will provide fusion-relevant spectra only up to ~ 50 dpa with a small irradiation volume [18, 19]. Predictive modeling is therefore not merely supplementary to the experimental effort but is an indispensable tool for bridging the current fission/ion database toward fusion-relevant conditions.

The objective of this work is to establish a high-fidelity, quantitatively validated predictive capability for microstructural evolution in Eurofer97 and related F/M steels by systematically comparing the RADCLUSTER cluster dynamics (CD) code against the full available experimental database of defect densities, mean sizes, and size distributions. RADCLUSTER is a generalized, modular CD framework that self-consistently tracks the hierarchical evolution of multi-species defect clusters (SIA clusters, vacancy clusters, He bubbles, and He–vacancy complexes), accounting for mixed 1D/3D transport, solute-trapped SIA migration, and generalized dissociation kinetics [20].

A fundamental challenge facing all CD and rate-theory models is the in-

herent uncertainty in their input parameters: formation, binding, and migration energies of point defects and small clusters; cascade production fractions and survival efficiencies; and reaction rate constants. These quantities are derived from atomistic simulations (DFT, MD) whose accuracy in complex alloy matrices is imperfect, and they have historically been tuned manually or against a very limited subset of experiments. To overcome this limitation, we employ a suite of complementary optimization strategies to navigate the high-dimensional parameter space of RADCLUSTER in a principled, data-driven manner. These include: (i) *Bayesian optimization*, which constructs a probabilistic surrogate of the objective function and intelligently selects parameter candidates to minimize the discrepancy between model and experiment while naturally quantifying parametric uncertainty; (ii) *surrogate-model-assisted optimization* using Gaussian process regression or polynomial chaos expansions trained on a Latin Hypercube Sampling of the parameter space; and (iii) *gradient-free global search* methods (genetic algorithms, differential evolution) to locate global optima in the presence of non-convex, multi-modal objective surfaces.

The importance of rigorous model-experiment comparison extends well beyond the calibration exercise itself. Quantitative agreement between CD predictions and the full experimental database simultaneously provides three forms of scientific value that manual or partial fitting cannot deliver. First, it tests the *physical completeness* of the model: a model that correctly reproduces loop density and mean size at one temperature but fails at another, or that matches loops but not voids, reveals missing physics—unaccounted mechanisms. Second, the inverse-calibration procedure yields a *probability distribution over the parameter space* rather than a single best-fit point, providing quantitative uncertainty bounds on all model predictions. Third, and most consequentially, a model that has been rigorously validated against diverse irradiation conditions—varying temperature, dose, dose rate, particle type, and He/dpa ratio—carries demonstrably higher predictive fidelity when *extrapolated* to conditions outside the calibration database. This last point is the central motivation for the present work. Fusion-relevant irradiation conditions—14 MeV neutron spectra, He/dpa ratios above 10 appm/dpa, and fluence exceeding 100 dpa at 300–550 °C—are not accessible by any currently operating or near-term fission facility. Moreover, the construction and operation of a dedicated fusion neutron source such as IFMIF-DONES will require billions of dollars and a decade. We therefore posit that validated predictive models represent the only near-term pathway to a quantitative

assessment of F/M steel performance under DEMO operating conditions. A RADCLUSTER code whose parameters have been constrained against the largest available fission and ion irradiation databases can extrapolate, with quantified uncertainty, to the high He/dpa ratios and displacement rates of fusion reactors. This predictive leverage—transforming an expensive experimental database into a physics-informed extrapolation tool—is the ultimate goal of the model-experiment integration described in this paper.

The remainder of the paper is organized as follows. Section 2 presents the RADCLUSTER cluster dynamics framework in detail. Subsection 2.1 sets out the coupled master equations governing the size- and composition-resolved cluster distributions, together with the treatment of defect production, energetics, diffusion, dissociation, sink interactions, and the loop-character conversion that couples the $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ populations. The subsequent subsections formalize which cluster reactions are physically permitted—first defining the admissible vertex set and then encoding the allowed transitions as a Reaction Admissibility Graph (RAG)—and Subsection 2.4 develops the uncertainty-quantification and calibration methodology, in which a multi-strategy pipeline (space-filling design and Sobol sensitivity screening, multi-fidelity Gaussian-process emulation, Bayesian and ensemble-Kalman inversion, and Wasserstein-based comparison of size distributions) is used to infer a posterior distribution over the model parameters and to elevate the calibrated model to a predictive digital twin with quantified uncertainty. Section 3 compares the calibrated model against the experimental database, benchmarking predicted loop and cavity number densities, mean sizes, size distributions, and dislocation-loop character against *in situ* and post-irradiation TEM measurements of Eurofer97 and related F/M steels across a range of doses and temperatures, and applies the validated model to extrapolate microstructural evolution toward fusion-relevant irradiation conditions. Section 4 summarizes the principal findings and their implications for fusion materials development.

2. RADCLUSTER Simulation Methodology

2.1. Model Equations and Parameters

The cluster dynamics framework employed in this work is RADCLUSTER, a spatially-averaged mean-field rate-theory code developed for the quantitative prediction of irradiation-induced microstructure evolution in chemically and structurally complex alloys [21]. RADCLUSTER integrates the

coupled master equations governing the full size- and composition-resolved cluster distributions, simultaneously evolving vacancy clusters, multiple self-interstitial loop populations of distinct Burgers-vector character, and gas-defect complexes, with defect production, energetics, diffusion, dissociation, and sink interactions treated on a common physical footing. Several features distinguish RADCLUSTER from conventional cluster dynamics solvers. The decomposition of each population into 3D-mobile, 1D-glissile, and sessile species with size-dependent rate kernels, together with the explicit treatment of loop-character conversion and helium trapping, allows the code to resolve the competing microstructural pathways that govern swelling and hardening in multi-component steels. The adaptive grouping and moment-based binning of the master equations render the computation tractable across the enormous range of cluster sizes that separates nascent defects from the micron-scale loops and voids that are accessible to electron microscopy. This is what makes RADCLUSTER uniquely suited to the central challenge addressed here: bridging, within a single self-consistent model, the gap between the atomistic scale at which defects are produced and the experimentally observable microstructure, so that the code can be rigorously calibrated against, and validated by, the full database of measured loop and void densities and size distributions. Crucially, because its parameters are physically grounded rather than fitted to a single condition, a validated RADCLUSTER model retains predictive fidelity when extrapolated beyond the reach of current experiments—to the high He/dpa ratios and the design-relevant displacement-damage levels of a fusion power plant—thereby providing a physics-informed description of the microstructure under DEMO-relevant conditions that no presently operating irradiation facility can reproduce. For mathematical and formulation details, the reader is referred to our recent publication [20].

For EUROFER-97 under neutron irradiation, the only resolved trapped solute is helium: $N_g = 1$, $g_1 = \text{He}$, $\ell = (\ell)$, so the gas reservoir $c_1^g(t) \equiv c_h(t)$ and the composition vector reduce to the scalar helium occupancy $\ell \in \mathbb{Z}_{\geq 0}$. EUROFER-97 carries two self-interstitial (SIA) loop populations of distinct Burgers-vector character on the positive polarity, together with a single vacancy population on the negative polarity:

$$\mathcal{P}_-^{\text{EUR}} = \{ \text{bulk} \}, \quad N_P^V = 1, \quad (1)$$

$$\mathcal{P}_+^{\text{EUR}} = \{ \text{bulk-111}, \text{bulk-100} \}, \quad N_P^I = 2. \quad (2)$$

The positive polarity resolves two distinct SIA loop characters as separate

populations: the glissile $\frac{1}{2}\langle 111 \rangle$ loops (bulk-111), which are 3D-mobile clusters for $n < 4$ and 1D-glissile prismatic loops for $n \geq 4$, and the sessile $\langle 100 \rangle$ loops (bulk-100), which form during microstructural evolution and act as essentially immobile sinks. Within each loop population the 3D/1D mobility distinction is encoded in size-dependent rate kernels; the two characters, by contrast, are tracked as separate population labels and are coupled by the $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion edges introduced below. Grain boundaries, the dislocation network, and MX/M₂₃C₆ precipitates are treated as fixed unresolved sinks. On the negative polarity a single vacancy population is retained and its label is suppressed, while on the positive polarity the loop-character label is kept explicit:

$$c_{n,\alpha}^I(t) \rightarrow c_n^{(111)}(t), c_n^{(100)}(t), c_{m,(\ell),\beta}^V(t) \rightarrow c_{m,\ell}(t), c_1^g(t) \rightarrow c_h(t). \quad (3)$$

Admissible vertex set.

$$\mathcal{V}_-^{\text{EUR}} = \{ (-1, m, \text{bulk}, (\ell)) : 1 \leq m \leq N_V, 0 \leq \ell \leq \ell_{\max}(m) \}, \quad (4)$$

$$\mathcal{V}_+^{\text{EUR}} = \{ (+1, n, p, \mathbf{0}) : p \in \{\text{bulk-111}, \text{bulk-100}\}, 1 \leq n \leq N_I \}, \quad (5)$$

$$\mathcal{V}_0^{\text{EUR}} = \{ g_1 = \text{He} \}, \quad (6)$$

comprising the size-resolved glissile $\frac{1}{2}\langle 111 \rangle$ and sessile $\langle 100 \rangle$ SIA loop populations, size-resolved voids ($\ell = 0$) and helium bubbles ($\ell \geq 1$), and the single matrix He reservoir. The sessile $\langle 100 \rangle$ vertices are admissible only for $n \geq n_{\text{conv}}$ (the conversion onset size, Eq. (32)).

Equations (4)–(6) yield three defect classes evolved by the master equations, with the SIA class resolved into two coupled loop-character populations: (1a) $\frac{1}{2}\langle 111 \rangle$ SIA loops $c_n^{(111)}(t)$, $n = 1, \dots, N_I$: glissile prismatic loops for $n \geq 4$ and 3D-mobile clusters for $n = 1, 2, 3$, fed by the cascade source and carrying no helium; (1b) $\langle 100 \rangle$ SIA loops $c_n^{(100)}(t)$, $n \geq n_{\text{conv}}$: sessile prismatic loops generated *only* by $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion (no cascade source, no self-coalescence), carrying no helium; (2) vacancy clusters / cavities $c_{m,\ell}(t)$, $m = 1, \dots, N_V$, $\ell = 0, \dots, \ell_{\max}(m)$: spherical cavities, pure voids when $\ell = 0$ and helium bubbles when $\ell \geq 1$; (3) free helium $c_h(t)$, the single gas-reservoir variable. The signed lattice-defect contents of both SIA characters are positive,

$$q(I_n^{(111)}) = +n, \quad q(I_n^{(100)}) = +n, \quad q(V_m \text{He}_\ell) = -m, \quad q(\text{He}) = 0, \quad (7)$$

as required by the conservation law; the conversion edges relabel SIA content between the two positive populations and therefore leave the total interstitial inventory unchanged.

All pairwise reaction rate constants are obtained from the Smoluchowski capture model in atomic-fraction units. For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the diffusion-limited form reduces to

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_{\alpha} [\text{s}^{-1}], \quad (8)$$

with the void, dislocation-loop, recombination, and mixed 1D/3D-glide capture constants ($\mathcal{K}^{\text{void}}$, $\mathcal{K}_{i,n}^{\text{loop}}$, $\mathcal{K}_{iv}$, $\mathcal{K}_{n,m}^{\text{mixed}}$), the 1D-glide reaction frequencies ω_n^{1D} , and the correlated 1D versus 3D. Thermal emission of species α from a size- n cluster is fixed by detailed balance against the capture rate,

$$\mathcal{E}_{\alpha}(n) = \mathcal{K}_{\alpha}(n-1) \frac{1}{\Omega} \exp\left(-\frac{E_b^{\alpha}(n)}{k_B T}\right), \quad (9)$$

with the corresponding loop-emission, vacancy/helium bubble-emission, and trap-mutation expressions also collected in Section ??, where Table E.23 lists the rate-constant prefactors, sink parameters, and state-space controls used throughout.

2.2. Reaction Admissibility Graph (RAG) & Master Equations

The dislocation network is a microstructural feature of irradiated F/M steels. Beyond serving as a static point-defect sink (process class P4), the network also captures glissile $\frac{1}{2}\langle 111 \rangle$ and sessile $\langle 100 \rangle$ loops that grow large enough to overlap or intersect a dislocation line—two additional processes that we designate P10 ($\frac{1}{2}\langle 111 \rangle$ loop absorbed by network) and P11 ($\langle 100 \rangle$ loop absorbed by network). Incorporating these edges into the $\mathcal{G}$ requires a dynamic network model in which the dislocation density ρ_{net} responds to the evolving loop population; the remainder of this paragraph develops that model, after which the full $\mathcal{G}$ for EUROFER-97 is assembled.

For a circular prismatic loop of n SIAs of character $c \in \{111, 100\}$ with Burgers-vector magnitude b_c , the platelet area is $n\Omega/b_c$, so the loop diameter is

$$d_n^{(c)} = 2 \left(\frac{n\Omega}{\pi b_c} \right)^{1/2}. \quad (10)$$

The mean loop-network spacing is

$$L_{\ell d} = \rho_{\text{net}}^{-1/2}, \quad s_{\ell d}(n) = \max[b_c, L_{\ell d} - \frac{1}{2}d_n^{(c)}], \quad (11)$$

the edge-to-edge gap between a loop and the nearest network line, clamped from below to prevent nonphysical negative gaps.

The elastic interaction range of a loop of size n is $R_{\text{int}}(n) = \chi d_n^{(c)}$, with $\chi = \mathcal{O}(1)$ as the dimensionless geometric range parameter, μ as the shear modulus, and T as the temperature. A smooth threshold gates the transfer based on the geometric proximity of the loop to the network.

$$P_{\ell d}(n) = \frac{1}{2} \left[1 + \tanh \left(\frac{R_{\text{int}}(n) - s_{\ell d}(n)}{\Delta_{\ell d}} \right) \right], \quad \Delta_{\ell d} \sim b_c, \quad (12)$$

so $P_{\ell d} \rightarrow 0$ in the dilute limit ($L_{\ell d} \gg R_{\text{int}}$) and $P_{\ell d} \rightarrow 1$ once the loop's elastic zone reaches the mean network spacing.

Loop incorporation proceeds by the network sweeping or climbing past a stationary loop, so the relevant kinetics are set by the *network*, not the loop. We write

$$\Lambda_n^{\text{net}} = \nu^{\text{net}} P_{\ell d}(n), \quad \nu^{\text{net}} = v_{\text{net}} \rho_{\text{net}} w_c, \quad (13)$$

where v_{net} is the effective network climb/glide velocity, ρ_{net} the network line density, and $w_c = \mathcal{O}(b_c)$ a capture width. The product $v_{\text{net}} \rho_{\text{net}}$ is the area swept by the network per unit volume per unit time; w_c converts it to a capture probability per loop. The climb velocity is itself irradiation-driven,

$$v_{\text{net}} = \frac{\Omega}{b} (Z_i \omega_i^{\text{eff}} c_i - Z_v \omega_v^{\text{eff}} c_v), \quad (14)$$

i.e. proportional to the net biased point-defect flux already computed for the P4 sink—so no new transport physics is introduced. Because Eq. (13) contains *no* factor of D_n^{eff} , the channel is active for sessile loops, as required. A glissile-loop enhancement may be added later as a separate, clearly labeled term; it is not needed for the saturation behavior and is omitted here to keep the parameter set minimal.

The loop population loses members to the network at

$$\left. \frac{dc_n}{dt} \right|_{\text{net}} = -\Lambda_n^{\text{net}} c_n, \quad (15)$$

applied to both the $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ populations for $n \geq n_{\text{inc}}$ (an incorporation-onset size; loops below n_{inc} are too small to register on the network and are excluded).

Each absorbed loop injects n SIAs into the network and adds approximately one loop circumference of dislocation line. In atom-fraction variables

(the code’s internal unit), number density is c_n/Ω , so line-length and SIA-content conservation require

$$\left. \frac{dN_I^{\text{net}}}{dt} \right|_{\text{gain}} = \sum_c \sum_{n \geq n_{\text{inc}}} n \Lambda_n^{\text{net}} c_n^{(c)}, \quad (16)$$

$$\frac{d\rho_{\text{net}}}{dt} = \frac{\pi}{\Omega} \sum_c \sum_{n \geq n_{\text{inc}}} d_n^{(c)} \Lambda_n^{\text{net}} c_n^{(c)} - \underbrace{K_{\text{rec}} \rho_{\text{net}}^{3/2}}_{\text{recovery}}. \quad (17)$$

The recovery sink $K_{\text{rec}} \rho_{\text{net}}^{3/2}$ is *mandatory*: with a gain term alone, ρ_{net} grows without bound, and the model merely relocates the non-saturation problem from loops to the network. K_{rec} sets the steady-state network density and is the second of the two primary calibration parameters.

The Frenkel-pair diagnostic subtracts the cumulative SIA content lost to fixed sinks, J_I^{fix} . The network *is* a fixed sink, so the SIAs carried away by absorbed loops, Eq. (16), are routed into the *same* integral:

$$\frac{dJ_I^{\text{fix}}}{dt} += \sum_c \sum_{n \geq n_{\text{inc}}} n \Lambda_n^{\text{net}} c_n^{(c)}. \quad (18)$$

Making ρ_{net} a fully implicit state variable would make the P4 sink diagonal state-dependent and break the banded-plus-low-rank Jacobian structure that the Woodbury preconditioner relies on, forcing a preconditioner/Jacobian rewrite. ρ_{net} evolves on the dose timescale—orders of magnitude slower than the point-defect relaxation that sets solver stiffness—so it is held *piecewise constant within each integration segment* and refreshed *between* segments by operator splitting. First, we integrate the stiff cluster system over a segment with ρ_{net} frozen, accumulating the gain integral $\int \frac{\pi}{\Omega} \sum d_n \Lambda_n c_n dt$ as an ordinary (non-stiff) quadrature. We then advance Eq. (17) explicitly (gain – recovery) to update ρ_{net} between segments. Finally, we rebuild the reaction rates so the P4 sink picks up the new ρ_{net} for the next segment. Within a segment, all kernels remain constant; the Jacobian sparsity, the band structure, and the Woodbury low-rank update are unchanged.

The network channel combines with the existing fixed sinks by the standard additive-sink-strength rule (not the discontinuous min of length scales used in the earlier draft): the total loss diagonal is the sum of the P4 fixed sink rate and Λ_n^{net} . This is smooth in the state and consistent with how P4d/P4gb/P4p are already summed.

The admissible edges of $\mathcal{G}^{\text{EUR}}$ ($\chi_{ab \rightarrow c} = 1$, & $\chi_{a \rightarrow b} = 1$) partition into eleven process classes, each mapped to an abstract edge class. Because the SIA class comprises two loop-character populations, the inter-population transfer edge (P9) is active and carries the $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion (Dudarev unary reorientation [22], Marian junction nucleation, and Marian absorption growth [23]); the defect production edge is treated as a source term that feeds the $\frac{1}{2}\langle 111 \rangle$ population alone. The explicit rate-kernel expressions corresponding to each of the eleven process classes (P1–P11) above — i.e., the actual form of every kernel $\mathcal{K}$, $\mathcal{D}$, ε , and Γ — are collected in Table 1.

Mobile clusters are bounded by size limits from Table E.23: $\frac{1}{2}\langle 111 \rangle$ SIA clusters are mobile (3D or 1D) for $1 \leq n \leq i_{\text{mobile}}$; larger $\frac{1}{2}\langle 111 \rangle$ clusters are sessile dislocation loops, acting only as fixed sinks. The $\langle 100 \rangle$ loops are sessile at every size ($D_n^{100} \equiv 0$) and grow or shrink only through single-monomer exchange. Vacancy clusters are 3D mobile for $1 \leq m \leq v_{\text{mobile}}$; larger cavities grow or shrink only through single-monomer absorption or emission. Under the mono-defect simplification ($i_{\text{mobile}} = v_{\text{mobile}} = 1$), the coalescence rate constants reduce to

$$\mathcal{K}_{n,1}^{ii} \big|_{n'=1} = \mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}, \quad (19)$$

$$\mathcal{K}_{m,1}^{vv} \big|_{m'=1} = \mathcal{K}_{v,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}, \quad (20)$$

$$\mathcal{K}_{m=1,n=1}^{vi} = \mathcal{K}_{iv} = 4\sqrt{3} \pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}). \quad (21)$$

The most complete description allows any two mobile clusters to react. The intra- and cross-polarity channels are $\frac{1}{2}\langle 111 \rangle$ SIA–SIA coalescence ($I_n + I_{n'} \rightarrow I_{n+n'}$), vacancy–vacancy coalescence ($V_m + V_{m'} \rightarrow V_{m+m'}$), and V–I annihilation ($V_m + I_n \rightarrow V_{m-n}, I_{n-m}$ or $\emptyset$). All three share the diffusion-limited two-mobile-cluster form

$$\mathcal{K}_{a,b} = 8\pi (\xi_a + \xi_b) (\omega_a^{\text{eff}} + \omega_b^{\text{eff}}), \quad \xi_a = (3a/8\pi)^{1/3}, \quad (22)$$

restricted to mobile sizes ($\leq i_{\text{mobile}}, v_{\text{mobile}}$); the channel-specific constants and size restrictions are given in Section 2.3. He–He coalescence is excluded because the interstitial He–He binding energy in bcc Fe is negligible ($\lesssim 0.05$ eV).

The abstract master equation is assembled algorithmically by looping over every admissible edge $r \in \mathcal{E}^{\text{EUR}}$, computing the flux J_r , and accumulating the signed contribution $S_{ar} J_r$ into each affected population a .

Table 1: Consolidated rate-kernel expressions for the eleven reaction process classes. Each kernel maps a process Pn to its abstract edge class and the explicit rate-constant expression used by RADCLUSTER. All rate constants are in s^{-1} and concentrations in at/at ; $\omega_\alpha^{\text{eff}}$ are effective reaction frequencies and $A_{\text{sph}}, A_{\text{loop}}, A_{1D}, B_{\text{rot}}$ the geometric prefactors of Table E.23.

Process	Abstract edge class	Kernel	Rate-constant expression
P1: V-SIA recombination	cross-polarity (recombination)	$\mathcal{K}_{iv}$	$4\sqrt{3}\pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$
P2: cavity absorption of mono- α	growth / shrinkage	$\mathcal{K}_{\alpha,m}^{\text{cav}}$	$A_{\text{sph}} m^{1/3} \omega_\alpha^{\text{eff}}$
He capture by bubble	growth / shrinkage	$\mathcal{K}_{h,m}$	$A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$
P3: loop absorption of mono- α (SIA, biased)	growth / shrinkage	$\mathcal{K}_{i,n}^{\text{loop}}$	$A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}$
(vacancy, unbiased)	growth / shrinkage	$\mathcal{K}_{v,n}^{\text{loop}}$	$A_{\text{loop}} n^{1/2} \omega_v^{\text{eff}}$
P4: absorption at fixed sink (dislocation)	sink edge	$\mathcal{D}_\alpha^d$	$Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}$
(grain boundary)	sink edge	$\mathcal{D}_\alpha^{gb}$	$(4\pi^2/d_g^2) a^2 \omega_\alpha^{\text{eff}}$
(precipitate)	sink edge	$\mathcal{D}_\alpha^p$	$(8\pi r_p/a) c_p^N \omega_\alpha^{\text{eff}}$
P5: thermal monomer emission (V)	dissociation	$\varepsilon_v(m, \ell)$	$A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} e^{-E_b^v(m, \ell)/k_B T}$
(SIA from loop)	dissociation	$\varepsilon_i(n)$	$A_{\text{loop}} (n-1)^{1/2} \omega_i^{\text{eff}} e^{-E_b^i(n)/k_B T}$
(He from bubble)	dissociation	$\varepsilon_h(m, \ell)$	$A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} e^{-E_B^{\text{He}}(m, \ell)/k_B T}$
P6: 1D-gliding SIA cluster $\rightarrow$ cavity (mixed 1D/3D)	cross-polarity	$\mathcal{K}_{n,m}^{\text{eff}}$	$A_{\text{sph}} m^{1/3} \omega_n^{1D} / (1 + B_{\text{rot}} \hat{L}^2 m^{-1/3})$
(pure 1D limit)	cross-polarity	$\mathcal{K}_{n,m}^{1D}$	$A_{1D} m^{4/3} \omega_n^{1D}$
P7: trap mutation	gas-pressure-driven	$\Gamma_{\text{TM}}(m, \ell)$	$\nu_0 e^{-E_{\text{TM}}(m, \ell)/k_B T}$
P8: radiation re-solution	solute detrapping	$\Gamma_{\text{res}}(m, \ell)$	$b_0 \ell \dot{\phi}$
P9: $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion	inter-population transfer	$\Gamma_{\text{uni}}, \varphi_{\text{junc}}, K^{\text{abs}}$	unary reorientation + junction nucleation; see Eqs. (26), (30), (36), (40)

Define the total cavity concentration at vacancy size m and the combined fixed-sink rate:

$$\bar{c}_m(t) \equiv \sum_{\ell=0}^{\ell_{\max}(m)} c_{m,\ell}(t), \quad \mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p. \quad (23)$$

Irradiated bcc Fe and F/M steels develop prismatic interstitial loops of two types: glissile $\frac{1}{2}\langle 111 \rangle$ loops, which are mobile to large sizes, and sessile square $\langle 100 \rangle$ loops [24, 25]. Elasticity favors the smaller $\frac{1}{2}\langle 111 \rangle$ vector, and cascades inject almost exclusively $\frac{1}{2}\langle 111 \rangle$ clusters [23], yet $\langle 100 \rangle$ loops dominate at elevated temperatures and must therefore be *generated* in situ. We model this with two additive channels feeding Eqs. (24)–(27): a thermodynamic single-loop transformation (Channel A) carrying the temperature trend [22], and a kinetic junction route (Channel B) carrying the dose- and size-dependence [23, 26]. The glissile population obeys the full SIA master equation, augmented by the $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ coupling (P9). Cascades feed this population alone, $G_n \equiv G_n^{(111)}$. The junction reaction of two comparable $\frac{1}{2}\langle 111 \rangle$ loops removes both reactants from this population (ordinary coalescence loss) but *redirects* a size-comparability fraction φ_{junc} of the coalescence *product* into the $\langle 100 \rangle$ population; the surviving fraction $(1 - \varphi_{\text{junc}})$ remains glissile. Hence, the coalescence gain below carries the $(1 - \varphi_{\text{junc}})$ weight, with the complementary product appearing in Eq. (27).

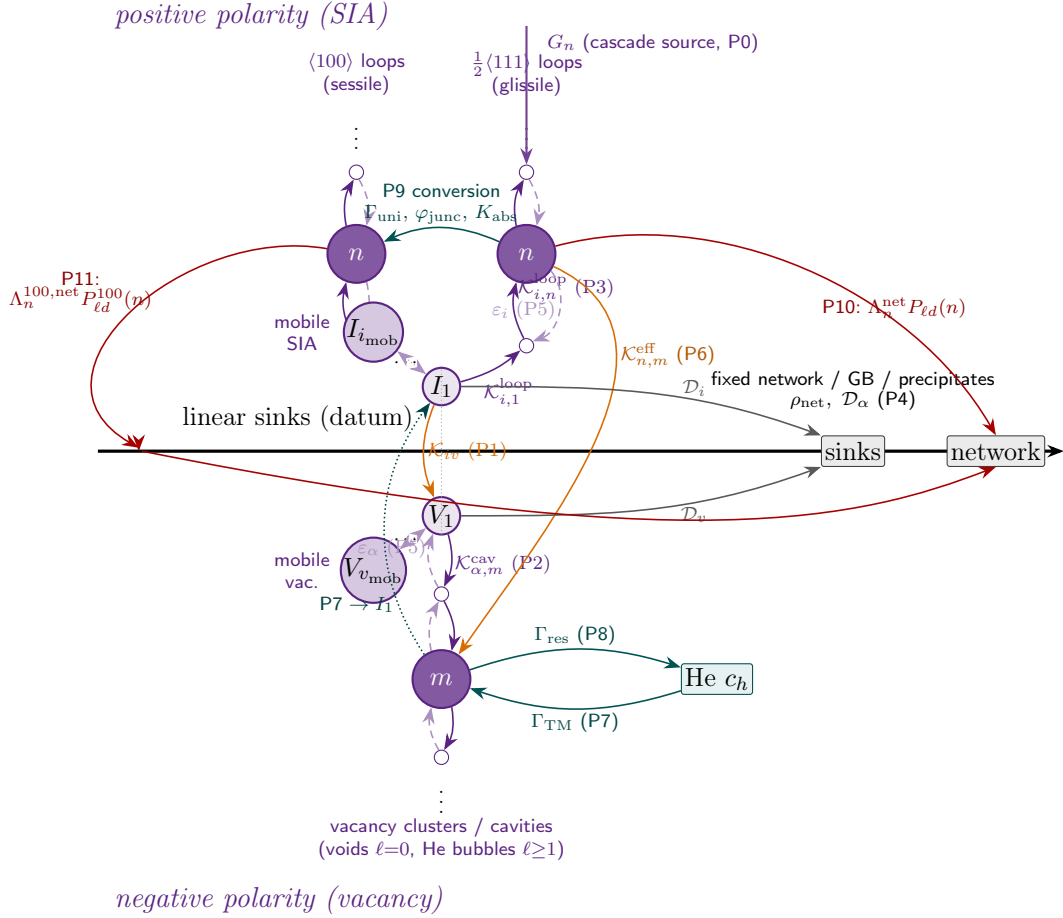


Figure 1: Reaction Admissibility Graph (RAG) for EUROFER-97. Vertices are the size-resolved defect populations: the sessile $\langle 100 \rangle$ and glissile $\frac{1}{2}\langle 111 \rangle$ SIA loop ladders on the positive polarity (above the datum), the vacancy cluster/cavity ladder on the negative polarity (below), and the fixed linear sinks (network, grain boundaries, precipitates) and helium reservoir on or near the datum. Flanking the datum are the *mobile* defect groups: small interstitial clusters $I_1, \dots, I_{i_{\text{mob}}}$ above and vacancy clusters $V_1, \dots, V_{v_{\text{mob}}}$ below, which are the migrating species feeding every reaction edge (the single monomers I_1, V_1 are recovered when $i_{\text{mob}}=v_{\text{mob}}=1$). A generic size n (SIA) and m (vacancy) is drawn per population. Directed edges are the admissible processes, each annotated with its reaction kernel: intra-ladder growth (P2, P3) and thermal emission (P5); cross-polarity recombination $\mathcal{K}_{iv}$ (P1) and 1D-glide capture $\mathcal{K}_{n,m}^{\text{eff}}$ (P6); fixed-sink absorption $\mathcal{D}_\alpha$ (P4); trap mutation Γ_{TM} (P7) and radiation re-solution Γ_{res} (P8); the $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion (P9); and network incorporation of $\frac{1}{2}\langle 111 \rangle$ (P10) and $\langle 100 \rangle$ (P11) loops. The cascade source G_n feeds the $\frac{1}{2}\langle 111 \rangle$ population alone. These edges are exactly the terms accumulated by the master equations below.

$$\begin{aligned}
\frac{dc_n^{(111)}}{dt} = & \underbrace{G_n}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} [1 - \varphi_{\text{junc}}(n', n-n')] \mathcal{K}_{n', n-n'}^{ii} c_{n'}^{(111)} c_{n-n'}^{(111)}}_{\text{SIA-SIA coalescence gain, glissile product}} \\
& - \underbrace{c_n^{(111)} \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n, n'}^{ii} c_{n'}^{(111)}}_{\text{SIA-SIA coalescence loss } (I_n + I_{n'} \rightarrow I_{n+n'})} + \underbrace{\varepsilon_i(n+1) c_{n+1}^{(111)} - \varepsilon_i(n) c_n^{(111)} \delta_{n \geq 2}}_{\text{thermal emission (SIA from loops, P5)}} \\
& + \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n+m'}^{vi} c_{m', 0} c_{n+m'}^{(111)}}_{\text{V-I annihilation gain } (V_{m'} + I_{n+m'} \rightarrow I_n)} - \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n}^{vi} c_{m', 0} c_n^{(111)}}_{\text{V-I annihilation loss } (V_{m'} + I_n \rightarrow I_{n-m'})} \\
& - \underbrace{\sum_{m=1}^{N_V} \mathcal{K}_{n, m}^{(\text{cav})} c_n^{(111)} \bar{c}_m}_{\text{SIA cluster} \rightarrow \text{cavity absorption (P2/P6)}} - \underbrace{\mathcal{D}_i(n) c_n^{(111)}}_{\text{fixed-sink loss (mobile clusters only, P4)}} \\
& + \underbrace{\delta_{n=1} \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \Gamma_{\text{TM}}(m, \ell) c_{m, \ell}}_{\text{trap-mutation SIA source at } n=1 \text{ (P7)}} + \left. \frac{dc_n^{(111)}}{dt} \right|_{\text{conv}}^{(\text{P9})}, - \underbrace{\Lambda_n^{\text{net}} P_{\ell d}(n) c_n^{(111)}}_{\text{network capture (P10)}}
\end{aligned} \tag{24}$$

where the cavity rate constant $\mathcal{K}_{n, m}^{(\text{cav})}$ is

$$\mathcal{K}_{n, m}^{(\text{cav})} = \begin{cases} A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}}, & 1 \leq n \leq 3 \text{ (3D, P2)}, \\ \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, & 4 \leq n \leq i_{\text{mobile}} \text{ (1D/3D, P6)}, \\ 0, & n > i_{\text{mobile}} \text{ (sessile)}, \end{cases} \tag{25}$$

and the glissile-side conversion coupling collects the unary depletion, the $\langle 100 \rangle$ absorption capture, and the monomer exchange with the sessile ladders (growth consumes a $\frac{1}{2} \langle 111 \rangle$ monomer; shrink of sub-onset loops and thermal

emission return one):

$$\begin{aligned}
\left. \frac{dc_n^{(111)}}{dt} \right|_{\text{conv}}^{\text{P9}} = & - \underbrace{\Gamma_{\text{uni}}(n) c_n^{(111)}}_{\text{unary reorientation}} - \underbrace{c_n^{(111)} \sum_{m \geq n_{\text{conv}}} K_{m,n}^{\text{abs}} c_m^{(100)}}_{\text{absorbed by sessile } \langle 100 \rangle} \\
& + \delta_{n=1} \sum_{n' \geq n_{\text{conv}}} \left[\varepsilon^{100}(n') c_{n'}^{(100)} - \mathcal{K}_g^{100}(n') c_{n'}^{(100)} c_1^{(111)} \right] \\
& + \underbrace{\mathcal{K}_s^{100}(n_{\text{conv}}) c_{n_{\text{conv}}}^{(100)} c_{1,0} \delta_{n=n_{\text{conv}}-1}}_{\text{shrink dissolves to } \frac{1}{2} \langle 111 \rangle}.
\end{aligned} \tag{26}$$

The $\langle 100 \rangle$ population has no cascade source and no self-coalescence (its loops are immobile, $D_n^{100} \equiv 0$). It is created by the three conversion channels (P9) and subsequently grows or shrinks only by exchanging single point defects with the matrix — absorbing $\frac{1}{2} \langle 111 \rangle$ SIA monomers ($\mathcal{K}_g^{100}$), absorbing vacancy monomers ($\mathcal{K}_s^{100}$, an SIA–vacancy annihilation that shrinks the loop), and emitting SIA monomers back into the $\frac{1}{2} \langle 111 \rangle$ pool (ε^{100}):

$$\begin{aligned}
\frac{dc_n^{(100)}}{dt} = & \underbrace{\Gamma_{\text{uni}}(n) c_n^{(111)}}_{\text{unary } \frac{1}{2} \langle 111 \rangle_n \rightarrow \langle 100 \rangle_n \text{ (A)}} + \underbrace{\frac{1}{2} \sum_{n'+n''=n} \varphi_{\text{junc}}(n', n'') \mathcal{K}_{n',n''}^{ii} c_{n'}^{(111)} c_{n''}^{(111)}}_{\text{junction nucleation (B)}} \\
& + \underbrace{\sum_{n'=1}^{i_{\text{mobile}}} K_{n-n',n'}^{\text{abs}} c_{n-n'}^{(100)} c_{n'}^{(111)}}_{\text{absorption growth gain } (\langle 100 \rangle_{n-n'} + I_{n'}^{(111)} \rightarrow \langle 100 \rangle_n)} - \underbrace{c_n^{(100)} \sum_{n'=1}^{i_{\text{mobile}}} K_{n,n'}^{\text{abs}} c_{n'}^{(111)}}_{\text{absorption growth loss}} \\
& + \underbrace{\mathcal{K}_g^{100}(n-1) c_{n-1}^{(100)} c_1^{(111)} - \mathcal{K}_g^{100}(n) c_n^{(100)} c_1^{(111)}}_{\text{SIA-monomer capture growth } (\langle 100 \rangle_n + I_1^{(111)} \rightarrow \langle 100 \rangle_{n+1})} \\
& + \underbrace{\mathcal{K}_s^{100}(n+1) c_{n+1}^{(100)} c_{1,0} - \mathcal{K}_s^{100}(n) c_n^{(100)} c_{1,0}}_{\text{vacancy-monomer capture shrinkage } (\langle 100 \rangle_n + V_1 \rightarrow \langle 100 \rangle_{n-1})} \\
& + \underbrace{\varepsilon^{100}(n+1) c_{n+1}^{(100)} - \varepsilon^{100}(n) c_n^{(100)}}_{\text{thermal SIA emission (P5)}}, \quad n \geq n_{\text{conv}}. - \underbrace{\Lambda_n^{100,\text{net}} P_{\ell d}^{100}(n) c_n^{(100)}}_{\text{network capture (P11)}}
\end{aligned} \tag{27}$$

The boundary fluxes are handled so that SIA content is conserved across the onset: an SIA monomer emitted or a vacancy captured at $n = n_{\text{conv}}$ returns the loop to the $\frac{1}{2}\langle 111 \rangle$ population [the δ -terms in Eq. (26)], and the corresponding vacancy monomer consumed by shrinkage is debited from $\dot{c}_{1,0}$ in Eq. (41). Because $\langle 100 \rangle$ loops are immobile, they contribute no $\mathcal{D}_i$ fixed-sink loss and act *as* biased sinks for the mobile $\frac{1}{2}\langle 111 \rangle$ flux.

2.3. Dislocation Loop Mechanisms

The conversion reactions of glissile-to-sessile dislocation loops are based on two established mechanisms.

1. Dudarev Channel: unary elastic instability conversion[22]: The free energy of a loop of n SIAs and character $X \in \{111, 100\}$ follows the anisotropic-elasticity form of [22],

$$E_l^X(n, T) = P_X(n) \left[\hat{F}_X(T) \ln \frac{4R_X^*(n)}{e\delta} + F_\delta^X(T) + F_c^X \right], \quad (28)$$

with P_X the perimeter, R_X^* the equivalent-circle radius, δ the core cut-off, $\hat{F}_X$ the prelogarithmic energy factor, and core constants in Table 2. The geometry follows from the platelet area $A = n\Omega/b_X$,

$$R_X^*(n) = \sqrt{A/\pi}, \quad P_{111}(n) = 6\sqrt{\frac{2A}{3\sqrt{3}}} \text{ (hexagon)}, \quad P_{100}(n) = 4\sqrt{A} \text{ (square)}, \quad (29)$$

and the pure-edge $\langle 100 \rangle[100]$ prelogarithmic factor is analytic and *vanishes* as the shear constant $c' = \frac{1}{2}(c_{11} - c_{12})$ softens:

$$\hat{F}_{001}([100]) = \frac{a^2}{4\pi}(c_{11}+c_{12}) \left[\frac{c_{44}(c_{11}-c_{12})}{c_{11}(c_{11}+c_{12}+2c_{44})} \right]^{1/2} \xrightarrow{c' \rightarrow 0} 0. \quad (30)$$

The softening is driven by spin fluctuations approaching the α - γ transition,

$$c'(T) \simeq 56.8 (1 - T/T_c)^{1/2} \text{ GPa}, \quad T_c = 1185 \text{ K}, \quad (31)$$

so the $\langle 100 \rangle$ elastic self-energy collapses at high T while the $\frac{1}{2}\langle 111 \rangle$ energy does not.

A loop of size n reorients once $\langle 100 \rangle$ is lower in energy, providing a per-loop driving force

$$\Delta F(n, T) = E_l^{111}(n, T) - E_l^{100}(n, T) = P(n) \Delta f(T), \quad (32)$$

parametrized through a per-length difference in which only the $\langle 100 \rangle$ prelogarithmic part softens, with the exponent fixed at $1/4$ by $\hat{F}_{001} \propto (1 - T/T_c)^{1/4}$ from Eq. (30):

$$f_{100}(T) = f_{100}^{\text{core}} + f_{100}^{\text{pre}} (1 - T/T_c)^{1/4}, \quad \Delta f(T) = f_{111} - f_{100}(T) \quad (33)$$

The single fit quantity is the crossover temperature T^* (where $\Delta f = 0$), chosen so that conversion turns on near 350°C and is strongly favored by 550°C , reproducing the three stability regions of [22]; conversion is admissible for $T > T^*$, and the onset size n_{conv} is the smallest n with $\Delta F(n, T) > 0$. The transformation is thermally activated and one-way; propagating segment-by-segment [23], its barrier scales with the perimeter,

$$\begin{aligned} \Gamma_{\text{uni}}(n, T) &= \nu_0 \exp\left[-\frac{E_a^{\text{uni}}(n)}{k_B T}\right] \max\left[0, 1 - \exp\left(-\frac{\Delta F(n, T)}{k_B T}\right)\right] \\ E_a^{\text{uni}}(n) &= E_a^0 + \gamma_a \frac{P(n)}{b_{111}} \end{aligned} \quad (34)$$

so the gating factor favors large loops, while the size-dependent barrier suppresses them, giving a preferred conversion-size window that reproduces the spontaneous transformation of *small* loops upon heating [26] and leaves large loops to convert via Channel B.

2. Marian Channel: junction nucleation and absorption growth.[23]: Two comparable mobile $\frac{1}{2}\langle 111 \rangle$ loops that collide can react through the junction [23, 27]

$$\frac{1}{2}[111] + \frac{1}{2}[\bar{1}\bar{1}\bar{1}] \rightarrow [100], \quad (36)$$

forming a single $\langle 100 \rangle$ loop; this occurs only for similarly sized partners (otherwise, the smaller simply rotates into the larger, i.e., ordinary coalescence). We branch the same-polarity kernel $\mathcal{K}^{ii}$ (Eq. 38) with a size-comparability fraction

$$\varphi_{\text{junc}}(n, n') = \varphi_{\text{max}} P_s(T) \exp\left[-\frac{(\ln(n/n'))^2}{2\sigma_s^2}\right] \Theta(\min(n, n') \geq n_{j,\text{min}}) \quad (37)$$

so that the product fraction φ_{junc} enters the $\langle 100 \rangle$ population [Eq. (27)] and the complement $(1 - \varphi_{\text{junc}})$ remains glissile [Eq. (24)], with $P_s(T)$

the Marian two-step success probability and

$$\mathcal{K}_{n,n'}^{ii} = \frac{8\pi}{\Omega^{2/3}} (\xi_n + \xi_{n'}) (D_n^{111} + D_{n'}^{111}), \quad \xi_n = \left(\frac{3n}{8\pi}\right)^{1/3}. \quad (38)$$

Once formed, $\langle 100 \rangle$ loops are stationary and act as biased sinks for mobile $\frac{1}{2}\langle 111 \rangle$ clusters [23],

$$[100]_m + \frac{1}{2}\langle 111 \rangle_n \rightarrow [100]_{m+n}, \quad (39)$$

the dominant route to TEM-visible $\langle 100 \rangle$ loops, with cross-character capture kernel driven by the mobile $\frac{1}{2}\langle 111 \rangle$ partner ($D_m^{100} \approx 0$),

$$K_{m,n}^{\text{abs}} = \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) (D_m^{100} + D_n^{111}) \approx \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) D_n^{111}. \quad (40)$$

The sessile point-defect kernels of Eq. (27) are the loop-absorption forms specialized for a $\langle 100 \rangle$ target: $\mathcal{K}_g^{100}(n) = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}$ (SIA-monomer capture) $\mathcal{K}_s^{100}(n) = A_{\text{loop}} n^{1/2} \omega_v^{\text{eff}}$ (vacancy-monomer capture), and $\varepsilon^{100}(n)$ the detailed-balance emission rate against $\mathcal{K}_g^{100}$ with $\langle 100 \rangle$ binding energy $E_b^{100}(n) = A_{100} n^{-B_{100}}$.

Every term in Eqs. (26)–(27) conserves the total SIA inventory $\sum_n n (c_n^{(111)} + c_n^{(100)})$: the unary and junction channels relabel content at fixed (or summed) sizes, the absorption channel carries the summed size into the product, and the growth/emission ladders only exchange single SIA monomers between the two populations. The only net SIA loss from the $\langle 100 \rangle$ population is the vacancy-capture shrinkage, which is a Frenkel-pair annihilation accounted for symmetrically in $\dot{c}_{1,0}$. The Frenkel-pair diagnostic is therefore unaffected by the split, provided S_I sums over both characters [Eq. (46)].

The numerical inputs to this two-population description are collected in two tables: Table 2 lists the zero-temperature core constants of the loop free energy, and Table 3 gives the conversion-model parameters—the softening temperature, shear-softening law, and related quantities—that are calibrated against the measured $\frac{1}{2}\langle 111 \rangle$ fraction.

Table 2: Zero-temperature core constants for the loop free energy (28), after [22] (eV/Å).

term	111[11 $\bar{2}$]	111[$\bar{1}$ 10]	001[100]	001[110]
F_c (nonlinear core)	0.46	0.47	0.33	—
F_δ (core-traction)	0.345	0.349	0.387	0.390

Table 3: Conversion-model parameters. Values marked “cal.” are calibrated against the measured $\frac{1}{2}\langle 111 \rangle$ fraction $f_{111}(T, \text{dose})$.

symbol	meaning	first-cut value	source
T_c	α - γ softening temperature	1185 K	[22]
$c'(T)$	shear softening law	$56.8(1-T/T_c)^{1/2}$ GPa	[22]
δ	dislocation core cutoff	0.4 nm	[22]
f_{111}, f_{100}^0	per-length loop energies	0.3–0.5 eV/Å	Tab. 2
softening exp.	exponent in $\Delta f(T)$	1/4 (fixed)	[22], Eq. (30)
T^*	crossover temperature	cal. ($\in [350, 550]$ °C)	[22]
ν_0	attempt frequency	10^{13} s^{-1}	Debye
E_a^0	unary barrier offset	0.5–2.0 eV	[23]
γ_a	unary barrier size slope	cal.	[26]
$\varphi_{\max}$	peak junction yield ($n=n'$)	0.5–1.0	[23]
σ_s	log-size tolerance	0.3–0.5	[23]
$n_{j,\min}$	minimum junction size	$\mathcal{O}(10)$ SIAs	[23]
A_{100}, B_{100}	$\langle 100 \rangle$ binding-energy fit	0.716, 0.358	(DFT)
n_{conv}	$\langle 100 \rangle$ onset size	<i>derived</i> ($\Delta F > 0$)	Eq. (32)

Voids ($\ell = 0$) and bubbles ($\ell \geq 1$) share a unified (m, ℓ) grid. Spherical cavities are unbiased ($Z_i^{\text{cav}} = Z_v^{\text{cav}} = 1$). The v-v coalescence terms are active only for $m \leq v_{\text{mobile}}$; SIA-induced shrinkage spans all mobile $\frac{1}{2}\langle 111 \rangle$ SIA sizes through $\mathcal{K}_{n,m}^{(\text{cav})}$. The vacancy monomer additionally feeds the sessile $\langle 100 \rangle$ shrinkage ladder of Eq. (27); that loss is the last term below.

$$\begin{aligned}
\frac{dc_{m,\ell}}{dt} = & \underbrace{G_{m,\ell}}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{m'=1}^{\min(m-1, v_{\text{mobile}})} \mathcal{K}_{m',m-m'}^{vv} c_{m',0} c_{m-m',\ell} \delta_{m \leq 2 v_{\text{mobile}}}}_{\text{V-V coalescence gain } (V_{m'}+V_{m-m'} \rightarrow V_m)} \\
& - \underbrace{c_{m,\ell} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m,m'}^{vv} c_{m',0}}_{\text{V-V coalescence loss } (V_m+V_{m'} \rightarrow V_{m+m'})} + \underbrace{\varepsilon_v(m+1, \ell) c_{m+1,\ell} - \varepsilon_v(m, \ell) c_{m,\ell}}_{\text{thermal emission (vacancy from cavities, P5)}} \\
& + \underbrace{\sum_{n=1}^{\min(i_{\text{mobile}}, N_V - m)} \mathcal{K}_{n,m+n}^{(\text{cav})} c_n^{(111)} c_{m+n,\ell}}_{\text{SIA-induced shrinkage gain } (I_n+V_{m+n} \rightarrow V_m)} - \underbrace{\sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n^{(111)} c_{m,\ell}}_{\text{SIA-induced shrinkage loss } (I_n+V_m \rightarrow V_{m-n})} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1} \delta_{\ell \geq 1} - A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell} \delta_{\ell < \ell_{\text{max}}}}_{\text{He absorption by cavities (P2, He-side)}} \\
& + \underbrace{\varepsilon_h(m, \ell+1) c_{m,\ell+1} - \varepsilon_h(m, \ell) c_{m,\ell} \delta_{\ell \geq 1}}_{\text{He emission from cavities (P5, He-side)}} + \underbrace{\Gamma_{\text{TM}}(m-1, \ell) c_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap mutation (P7)}} \\
& + \underbrace{b_0 \dot{\phi}(\ell+1) c_{m,\ell+1} - b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}} - \underbrace{\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mobile}}}}_{\text{fixed-sink loss (mobile cavities only, P4)}} \\
& - \underbrace{\delta_{m=1} \delta_{\ell=0} c_{1,0} \sum_{n \geq n_{\text{conv}}} \mathcal{K}_s^{100}(n) c_n^{(100)}}_{\text{vacancy monomer consumed by } \langle 100 \rangle \text{ shrinkage (P9)}}.
\end{aligned} \tag{41}$$

He-He coalescence is excluded, and helium interacts with the vacancy-cluster grid only through mono-He trapping and emission; the $\langle 100 \rangle$ popula-

tion carries no helium and does not appear here.

$$\begin{aligned}
\frac{dc_h}{dt} = & \underbrace{G_{\text{He}}}_{\text{production}} - \underbrace{\sum_{m=1}^{N_V} A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \sum_{\ell=0}^{\ell_{\text{max}}(m)-1} c_{m,\ell}}_{\text{He absorption by cavities (P2)}} \\
& - \underbrace{\mathcal{D}_h c_h}_{\text{fixed-sink loss (P4)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \varepsilon_h(m, \ell) c_{m,\ell}}_{\text{He emission (P5)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}}.
\end{aligned} \tag{42}$$

The master equations (24)–(42) satisfy two exact conservation identities. Their derivation rests on three facts built into the RADCLUSTER code:

- (i) cascade production deposits SIA and vacancy atoms in a balanced Frenkel-pair ratio (and feeds only the $\frac{1}{2}\langle 111 \rangle$ character), $\sum_{n=1}^{N_I} n P_n^{(i)} = \sum_{m=1}^{N_V} m P_m^{(v)} = \eta G$;
- (ii) every mutual reaction destroys $\min(n, m)$ SIA atoms *and* $\min(n, m)$ vacancy atoms with the residual fragment surviving in its population; this holds for the mobile $\frac{1}{2}\langle 111 \rangle$ –vacancy annihilations *and* for the sessile $\langle 100 \rangle$ vacancy-capture shrinkage (each event removes one SIA from $c^{(100)}$ and one vacancy from $c_{1,0}$), so the two destruction sums cancel identically in $d(S_I - S)/dt$; the $\frac{1}{2}\langle 111 \rangle \leftrightarrow \langle 100 \rangle$ conversion edges only relabel SIA content and drop out;
- (iii) only mobile clusters with $n \leq i_{\text{mobile}}$ (glissile SIAs) or $m \leq v_{\text{mobile}}$ (vacancies) reach fixed sinks. The sessile $\langle 100 \rangle$ loops are sinks, not sink-bound species, and carry no $\mathcal{D}_i$ term; the rate-equation losses are summed over the full mobile range, not the monomer alone.

Table 4 lists every edge class, its reaction, its flux, and its non-zero stoichiometric contributions to the four EUROFER-97 population equations $\dot{c}_n^{(111)}$, $\dot{c}_n^{(100)}$, $\dot{c}_{m,\ell}$, and $\dot{c}_h$. Where a process connects adjacent nodes on a ladder (e.g. cavity or sessile-loop growth/shrinkage), the gain term at a and the loss term at a appear on separate rows; the net effect is the tridiagonal ladder operator visible in the equations. Whereas Table 4 lists the abstract edge classes and their signed contributions, the graph walker requires the *concrete*

Table 4: Graph-walker contribution table for EUROFER-97 (mono-defect, $i_{\text{mobile}} = v_{\text{mobile}} = 1$). Each row shows one edge class, its reaction, the corresponding flux J_r , and the sign of the contribution $S_{ar}J_r$ to each population. The symbol $\Delta_m^\pm$ denotes a ladder step: $\Delta_m^+ f_m \equiv f_{m-1} - f_m$ (gain from smaller cavity minus loss at m) and $\Delta_m^- f_m \equiv f_{m+1} - f_m$ (gain from larger cavity minus loss at m). Sums over repeated size indices are implied

P	Reaction	Flux J_r	$\Delta \dot{c}_n$	$\Delta \dot{c}_{m,\ell}$	$\Delta \dot{c}_h$
P1	$I_1 + V_1 \rightarrow 0$	$\mathcal{K}_{iv} c_1 c_{1,0}$	$-J$ at $n=1$	$-J$ at $(1,0)$	0
P2v (growth)	$V_1 + V_m \rightarrow V_{m+1}$	$\mathcal{K}_{v,m}^{\text{cav}} c_{1,0} c_{m,\ell}$	0	$\Delta_m^+ J$ ladder	0
P2i (shrinkage)	$I_n + V_m \rightarrow V_{m-n}$	$\mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell}$	$-\sum_m J$ at n	SIA-shrinkage ladder	0
P2h (He trapping)	$\text{He} + V_{m,\ell} \rightarrow V_{m,\ell+1}$	$\mathcal{K}_{h,m}^{\text{cav}} c_h c_{m,\ell}$	0	He-occ. ladder in ℓ	$-\sum_{m,\ell < \ell_{\text{max}}} J$
P3 (loop growth)	$I_1 + I_n \rightarrow I_{n+1}$	$\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$	i-i ladder in n	0	0
P3 (loop shrink)	$V_1 + I_n \rightarrow I_{n-1}$	$\mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n$	v-i ladder in n	0	0
P4 (sink loss)	$X_\alpha \rightarrow \emptyset$	$\mathcal{D}_\alpha c_\alpha$	$-\mathcal{D}_i(n) c_n$	$-\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mob}}}$	$-\mathcal{D}_h c_h$
P5v (emission)	$V_m \rightarrow V_{m-1} + V_1$	$\varepsilon_v(m, \ell) c_{m,\ell}$	0	ε_v ladder in m	0
P5i (emission)	$I_n \rightarrow I_{n-1} + I_1$	$\varepsilon_i(n) c_n$	ε_i ladder in n	0	0
P5h (emission)	$V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}$	$\varepsilon_h(m, \ell) c_{m,\ell}$	0	He-occ. ladder in ℓ	$+\sum_{m,\ell \geq 1} \varepsilon_h c_{m,\ell}$
P6 (1D glide)	$I_n^{1D} + V_m \rightarrow V_{m-n}$	$\mathcal{K}_{n,m}^{\text{eff}} c_n c_{m,\ell}$	$-\sum_m J$ at $n \geq 4$	1D-shrinkage ladder	0
P7 (trap mutation)	$\text{He}_\ell V_m \rightarrow \text{He}_\ell V_{m+1} + I_1$	$\Gamma_{\text{TM}}(m, \ell) c_{m,\ell}$	$+\sum_{m,\ell} \Gamma_{\text{TM}} c_{m,\ell}$ at $n=1$	TM ladder in m	0
P8 (re-solution)	$\text{He}_\ell V_m \rightarrow \text{He}_{\ell-1} V_m + \text{He}$	$b_0 \phi \ell c_{m,\ell}$	0	re-sol. ladder in ℓ	$+\sum_{m,\ell \geq 1} b_0 \phi \ell c_{m,\ell}$
P10 (network capture, $\frac{1}{2}(111)$)	$I_n \rightarrow \text{dislocation network}$	$\Lambda_n^{\text{net}} P_{d(n)} c_n^{(111)}$	$-\Lambda_n^{\text{net}} P_{d(n)} c_n^{(111)}$ (P10 loss)	0	0
P11 (network capture, (100))	$[100]_n \rightarrow \text{dislocation network}$	$\Lambda_n^{100,\text{net}} P_{d(n)}^{100} c_n^{(100)}$	$-\Lambda_n^{100,\text{net}} P_{d(n)}^{100} c_n^{(100)}$ (P11 loss)	0	0
G_a	cascade sources	$G_n, G_{m,\ell}, G_{\text{He}}$	$+G_n$	$+G_{m,\ell}$	$+G_{\text{He}}$

vertices (population nodes) and directed edges that instantiate each master equation. Table 5 enumerates, for each of the four EUROFER-97 master equations, its graph vertices and the incident reaction edges, each tagged with the originating process P_n of Table 1.

The total fixed-sink rate for species $\alpha \in \{i, v, h\}$ collects all three sink channels, $\mathcal{D}_\alpha(n) = \mathcal{D}_\alpha^d(n) + \mathcal{D}_\alpha^{gb}(n) + \mathcal{D}_\alpha^p(n)$, and vanishes outside the mobile range. Define the cumulative atomic content delivered to fixed sinks (only the glissile SIAs are mobile):

$$\Delta J_i^{\text{fix}}(t) \equiv \int_0^t \sum_{n=1}^{i_{\text{mobile}}} n \mathcal{D}_i(n) c_n^{(111)}(t') dt' \geq 0, \quad (43)$$

$$\Delta J_v^{\text{fix}}(t) \equiv \int_0^t \sum_{m=1}^{v_{\text{mobile}}} m \mathcal{D}_v \bar{c}_m(t') dt' \geq 0, \quad (44)$$

$$\Delta J_h^{\text{fix}}(t) \equiv \int_0^t \left[\mathcal{D}_h c_h + \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \sum_{\ell} \ell c_{m,\ell} \right] dt' \geq 0. \quad (45)$$

These three integrals are appended to the state vector and evolved as aux-

Table 5: Graph vertices and directed edges instantiating each EUROFER-97 master equation, consistent with the abstract processes/edges of Table 1 (A.14) and the contribution table 4. Vertices are population nodes; an edge $u \rightarrow w$ is a directed reaction from source node u to target node w , driven by the mobile reactant shown. Mobile monomers are I_1 ($\frac{1}{2}\langle 111 \rangle$ SIA), V_1 (vacancy), and He; $\emptyset$ denotes the vacuum/sink node and G the external cascade source. Process tags Pn refer to Table 1.

Master equation (population)	Graph vertices (nodes)	Incident directed edges (reaction)	Process
$\dot{c}_n^{(111)}$ ($\frac{1}{2}\langle 111 \rangle$ glissile loops, size n)	$\{I_n^{(111)}\}_{n \geq 1}$; mobile I_1 , V_1 ; source G	$I_n^{(111)} + I_1 \rightarrow I_{n+1}^{(111)}$ (growth); $I_n^{(111)} + V_1 \rightarrow I_{n-1}^{(111)}$ (shrink); $I_n^{(111)} + V_m \rightarrow V_{m-n}$ (cavity capture); $I_n^{(111)} \rightarrow I_{n-1}^{(111)} + I_1$ (emission); $I_n^{(111)} \rightarrow \langle 100 \rangle$ (reorient/junction); $I_n^{(111)} \rightarrow \emptyset$ (sink); $G \rightarrow I_n^{(111)}$	P3, P5i, P2i, P9, P4
$\dot{c}_n^{(100)}$ ($\langle 100 \rangle$ sessile loops, size n)	$\{I_n^{(100)}\}_{n \geq 1}$; mobile I_1 , V_1	$\frac{1}{2}[111] + \frac{1}{2}[\bar{1}\bar{1}1] \rightarrow [100]$ (junction nucleation); $[100]_m + \frac{1}{2}\langle 111 \rangle \rightarrow [100]_{m+n}$ (absorption growth); $I_n^{(100)} + V_1 \rightarrow I_{n-1}^{(100)}$ (shrink)	P9, P3, P5i
$\dot{c}_{m,\ell}$ (cavities, vacancy size m , He content ℓ)	$\{V_{m,\ell}\}_{m \geq 1, 0 \leq \ell \leq \ell_{\max}}$; mobile V_1 , I_1 , He	$V_1 + V_{m,\ell} \rightarrow V_{m+1,\ell}$ (growth); $I_n + V_{m,\ell} \rightarrow V_{m-n,\ell}$ (shrink); $\text{He} + V_{m,\ell} \rightarrow V_{m,\ell+1}$ (He trapping); $V_{m,\ell} \rightarrow V_{m-1,\ell} + V_1$ (V emission); $V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}$ (He emission/re-solution); $\text{He}_\ell V_m \rightarrow \text{He}_\ell V_{m+1} + I_1$ (trap mutation); $I_n^{1D} + V_m \rightarrow V_{m-n}$ (1D capture); $G \rightarrow V_{m,\ell}$	P2v, P2i, P2h, P5v, P5h/P8, P7, P6
$\dot{c}_h$ (mobile He monomer)	$\{\text{He}\}$; couples to cavity nodes $V_{m,\ell}$	$\text{He} + V_{m,\ell} \rightarrow V_{m,\ell+1}$ (capture, loss of He); $V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}$ (emission, gain of He); $\text{He}_\ell V_m \rightarrow \text{He}_{\ell-1} V_m + \text{He}$ (re-solution); $G \rightarrow \text{He}$ (source)	P2h, P5h, P8

iliary ODEs so that they share the integrator's local error control with the physical concentrations.

With total inventories summed over *both* SIA characters,

$$S_I(t) \equiv \sum_{n=1}^{N_I} n [c_n^{(111)}(t) + c_n^{(100)}(t)], \quad S(t) \equiv \sum_{m=1}^{N_V} m \bar{c}_m(t), \quad (46)$$

facts (i)–(iii) combine into the exact Frenkel-pair conservation law

$$S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t) = S(0) - S_I(0) = \text{const.} \quad (47)$$

For a defect-free initial condition, this is the *swelling identity*

$$S(t) = S_I(t) + \Delta J_i^{\text{fix}}(t) - \Delta J_v^{\text{fix}}(t), \quad (48)$$

i.e. the total vacancy content of all cavities equals the SIA inventory in loops (both characters) plus the *net* atomic flux delivered to fixed sinks. Differentiating gives the rate form

$$\frac{dS}{dt} = \frac{dS_I}{dt} + \sum_{n=1}^{i_{\text{mobile}}} n \mathcal{D}_i(n) c_n^{(111)} - \sum_{m=1}^{v_{\text{mobile}}} m \mathcal{D}_v \bar{c}_m. \quad (49)$$

Once loops saturate ($dS_I/dt \approx 0$), swelling grows at the rate at which the biased dislocation channel ($Z_i^d \neq Z_v^d$) drains the surplus SIA inventory faster than the vacancy inventory; the unbiased grain-boundary and precipitate channels contribute equally to both sums and cancel in $dS/dt - dS_I/dt$ only insofar as the mobile SIA and vacancy populations are equal.

The total helium inventory is the sum of free and trapped helium,

$$S_{\text{He}}(t) \equiv c_h(t) + \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \ell c_{m,\ell}(t). \quad (50)$$

Trap mutation (P7), radiation re-resolution (P8), and thermal He emission (P5h) only redistribute helium between c_h and the trapped pool, so their contributions to dS_{He}/dt cancel pairwise. The only net helium sinks are free helium captured at fixed sinks ($\mathcal{D}_h c_h$) and helium carried to fixed sinks by mobile cavities ($m \leq v_{\text{mobile}}$); immobile bubbles retain their helium. Hence

$$\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - \mathcal{D}_h c_h - \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \sum_{\ell} \ell c_{m,\ell}, \quad (51)$$

i.e. $dS_{\text{He}}/dt = G_{\text{He}} - d\Delta J_h^{\text{fix}}/dt$. In particular, S_{He} is *not* conserved against $\int G_{\text{He}} dt'$ alone — the fixed-sink loss must be tracked explicitly. Both loss terms are typically small (mobile clusters carry little He) but they are non-zero and must appear in the diagnostic below for it to converge to zero.

The swelling identity (48) and the helium balance (51) provide two inexpensive a-posteriori checks, neither of which requires post-processing because the auxiliary integrals (43)–(45) are already state variables:

$$\delta_{\text{FP}}(t) = \frac{|S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)|}{S(t) + S_I(t) + \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)}, \quad (52)$$

$$\delta_{\text{He}}(t) = \frac{|S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) - \int_0^t G_{\text{He}} dt' - S_{\text{He}}(0)|}{S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) + \int_0^t G_{\text{He}} dt'}. \quad (53)$$

Both should remain below $\sim 10^{-6}$ in a correctly implemented code; values above 10^{-3} almost always signal a coding error, missing or incorrect reactions, or an overly loose integrator tolerance. We have verified that the two-population split preserves these bounds: with the $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ conversion active, δ_{FP} remains at the 10^{-7} level in the bin-moment representation (and at the round-off level $\sim 10^{-14}$ in the fully discrete representation), confirming that the conversion edges and the sessile $\langle 100 \rangle$ ladders are SIA-content conserving by construction. Despite the importance of these conservation identities, they have seldom been discussed or tested in the cluster dynamics literature.

2.4. Uncertainty Quantification

The parameters that enter the RADCLUSTER rate equations —migration and binding energies, capture efficiencies, loop and cavity bias factors, helium–vacancy association parameters, sink strengths, and coalescence coefficients—are known only to within physically motivated bounds, while the experimental targets (loop and cavity number densities, mean sizes, and full size distributions at selected doses and temperatures) carry their own measurement uncertainty. Quantifying how these two sources of uncertainty propagate to the predicted microstructure is therefore not an optional post-processing step but an intrinsic part of using the model predictability.

A range of established methodologies addresses this problem. The canonical framework for confronting an expensive mechanistic simulator with noisy

data is the Bayesian calibration of computer models, in which a prior over the uncertain parameters is updated into a posterior through a likelihood that accounts for both measurement noise and model discrepancy [28, 29]. Because each RADCLUSTER evaluation is costly, the posterior is rarely explored on the simulator directly; instead, a statistical surrogate—most commonly a Gaussian-process emulator—is trained on a designed set of runs and substituted for the forward model [30, 31]. The informativeness of the data is assessed through variance-based global sensitivity analysis, whose Sobol indices reveal which parameters and parameter combinations actually control the observables [32, 33]; this is also the natural language for diagnosing the parameter “sloppiness” and partial non-identifiability that are endemic to multi-parameter mechanistic models [34]. When data are available sequentially along a dose or temperature trajectory, ensemble Kalman inversion offers a derivative-free, gradient-free route to approximate posteriors that scales gracefully to the ten-to-twelve parameters of interest here [35]. Finally, because densities span several orders of magnitude and because full size distributions are available, distributional discrepancy is most faithfully measured by an optimal-transport (Wasserstein) distance rather than by moments alone [36].

Workflow. Rather than applying these techniques as an undifferentiated toolbox, we adopt a single reasoned pipeline whose guiding principle is that *physical inference must never be contaminated by numerical approximation*. We therefore fix verified integrator tolerances throughout the calibration and treat uncertainty quantification as a four-stage sequence, each stage feeding into the next. (i) *Design and screening*: a small set of uncertain parameters $\boldsymbol{\theta} \in \mathbb{R}^p$ ($p \approx 10\text{--}12$), each given a physically bounded prior $p_i(\theta_i)$, is sampled by a space-filling Sobol design and screened with global sensitivity analysis so that sloppy or regime-inactive parameters are demoted before they can absorb spurious degrees of freedom. (ii) *Emulation*: the retained parameters are mapped to the observables $\mathbf{y}_{\text{sim}} = \mathcal{M}(\boldsymbol{\theta}, T, d)$ by a *multi-fidelity* Gaussian-process surrogate $\widehat{\mathcal{M}}$ that reports its own predictive variance. Because a single fidelity forces an unfavorable trade-off between the number of training runs needed to map the $p \approx 10\text{--}12$ -dimensional parameter response and the cost of integrating each run to fusion-relevant dose, we train the emulator on two fidelities. A large, space-filling ensemble of *low-fidelity* runs—coarse bin-moment resolution and integration truncated at $d_{\text{LF}} \approx 0.1$ dpa—cheaply charts the shape of the response surface and the early-time, identifiable ob-

servables (incipient loop number density, He-stabilized bubble population). A sparse set of *high-fidelity* runs—full bin resolution integrated to $d \sim 100$ dpa, capturing the cavity bifurcation and the onset of supercritical void swelling—anchors the absolute level. The two levels are fused in an auto-regressive multi-fidelity GP [37, 38], $\widehat{\mathcal{M}}_{\text{HF}}(\boldsymbol{\theta}) = \rho \widehat{\mathcal{M}}_{\text{LF}}(\boldsymbol{\theta}) + \delta(\boldsymbol{\theta})$, so that the inexpensive runs ensure parameter-space coverage while the few expensive runs correct the dose-extrapolation bias rather than a polynomial doing so blindly. All runs, at both fidelities, use verified integrator tolerances, preserving the principle that physical inference is never contaminated by numerical approximation. (iii) *Calibration*: the posterior

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}}) \propto p(\mathbf{y}_{\text{exp}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) \quad (54)$$

is obtained on the surrogate, with a log-space likelihood for the densities and a Wasserstein term for the size distributions.

$$-\log p(\mathbf{y}_{\text{exp}} \mid \boldsymbol{\theta}) = \sum_j w_j [\log y_{j,\text{sim}}(\boldsymbol{\theta}) - \log y_{j,\text{exp}}]^2 + w_{\text{dist}} W_1(p_{\text{sim}}(R), p_{\text{exp}}(R)). \quad (55)$$

(iv) *Active refinement*: new RADCLUSTER runs are requested only where they most reduce posterior uncertainty per unit of predicted CPU cost, closing the loop back to stage (ii). The end product is not a single “best fit” but the calibrated posterior $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}})$ together with its propagated predictive uncertainty on loop and cavity densities, sizes, and distributions.

The workflow is summarized in Fig. 2.

From calibrated model to digital twin. Once the posterior in Eq. (54) is available, RADCLUSTER ceases to be a one-shot forward model and becomes a *digital twin* of the evolving irradiated microstructure: a living, uncertainty-aware representation whose state $\mathbf{x}(d, T)$ —the SIA, vacancy, and helium–vacancy cluster populations and their derived loop and cavity descriptors—is kept consistent with the experimental record as it accumulates. The key point is that the twin is defined not by a single tuned parameter set but by the calibrated distribution $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}})$ and the predictive uncertainty it induces. When measurements arrive at successive doses, $\mathbf{z}_k = H(\mathbf{x}_k) + \boldsymbol{\epsilon}_k$, the same Bayesian machinery is run sequentially—naturally as an ensemble Kalman or smoother update [35]—so that the twin assimilates the full 0.1–100 dpa trajectory rather than isolated endpoints. In this view, uncertainty quantification is not an appendage to the digital twin but its defining

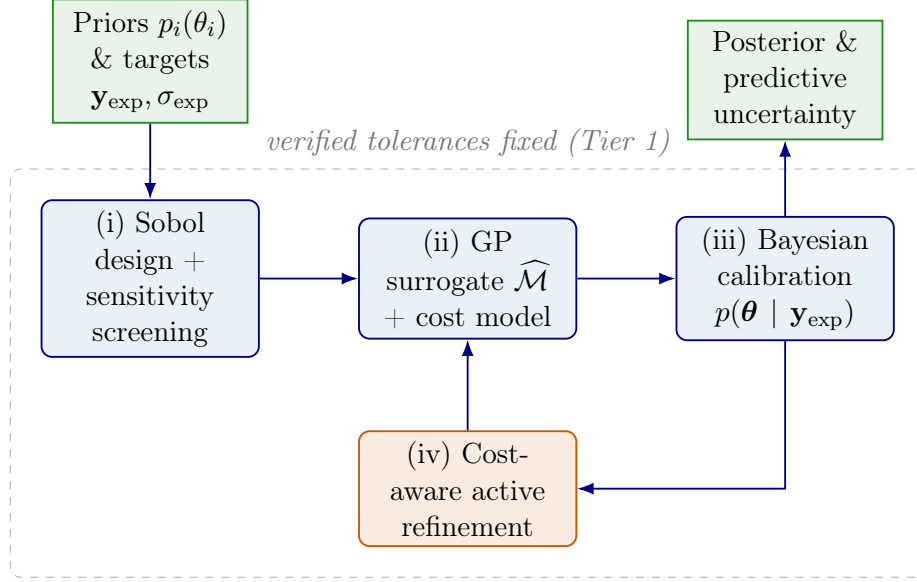


Figure 2: Uncertainty-quantification workflow for RADCLUSTER. Physical inference (stages i–iv) is carried out at verified integrator tolerances so that numerical error cannot masquerade as physics; the cost-aware active loop returns new simulations to the emulator where they are most informative per unit of CPU time.

mechanism: it is what lets the twin state which predictions are trustworthy, where the data are silent, and which additional irradiation experiments would most sharpen the picture. A description of the computational algorithm is given in Appendix F.

3. Comparison Between Modeling and Experiment

In situ TEM observations provide a complementary view of the defect evolution that the present model aims to capture. Zheng et al. [39] irradiated HT9 to 20 dpa with 1 MeV Kr ions at 420, 440, and 470°C and tracked loop formation dynamically. Both $a/2\langle 111 \rangle$ and $a\langle 100 \rangle$ loops were resolved, and at all three temperatures, the loop number density first rose with dose (up to ~ 4 dpa) and then fell at higher doses. The dislocation line density followed the opposite trend: pre-existing lines were initially lost by climb to the foil surfaces, while beyond ~ 4 dpa a new dislocation network built up as loops were absorbed into it, with the $a/2\langle 111 \rangle$ population most affected by this decline. Notably, the resulting $a\langle 100 \rangle$ to $a/2\langle 111 \rangle$ loop ratio was consistent with bulk 5 MeV Fe^{2+} irradiation of the same alloy to comparable doses and temperatures, supporting the use of these data as a benchmark for the simulated loop populations.

The quantity of interest is the content-weighted $\langle 100 \rangle$ fraction

$$f_{\langle 100 \rangle}(t) = \frac{\sum_n n c_n^{\langle 100 \rangle}}{\sum_n n c_n^{\langle 100 \rangle} + \sum_n n c_n^{\langle 111 \rangle}}, \quad (56)$$

reported by post-processing as `1 - f_111_loop`. The model (see `loop_111_to_100_conversion.t`) provides three additive conversion routes grouped into two mechanisms.

Mechanism A — Dudarev unary (thermodynamic).. A single loop reorients at

$$\Gamma_{\text{uni}}(n, T) = \nu_0 \exp\left[-\frac{E_a^{\text{uni}}(n)}{k_B T}\right] \max[0, 1 - e^{-\Delta F(n, T)/k_B T}], \quad E_a^{\text{uni}}(n) = E_a^0 + \gamma_a \frac{P(n)}{b_{111}}, \quad (57)$$

where $P(n) \propto \sqrt{n}$ is the loop perimeter and ΔF is the thermodynamic driving force, positive only above the crossover temperature T^* (`T_star_conv_C`).

Mechanism B — Marian junction + absorption (kinetic).. Two comparable $\langle 111 \rangle$ loops may condense into a $\langle 100 \rangle$ loop, and an existing $\langle 100 \rangle$ loop may absorb mobile $\langle 111 \rangle$ clusters. Both are gated by the two-step success probability through the metastable $\frac{1}{2}\langle 110 \rangle$ intermediate,

$$P_{\text{succ}}(T) = \frac{e^{-\Delta H_2/k_B T}}{e^{-\Delta H_2/k_B T} + e^{-\Delta H_{\text{rev}}/k_B T}} = \frac{1}{1 + \exp[(\Delta H_2 - \Delta H_{\text{rev}})/k_B T]}. \quad (58)$$

Critically, P_{succ} multiplies *both* the junction yield φ_{junc} and the absorption rate; it is the *only* explicit temperature dependence of Mechanism B.

Calibration campaign: what was varied and what happened

All runs use the C++ solver, fission spectrum, $G = 10^{-6}$ dpa/s, `full_CD_fission` on a reduced grid ($I = 200$, $i_{\text{mobile}} = 10$), loop conversion ON and the network-loss channel OFF, driven through `run_adaptive`. Driver: `codes/Python_Testing/sweep_`

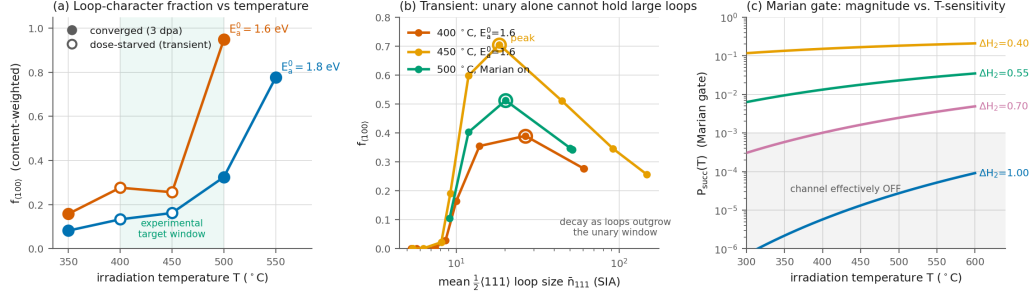


Figure 3: Summary of the measured campaign. (a) $f_{\langle 100 \rangle}(T)$ for the two unary calibrations; filled markers are converged (full 3 dpa), open markers are dose-starved transients (see §3.2). (b) The peak-then-decay signature, plotted against the growing mean $\langle 111 \rangle$ loop size: $f_{\langle 100 \rangle}$ rises while loops are small, then falls once they outgrow the unary window (open rings mark the peak). (c) The Marian gate $P_{\text{succ}}(T)$, Eq. (58): at the shipped $\Delta H_2 = 1.0$ eV the channel sits in the “effectively off” band, and raising its magnitude flattens its temperature slope.

The complete numerical output of this campaign is tabulated in Table 6, which lists every measured $f_{\langle 100 \rangle}(T)$ value together with the loop and cavity densities, mean sizes, and convergence status of each run. Table 7 then summarizes how the target $f_{\langle 100 \rangle}(500 \text{ °C})$ responds to each candidate calibration lever, isolating the effect of dose, gate temperature, and stiffness.

3.1. Failure mode I — the unary channel is small-loop biased

Because E_a^{uni} grows as $\sqrt{n}$ via Eq. (57), the unary rate collapses for large loops. The per-size conversion fraction measured in-run tracks the unary timescale exactly (Table 8), confirming the implementation is correct and the limitation is physical, not numerical.

The dynamical consequence is a **peak-then-decay** of $f_{\langle 100 \rangle}$: early in the irradiation the loops are small and convert efficiently, but as they grow past the unary window the un-converted $\langle 111 \rangle$ content accumulates and $f_{\langle 100 \rangle}$ falls back. This is observed directly in the per-segment traces:

Table 6: **Master result table** — every $f_{\langle 100 \rangle}(T)$ measurement obtained. “conv.” marks runs that reached the full 3 dpa and whose trace had gone flat; the remainder stopped at the quoted dose on the solver time limit and are transient snapshots. All runs: $T^* = 250^\circ\text{C}$, $\gamma_a = 0.02$, $I = 200$, conversion ON, network OFF.

Calibration	T ($^\circ\text{C}$)	$f_{\langle 100 \rangle}$	dose (dpa)	$\bar{n}_{111}$	$\bar{n}_{100}$	status
baseline $T^* = 450$	450	0.0002	10.0	206.3	7.9	conv. (no conversion)
$E_a^0 = 1.8$	350	0.0820	3.00	38.0	11.4	conv.
$E_a^0 = 1.8$	400	0.1332	0.50	50.3	11.8	dose-starved
$E_a^0 = 1.8$	450	0.1620	0.05	60.0	11.2	dose-starved
$E_a^0 = 1.8$	500	0.3249	3.00	52.0	9.3	conv.
$E_a^0 = 1.8$	500	0.3249	30.0	52.0	9.3	conv. (dose test)
$E_a^0 = 1.8$	550	0.7762	3.00	23.9	8.5	conv.
$E_a^0 = 1.6$	350	0.1579	3.00	55.0	10.9	conv.
$E_a^0 = 1.6$	400	0.2767	0.01	60.8	9.1	dose-starved
$E_a^0 = 1.6$	450	0.2562	0.00	148.4	8.4	dose-starved (still falling)
$E_a^0 = 1.6$	500	0.9483	0.50	13.5	8.1	flat/converged
$E_a^0 = 1.4$	500	0.9973	0.01	10.0	7.9	dose-starved
$E_a^0 = 1.8, \Delta H_2 = 0.55$	500	0.3421	10.0	51.6	9.3	conv. (282 s)

The last row of Table 9 is the exception that proves the mechanism: at 500°C with $E_a^0 = 1.6\text{ eV}$ the unary is fast enough that loops convert *while still small* ($\bar{n}_{111}$ saturates at 13.5), so they never enter the declining regime and $f_{\langle 100 \rangle}$ is stable. The decay appears precisely when loop growth outruns conversion.

Attempting to defeat this by lowering E_a^0 or γ_a pushes the system into a numerically stiff, actively-converting state in which the integration dose-starves — the crossover region (400–450 $^\circ\text{C}$) is by far the most expensive part of the parameter space.

3.2. Failure mode II — the Marian gate cannot generate a crossover

Equation (58) contains an intrinsic trade-off: the same parameter combination $\Delta = \Delta H_2 - \Delta H_{\text{rev}}$ sets *both* the magnitude and the temperature sensitivity of P_{succ} . Raising the magnitude to a useful level necessarily flattens the temperature response (Table 10).

At the shipped value $\Delta H_2 = 1.0\text{ eV}$ the Marian channels are effectively *switched off* ($P_{\text{succ}} \sim 10^{-5}$), which is why the unary has been performing

Table 7: Response of $f_{\langle 100 \rangle}(500^\circ\text{C})$ to each candidate lever.

Lever	Result	Interpretation
Dose $3 \rightarrow 30$ dpa	$0.3249 \rightarrow 0.3249$	Bit-identical. $f_{\langle 100 \rangle}$ is <i>efficiency</i> -limited, not dose-limited.
$T^* : 250 \rightarrow 180^\circ\text{C}$	0.3249 (unchanged)	The $\Delta F > 0$ cutoff rises $255 \rightarrow 493$, but the loops live at $\bar{n} \approx 52$ and never reach it, so the thermodynamic cutoff is irrelevant at this temperature.
$\gamma_a : 0.02 \rightarrow 0.015$	stiffness wall	γ_a scales with $P(n)$ and so only alters large- n rates where no population exists; it drags the actively-converting (stiff) regime onto 500°C , and the run dose-starves at 0.27 dpa.
$E_a^0 : 1.8 \rightarrow 1.6$ eV	$0.325 \rightarrow 0.948$	The effective lever. Size- <i>independent</i> , so it acts at $n \approx 10$ –50 where the loops actually are. $\bar{n}_{111}$ collapses $52 \rightarrow 13.5$.
$\Delta H_2 : 1.0 \rightarrow 0.55$ eV	$0.325 \rightarrow 0.342$	Marian engaged but weak ($P_{\text{succ}} \approx 2 \times 10^{-2}$); +5% only. Converged and fast (282 s to 10 dpa).

all of the conversion. Conversely, at $\Delta H_2 \lesssim 0.40$ eV the channels become strong but nearly temperature-independent, so they raise $f_{\langle 100 \rangle}$ almost *uniformly* at all temperatures rather than producing a crossover. Mechanism B therefore cannot, by construction, be the origin of the observed temperature dependence.

Why both mechanisms are necessary

The experimental observation imposes *two logically independent* requirements, and each mechanism satisfies exactly one of them.

1. **A steep temperature grading.** $f_{\langle 100 \rangle}$ must rise from $\lesssim 0.1$ at 350°C to ≈ 1 at 500°C . Only an Arrhenius rate competing against loop growth can produce this. The unary channel supplies it through $\exp[-E_a^{\text{uni}}/k_B T]$ together with the ΔF gate; the measured response ($f_{\langle 100 \rangle} = 0.16$ at 350°C versus 0.95 at 500°C for $E_a^0 = 1.6$ eV) is precisely this competition. Mechanism B cannot supply it (Table 10).
2. **Dose stability of the converted fraction.** The observed $\langle 100 \rangle$

Table 8: In-run conversion fraction per size at 500 °C ($E_a^0 = 1.8$, $\gamma_a = 0.02$). The C++ unary drains precisely as its rate predicts.

n	τ_{uni}	$c^{\langle 100 \rangle} / (c^{\langle 100 \rangle} + c^{\langle 111 \rangle})$
4	0.4 s	0.95
15	2.4 s	0.73
30	12 s	0.36
67	166 s	0.05
149	8600 s	0.001

Table 9: Measured peak-then-decay. $f_{\langle 100 \rangle}$ overshoots while the loops are still small, then relaxes as $\bar{n}_{111}$ grows past the unary window. Plotted in Fig. 3(b).

T	calibration	peak $f_{\langle 100 \rangle}$	at $\bar{n}_{111}$	final $f_{\langle 100 \rangle}$	final $\bar{n}_{111}$
400 °C	$E_a^0 = 1.6$	0.390	26.6	0.277	60.8
450 °C	$E_a^0 = 1.6$	0.705	18.4	0.256	148.4 (still falling)
500 °C	$E_a^0 = 1.8$, $\Delta H_2 = 0.55$	0.512	20.1	0.342	51.6 (flat to 10 dpa)
500 °C	$E_a^0 = 1.6$	0.949	13.3	0.948	13.5 (no decay)

dominance is a *steady* feature of the irradiated microstructure, not a transient. Because the unary barrier grows as $\sqrt{n}$, Mechanism A alone *cannot* hold loops that have grown past its window, and $f_{\langle 100 \rangle}$ decays after an early peak (450 °C: $0.705 \rightarrow 0.256$). Sustaining $f_{\langle 100 \rangle}$ requires a channel whose rate does *not* carry the $\sqrt{n}$ penalty and which acts on the escaping population: the Marian junction (which requires $\min(n, n') \geq n_{j,\text{min}} = 30$, i.e. it targets exactly the large loops) and the absorption route (which continuously drains mobile $\langle 111 \rangle$ into $\langle 100 \rangle$). Mechanism A cannot supply this.

Stated compactly: **Mechanism A sets *where* (in temperature) the conversion turns on; Mechanism B sets *whether it survives* to technologically relevant dose.** A model containing only the unary channel reproduces the crossover but predicts a spurious decay of $f_{\langle 100 \rangle}$ with dose; a model containing only the Marian channels reproduces a dose-stable $\langle 100 \rangle$ population but with essentially no temperature dependence, contradicting the observed low-temperature $\langle 111 \rangle$ dominance. Only the combination is consistent with both the temperature trend and the dose behavior reported

Table 10: The magnitude–sensitivity trade-off of the Marian gate, Eq. (58), with $\Delta H_{\text{rev}} = 0.30 \text{ eV}$.

$\Delta H_2 \text{ (eV)}$	$P_{\text{succ}}(400^\circ\text{C})$	$P_{\text{succ}}(500^\circ\text{C})$	ratio 500/400
1.00 (shipped)	5.7×10^{-6}	2.7×10^{-5}	$4.76\times$
0.70	1.0×10^{-3}	2.5×10^{-3}	$2.44\times$
0.55	1.3×10^{-2}	2.3×10^{-2}	$1.73\times$
0.40	1.5×10^{-1}	1.8×10^{-1}	$1.20\times$
0.35	2.97×10^{-1}	3.21×10^{-1}	$1.08\times$

experimentally.

This also rationalizes the historical finding recorded in `loop_conversion_calibration.md` that the model “over-converted at all temperatures” before P_{succ} was introduced: without the gate, the kinetic channels were both strong *and* temperature-blind, so they masked the unary temperature trend entirely.

Recommended parameter set and status Table 11 lists the current recommended calibration, giving each parameter’s adopted value and its role, with the $f_{\langle 100 \rangle}$ entries content-weighted at the stated dose.

Table 11: Current calibration. $f_{\langle 100 \rangle}$ values are content-weighted at the stated dose.

Parameter	Value	Role
T_star_conv_C	250 °C	ΔF gate; must lie below the desired crossover
E_a0_conv	1.6 eV	sets the crossover temperature
gamma_a_conv	0.02	size penalty; leave at the physical value
dH2_conv	0.55 $\rightarrow$ 0.40 eV (under test)	engages Mechanism B for dose stability

Measured response at $E_a^0 = 1.6 \text{ eV}$: 350 °C: 0.158 (3.0 dpa, converged); 400 °C: 0.277; 450 °C: 0.256; 500 °C: 0.948 (flat, $\bar{n}_{111} = 13.5$).

Numerical caveat. The 400–450 °C points are *dose-starved*: they reached only 0.014 and 0.001 dpa, respectively, before the solver time limit, and their traces were still evolving. They are transient snapshots, *not* converged values, and the apparent non-monotonicity ($450 < 400$) is an artifact of this. The 350 °C and 500 °C anchors are converged and trustworthy. A full-dose crossover curve remains a cluster-scale computation.

Outstanding work.. (i) Complete the strong-Marian test ($\Delta H_2 = 0.40$, $\varphi_{\max} = 1$) at 450 and 500 °C to confirm that Mechanism B removes the peak-then-decay; (ii) with Mechanism B active, re-fit E_a^0 (it will need to *rise* again since Marian now contributes to conversion); (iii) run the converged $f_{\langle 100 \rangle}(T)$ curve at experimental dose offline.

Companion result: the loop $\rightarrow$ network channel (ρ_{net})

The same campaign commissioned the dislocation-evolution channel (`LOOP_NETWORK_LOSS`, see `loop_network_loss.tex`), which was previously dormant. Two prerequisites had to be met before ρ_{net} would move at all, and both are recorded here because they are easy to re-trip:

1. The driver must call `sim.run_adaptive()`, *not* `sim.run()` — the operator-split update of ρ_{net} executes only *between* integration segments, so a single-shot run leaves it exactly constant even with the flag enabled.
2. The geometric capture switch $P_{\ell d}$ only opens when $\chi d_{\text{loop}} \gtrsim L_{\ell d} = \rho_{\text{net}}^{-1/2}$. With few-nm loops against a ~ 100 nm network spacing, this requires $\chi \gtrsim 30$; at the physical $\chi \sim 1$, the gain is *identically zero*. Equivalently, reduced- I test grids (whose loops stay small) silently disable the channel.

Table 12 records the resulting network-density response at 450 °C and 10 dpa, showing how the dislocation-network density ρ_{net} evolves from its initial to its final value as the capture width w_c is varied.

Table 12: Network-density response at 450 °C, 10 dpa, $I = 1000$ (bin-moment, conversion off), $\chi = 50$, $K_{\text{rec}} = 0$, $\rho_{\text{max}} = 5 \times 10^{14} \text{ m}^{-2}$. $\dot{\rho} \propto w_c$ was verified linear.

capture width w_c	ρ_{net} initial	ρ_{net} final	note
b_{111} (physical)	5×10^{13}	—	$\sim 0.1\%$ /dpa; would saturate at ~ 160 dpa
$50 b_{111}$	5.00×10^{13}	1.85×10^{14}	$\times 3.7$
$150 b_{111}$	5.00×10^{13}	4.56×10^{14}	$\times 9.1$, reaches the plateau

The physical capture width gives a real but negligible drift because the net climb velocity is small ($v_{\text{net}} \approx 5.5 \times 10^{-13} \text{ m/s}$: the SIA and vacancy fluxes to the climbing network very nearly cancel). w_c is therefore used as the designated amplification knob, exactly as in `check_loop_network_loss.py`. At $w_c = 150 b_{111}$, the network density rises by an order of magnitude and saturates at the experimental EUROFER plateau — the intended “loop density

saturates with dose” behavior. The precise physical (w_c, K_{rec}) pair remains an offline calibration.

4. Conclusions

Supplementary Material

Appendix A. Production

Let G denote the *actual* (survival-corrected) defect production rate per unit volume, i.e.

$$G = \eta G_{\text{NRT}}, \quad (\text{A.1})$$

where $\eta = N_d/N_{\text{NRT}}$ is the survival efficiency, and G_{NRT} is the NRT displacement rate. The production rate of interstitial clusters of size m is $P_m^{(i)} = \epsilon_m^{(i)} G$, and the mono-interstitial production rate is

$$P_1^{(i)} = G \left(1 - \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)} \right). \quad (\text{A.2})$$

Analogous expressions hold for vacancies with superscript (v) , cluster size index n , and upper cutoff v_{cascade} . The total fraction of surviving SIAs born in clusters is $f_i^{\text{cl}} = \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)}$, with the corresponding vacancy quantity f_v^{cl} . Note that the mono-defect fraction is simply $1 - f^{\text{cl}}$. The fraction of NRT-predicted Frenkel pairs that survive in-cascade recombination is well characterized by MD simulations [40–42]. For α -Fe at reactor-relevant temperatures, the survival efficiency is approximately

$$\eta(E_{\text{PKA}}) \approx 0.84 \left(\frac{E_{\text{PKA}}}{1 \text{ keV}} \right)^{-0.20}, \quad E_{\text{PKA}} \gtrsim 1 \text{ keV}, \quad (\text{A.3})$$

saturating near $\eta \approx 0.30$ above ~ 10 keV. MD cascade simulations in α -Fe consistently show that the *number* of in-cascade clusters of size m follows an approximate power law [40, 43–45]: $N_{\text{cl}}(m) \propto m^{-s}$, with distinct exponents for SIA clusters (s_i) and vacancy clusters (s_v). The steeper vacancy exponent reflects the well-known inhibition of true vacancy clustering in bcc Fe, where vacancy “clusters” are often loose, correlated arrangements rather than compact loops [43, 46]. The production fraction is therefore modeled as $\epsilon_m^{(i)} = C_i m^{-s_i}$, $m = 2, \dots, i_{\text{cascade}}$, where the normalization constant C_i is fixed by the constraint

$$C_i = \frac{f_i^{\text{cl}}}{\sum_{m=2}^{m_{\text{cascade}}} m^{1-s_i}}. \quad (\text{A.4})$$

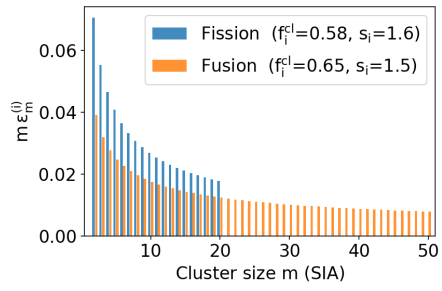
An identical prescription applies to vacancies with parameters $(C_v, s_v, v_{\text{cascade}}, f_v^{\text{cl}})$.

The parameters that distinguish fission and fusion spectra in terms of cascade damage are summarized in Table A.13. Fission-spectrum neutrons produce a PKA distribution weighted toward lower energies ($\langle E_{\text{PKA}} \rangle \approx 10\text{--}15\text{ keV}$), whereas 14 MeV fusion neutrons shift the average to $\sim 30\text{--}50\text{ keV}$, leading to subcascade fragmentation [44, 47]. Higher PKA energies produce larger in-cascade clusters, so the maximum cluster sizes i_{cascade} and v_{cascade} and the power-law exponents all shift [40, 45]. Helium generation is 0.5–1 appm He/dpa for fission spectra and ~ 10 appm He/dpa for fusion, affecting vacancy cluster (bubble) nucleation at longer times but not the primary cascade spectrum itself.

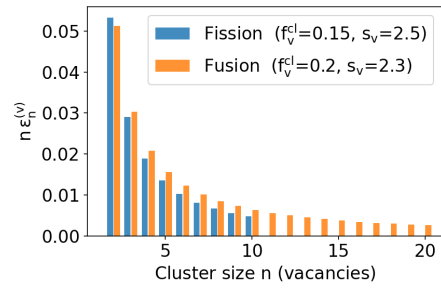
Table A.13: Cascade-damage parameters for α -Fe under fission and fusion neutron spectra, synthesized from [40], [41], [44], and [48].

Parameter	Fission	Fusion
$\langle E_{\text{PKA}} \rangle$ (keV)	~ 15	~ 40
η (spectrum-averaged)	0.30	0.28
f_i^{cl} (SIA clustering fraction)	0.58	0.65
f_v^{cl} (vacancy clustering fraction)	0.15	0.20
s_i (SIA power-law exponent)	1.6	1.5
s_v (vacancy power-law exponent)	2.5	2.3
i_{cascade} (max SIA cluster size)	20	50
v_{cascade} (max vacancy cluster size)	10	20
He production (appm He/dpa)	0.5–1	~ 10

Figure A.4a shows the size distribution of SIA clusters produced in cascades in typical fission and fusion neutron spectra, while Figure A.4b displays similar information for vacancy clusters.



(a)



(b)

Figure A.4: (A.4a): Interstitial cluster production spectrum under fission and fusion conditions, (A.4b): Vacancy cluster production spectrum under fission and fusion conditions

Appendix B. Energetics

Appendix B.1. Formation and Migration

In bcc α -iron (lattice constant $a = 2.87 \text{ \AA}$) defects migrate by thermally activated jumps between lattice sites. Within harmonic transition-state theory [49] the jump rate is $\nu = \nu_0 \exp(-E_m/k_B T)$, where E_m is the migration energy, ν_0 the attempt frequency, k_B Boltzmann’s constant, and T the temperature; the resulting diffusion coefficients are derived in Section Appendix C. Table E.25 summarizes the key energetic parameters for the three defect species.

The attempt frequency of a monovacancy spans a wide range in the literature. The Debye-based estimate $\nu_0^v \approx 6 \times 10^{12} \text{ s}^{-1}$ is used in rate-theory models [50, 51]. Full DFT calculations give $\sim 5 \times 10^{13} \text{ s}^{-1}$ [52], while values as large as $5 \times 10^{15} \text{ s}^{-1}$ are obtained from fitting self-diffusion data [53] because the experimental pre-exponential D_0 folds in the vacancy formation entropy $\Delta S_f^v \approx 4 k_B$ [54].

The ground-state SIA in ferromagnetic bcc Fe is the $\langle 110 \rangle$ dumbbell (unique among bcc metals owing to magnetism) [55]. Its translation-rotation migration has $E_m = 0.34 \text{ eV}$ and $\nu_0 \approx 4 \times 10^{12} \text{ s}^{-1}$ from DFT [52, 56]. The $\langle 111 \rangle$ crowdion has a negligible barrier ($\sim 0.013 \text{ eV}$) and migrates nearly athermally [55]. The $\langle 110 \rangle$ dumbbell creates a *high-frequency localized mode*: the two dumbbell atoms oscillate against each other along the $\langle 110 \rangle$ bond direction, giving a stretch frequency of $\approx 60\text{--}80 \text{ meV}$ (14–19 THz) [56, 57].

Interstitial He occupies the tetrahedral site in bcc Fe, which is more stable than the octahedral site by 0.2 eV (DFT) [58]. He migrates via tetrahedral-to-tetrahedral hops through the octahedral saddle with a very low barrier of $E_m^{He} = 0.06 \text{ eV}$ [58] (0.078–0.086 eV from MD and other DFT methods [59, 60]), making He diffusion essentially athermal above room temperature. Because $m_{He}/m_{Fe} \approx 0.07$, He rattles inside the tetrahedral cage at a frequency far above the host phonon band. This localized optical mode has been confirmed by DFT phonon calculations [58, 61]. The ordering of attempt frequencies mirrors the Debye frequency: $\nu_0^{He} \lesssim \nu_0^{SIA} \lesssim \nu_0^V \sim \nu_D$. All lie within roughly an order of magnitude of $6 \times 10^{12} \text{ s}^{-1}$, and the Debye frequency is an adequate approximation when full Vineyard calculations are unavailable. The strongly differing migration energies — He ($< 0.1 \text{ eV}$) $\ll$ SIA (0.34 eV) $\ll$ V (0.55–0.62 eV) — dominate the diffusivity hierarchy over many orders of magnitude.

Appendix B.2. Binding Energies

The equilibrium thermal vacancy concentration in the bulk is approximated by

$$C_v^0 = \exp\left(-\frac{E_v^f}{k_B T}\right), \quad (\text{B.1})$$

where E_v^f is the vacancy formation free energy, or, in the usual approximation, the vacancy formation energy. At the surface of a vacancy cluster containing α vacancies, the local equilibrium vacancy concentration is elevated above C_v^0 according to

$$C_v^e(\alpha) = C_v^0 \exp\left(\frac{\mu(\alpha)}{k_B T}\right) = \exp\left[-\frac{E_v^f - \mu(\alpha)}{k_B T}\right] \quad (\text{B.2})$$

where $\mu(\alpha)$ is the vacancy chemical potential of the cluster. For an interstitial cluster containing β interstitials, the equilibrium vacancy concentration at the surface is lowered below C_v^0 :

$$C_v^e(\beta) = C_v^0 \exp\left(-\frac{\mu(\beta)}{k_B T}\right) = \exp\left[-\frac{E_v^f + \mu(\beta)}{k_B T}\right]. \quad (\text{B.3})$$

We can have a unified expression for both types of clusters by defining the binding energy of a vacancy to a vacancy cluster as

$$E_b^{(v)}(\alpha) = E_v^f - \mu(\alpha) \quad (\text{B.4})$$

and the binding energy of a vacancy to an interstitial cluster of size β is

$$E_b^{(i)}(\beta) = E_v^f + \mu(\beta) \quad (\text{B.5})$$

Then, the equilibrium vacancy concentration at the surface of a vacancy or interstitial cluster can be written as

$$C_v^e = \exp\left[-\frac{E_b}{k_B T}\right] \quad (\text{B.6})$$

The binding energy can be found from atomistic calculations (e.g., DFT or MD), from experiments, or by using continuum theory for large clusters.

Appendix B.2.1. Bubbles

For a spherical cavity of radius

$$R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3}, \quad (\text{B.7})$$

the defect chemical potential for vacancy exchange may be written as

$$\mu(\alpha) = \Omega \left(\frac{2\gamma}{R(\alpha)} - P_g - \sigma_h \right), \quad (\text{B.8})$$

where γ is the surface energy, Ω is the atomic volume, σ_h is the hydrostatic *tension* component of the stress, and P_g is the internal gas pressure. The gas pressure inside a helium bubble may be represented more accurately by a virial equation of state rather than by a hard-core Van der Waals approximation. Writing the helium number density in the bubble as

$$n_{\text{He}} = \frac{m}{V_b} = \frac{m}{\alpha\Omega}, \quad (\text{B.9})$$

where m is the number of helium atoms in the bubble, α is the number of vacancies, Ω is the atomic volume, and $V_b = \alpha\Omega$ is the bubble volume, the pressure may be written as

$$P_g = n_{\text{He}} k_B T \left[1 + B_2(T) n_{\text{He}} + B_3(T) n_{\text{He}}^2 + \dots \right]. \quad (\text{B.10})$$

Equivalently,

$$P_g = \frac{mk_B T}{\alpha\Omega} \left[1 + B_2(T) \frac{m}{\alpha\Omega} + B_3(T) \left(\frac{m}{\alpha\Omega} \right)^2 + \dots \right]. \quad (\text{B.11})$$

Here $B_2(T)$, $B_3(T)$, $\dots$ are the second, third, and higher virial coefficients of helium, which are functions of temperature. At temperatures in the range 600–1000 K, representative average virial coefficients for ^4He are

$$B_2 \approx 1.67 \times 10^{-29} \text{ m}^3/\text{atom}, \quad (\text{B.12})$$

$$B_3 \approx 1.84 \times 10^{-58} \text{ m}^6/\text{atom}^2, \quad (\text{B.13})$$

where these units are to be interpreted consistently with n_{He} in atoms m^{-3} . These values correspond to $\bar{B}_2 \approx 10.06 \text{ cm}^3 \text{ mol}^{-1}$ and $\bar{B}_3 \approx 66.72 \text{ cm}^6 \text{ mol}^{-2}$ for ^4He over 600–1000 K [62, 63].

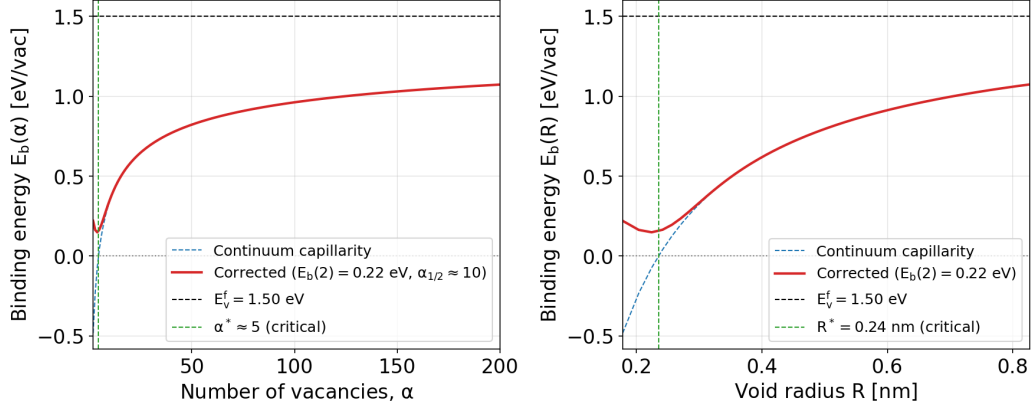


Figure B.5: Binding energy of a vacancy to a void as a function of the number of vacancies in the void, as a function of the number of vacancies (left), and the radius (right).

The binding energy of a vacancy to a void is expressed as

$$E_b^{\text{mod}}(\alpha) = E_b^{\text{cont}}(\alpha) + [E_b^{(v)}(2) - E_b^{\text{cont}}(2)] \exp(-\lambda(\alpha - 2)) \quad (\text{B.14})$$

With $E_b^{(v)}(2)$ the divacancy binding energy from DFT, and without external stress, the continuum capillary term is

$$E_b^{\text{cont}}(\alpha) = E_v^f - \Omega \frac{2\gamma}{R(\alpha)}, \quad R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3} \quad (\text{B.15})$$

The decay constant is chosen so that the correction is 1% at $\alpha = 10$: $\lambda = \frac{\ln 100}{8} \approx 0.575 \text{ vac}^{-1}$. The binding energy of a vacancy to a void as a function of the void size is shown in Fig. B.5, using the capillary approximation for large sizes and transitioning to atomistic data for small sizes using Eq. B.14.

The defining feature of helium in irradiated metals is its extremely strong binding to vacancies. DFT calculations give a binding energy of $E_b^{h-V} \approx 2.0$ – 2.4 eV for the first He atom entering a monovacancy [64–66]. The binding remains strong as additional He atoms are added: up to 4–6 He atoms can be accommodated in a single vacancy with cumulative binding energies exceeding 4 eV [64]. These He_nV_m complexes are essentially immobile (substitutional He migration requires the vacancy mechanism with a barrier of ~ 1.1 eV) and serve as the nucleation sites for helium bubbles.

The consequence for transport is that the *free* interstitial helium concentration c_h available for long-range diffusion is not the total implanted He

content but rather the small fraction that has not yet been trapped by vacancies, vacancy clusters, or other deep traps. In a vacancy-rich irradiation environment, the free He concentration can be orders of magnitude below the total He inventory. A helium–vacancy cluster $\text{He}_m\text{V}_\alpha$ is characterized by two integers: α (the number of vacancies) and m (the number of trapped helium atoms). Cluster dynamics requires the binding energy of both a vacancy and a helium atom to such a cluster, since these two binding energies govern, respectively, the vacancy emission rate (cavity growth/shrinkage) and the helium emission rate (gas release). We treat each in turn.

Table B.14 collects representative DFT/MD binding energies for $\text{He}_m\text{V}_\alpha$ clusters in α -Fe. The values are recast from the dissociation energies tabulated by Fu and Willaime [67] and Caturla et al. [68], using the definitions:

$$E_B^V(\alpha, m) = E_f(\text{He}_m\text{V}_{\alpha-1}) + E_f(V) - E_f(\text{He}_m\text{V}_\alpha), \quad (\text{B.16})$$

$$E_B^{\text{He}}(\alpha, m) = E_f(\text{He}_{m-1}\text{V}_\alpha) + E_f(\text{He}) - E_f(\text{He}_m\text{V}_\alpha). \quad (\text{B.17})$$

Table B.14: Binding energies for vacancy and helium addition to $\text{He}_m\text{V}_\alpha$ clusters in α -Fe. Data from [67] and [68].

α	m	$E_B^V(\alpha, m)$ (eV)	$E_B^{\text{He}}(\alpha, m)$ (eV)
1	1	2.30	—
2	1	0.78	2.85
3	1	0.83	3.30
4	1	1.16	3.84
1	2	3.71	1.84
2	2	1.61	2.75
3	2	1.04	2.96
4	2	1.32	3.12
1	3	4.59	1.83
2	3	1.85	2.07
4	3	1.57	3.16
1	4	5.52	1.91
2	4	2.30	2.36
3	4	2.03	2.57
4	4	1.97	3.05

Two important trends are visible in Table B.14:

1. $E_B^V(\alpha, m)$ **increases** with m at fixed α : more helium inside a cavity raises its internal pressure, making it harder for the cavity to emit a vacancy. This is the mechanism by which helium stabilizes bubbles against thermal dissolution.
2. $E_B^{\text{He}}(\alpha, m)$ **decreases** with m at fixed α : as the cavity fills with helium, the incremental binding of each additional He atom decreases because the available trapping volume shrinks and the He–He repulsion increases.

For the vacancy binding energy, we use the same atomistic–continuum transitional model developed for voids, extended to include the internal gas pressure:

$$E_B^v(\alpha, m) = \underbrace{E_f^v - \Omega \left[\frac{2\gamma}{R(\alpha)} - P_g(\alpha, m) \right]}_{E_B^{v, \text{cont}}(\alpha, m)} + A(m) e^{-\lambda(\alpha-1)}, \quad (\text{B.18})$$

with subsidiary expressions:

$$R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3}, \quad (\text{B.19})$$

$$P_g(\alpha, m) = \frac{m}{\alpha \Omega} k_B T \left[1 + B_2 \frac{m}{\alpha \Omega} + B_3 \left(\frac{m}{\alpha \Omega} \right)^2 \right], \quad (\text{B.20})$$

$$\lambda = \frac{\ln 100}{8} \approx 0.5756 \text{ vac}^{-1}. \quad (\text{B.21})$$

The continuum term $E_B^{v, \text{cont}}$ captures the competition between surface tension ($2\gamma/R$, which favours vacancy emission) and gas pressure (P_g , which opposes it). The exponential correction $A(m) \exp[-\lambda(\alpha - 1)]$ bridges the atomistic small-cluster binding energies to the continuum limit, decaying to less than 1% by $\alpha = 10$.

The fitted amplitudes $A(m)$, obtained by least-squares regression of Eq. (B.18) against the atomistic data of [69] ($m = 0$) and [68] ($m \geq 1$), are:

The amplitude at each integer m_k is determined by:

$$A(m_k) = \frac{\sum_{i \in \mathcal{D}_k} \delta_i e^{-\lambda(\alpha_i-1)}}{\sum_{i \in \mathcal{D}_k} e^{-2\lambda(\alpha_i-1)}}, \quad \delta_i = E_B^{v, \text{atom}}(\alpha_i, m_k) - E_B^{v, \text{cont}}(\alpha_i, m_k), \quad (\text{B.22})$$

Table B.15: Least-squares fitting amplitudes for the vacancy binding energy to bubbles (Eq. B.18).

m	$A(m)$ [eV]
0	1.2353
1	2.9064
2	3.4147
3	2.1504
4	-0.1590

where $\mathcal{D}_k$ is the subset of atomistic data at fixed m_k . The Binding energy of a vacancy to a bubble as a function of the number of vacancies for several He-to-vacancy ratios is shown in Figure B.6.

Contour maps of vacancy binding energy are shown in Figure B.7. The dashed lines $E_B^v = E_f^v$ separate over-pressurized bubbles (above, which grow by absorbing thermal vacancies) from under-pressurized cavities (below, which tend to shrink)

The helium binding energy $E_B^{\text{He}}(\alpha, m)$ —the energy required to remove one He atom from a $\text{He}_m\text{V}_\alpha$ cluster and place it at a tetrahedral interstitial site in the bulk—is needed for the helium thermal emission (detrapping) rate and for determining the maximum stable He loading $\ell_{\text{max}}(\alpha)$.

When a He atom is removed from a bubble of α vacancies containing m helium atoms, two energy contributions change: (i) the He–cavity binding interaction (governed by the attractive potential well of the vacant volume) is lost, and (ii) the internal gas pressure drops, slightly reducing the He–He repulsion among the remaining $m - 1$ atoms. For small clusters ($\alpha \leq 4$, $m \leq 4$), both contributions are highly configuration-dependent; for large bubbles, they are captured by continuum thermodynamics.

For a spherical bubble of radius $R(\alpha)$ containing m He atoms, the chemical potential of a helium atom inside the bubble is:

$$\mu_{\text{He}}^{\text{bubble}} = k_B T \ln\left(\frac{m}{\alpha\Omega} \Lambda_h^3\right) + k_B T \left[2B_2 \frac{m}{\alpha\Omega} + 3B_3 \left(\frac{m}{\alpha\Omega}\right)^2 \right], \quad (\text{B.23})$$

where $\Lambda_h = h/\sqrt{2\pi m_{\text{He}} k_B T}$ is the thermal de Broglie wavelength of ^4He . The helium binding energy in the continuum limit is the energy difference between this chemical potential and the energy of an interstitial He atom in

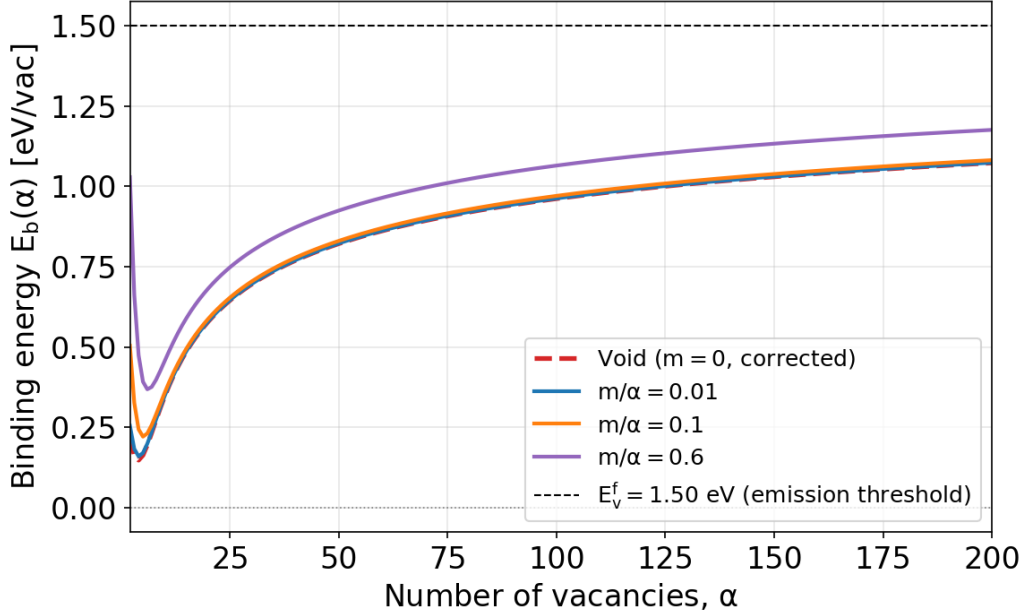


Figure B.6: Binding energy of a vacancy to a bubble as a function of the number of vacancies α , for several He-to-vacancy ratios m/α . The binding energy increases with helium content because the internal gas pressure opposes vacancy emission.

the bulk:

$$E_B^{\text{He,cont}}(\alpha, m) = E_s^{\text{He}} - k_B T \ln\left(\frac{m}{\alpha\Omega} \Lambda_h^3\right) - k_B T \left[2B_2 \frac{m}{\alpha\Omega} + 3B_3 \left(\frac{m}{\alpha\Omega}\right)^2 \right], \quad (\text{B.24})$$

where $E_s^{\text{He}} \approx 2.3\text{--}2.4$ eV is the solution energy of interstitial He in bulk bcc Fe (the energy cost of inserting a He atom at a tetrahedral site). The first correction term is the ideal-gas entropy contribution, and the second accounts for the non-ideal (hard-core repulsive) He–He interactions through the virial coefficients.

Equation (B.24) predicts two important limits:

- **Low He density** ($m/\alpha \ll 1/\Omega B_2$): the binding energy is large ($\sim E_s^{\text{He}}$), since the He atom gains the full solution-energy benefit of entering the open volume.
- **High He density** (m/α approaching the close-packing limit): the binding energy decreases toward zero and eventually becomes nega-

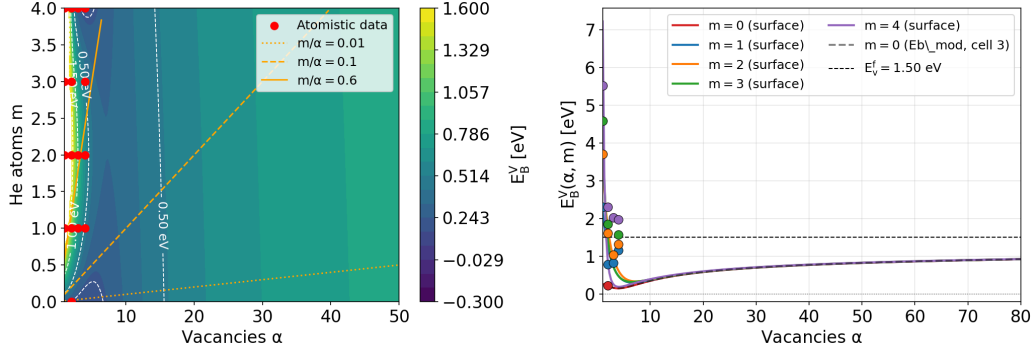


Figure B.7: Contour maps of vacancy binding energy $E_B^v(\alpha, m)$. Left: surface plot as a function of α and m . Right: cuts at fixed m showing the atomistic-continuum transition.

tive, meaning it is energetically favorable for the He atom to leave the overpressurized bubble. This defines the maximum stable He loading.

By direct analogy with the vacancy binding energy model (Eq. B.18), we write:

$$E_B^{\text{He}}(\alpha, m) = E_B^{\text{He,cont}}(\alpha, m) + A^{\text{He}}(\alpha) e^{-\mu(m-1)}, \quad (\text{B.25})$$

where $A^{\text{He}}(\alpha)$ is a fitted amplitude at each vacancy count α , and μ is a decay constant governing how quickly the atomistic correction vanishes as m increases. The exponential correction ensures that the blended expression reproduces the DFT data at small m and converges to the continuum model for large m .

The decay constant is chosen by the same criterion as for the vacancy case (correction $< 1\%$ at $m = 8$):

$$\mu = \frac{\ln 100}{7} \approx 0.658 \text{ He}^{-1}. \quad (\text{B.26})$$

The amplitudes $A^{\text{He}}(\alpha)$ are obtained by least-squares regression against the atomistic data of Table B.14:

$$A^{\text{He}}(\alpha_k) = \frac{\sum_{j \in \mathcal{E}_k} \delta_j^{\text{He}} e^{-\mu(m_j-1)}}{\sum_{j \in \mathcal{E}_k} e^{-2\mu(m_j-1)}}, \quad \delta_j^{\text{He}} = E_B^{\text{He,atom}}(\alpha_k, m_j) - E_B^{\text{He,cont}}(\alpha_k, m_j), \quad (\text{B.27})$$

where $\mathcal{E}_k$ is the subset of the atomistic data at fixed α_k .

Table B.14 and Eq. (B.24) reveal the following trends:

Table B.16: Least-squares fitting amplitudes for the helium binding energy to bubbles (Eq. B.25), obtained from the data of [67] and [68].

α	$A^{\text{He}}(\alpha)$ [eV]
1	~ 0 (He_1V_1 is the anchor; E_B^{He} undefined)
2	0.55
3	0.40
4	0.75

1. At fixed α , E_B^{He} **decreases** as m increases (the cavity fills up and the marginal binding drops). For example, at $\alpha = 2$: $E_B^{\text{He}} = 2.85$ eV ($m = 1$), 2.75 eV ($m = 2$), 2.07 eV ($m = 3$), 2.36 eV ($m = 4$). The non-monotonic behavior at $m = 4$ reflects configuration-dependent effects in the small DFT clusters.
2. At fixed m , E_B^{He} **increases** as α increases (larger cavities have more open volume, so each He atom is more strongly bound). For example, at $m = 1$: $E_B^{\text{He}} = 2.85$ eV ($\alpha = 2$), 3.30 eV ($\alpha = 3$), 3.84 eV ($\alpha = 4$).
3. The He binding energy at $\alpha = 1$ (substitutional He) is ~ 1.8 – 1.9 eV for $m \geq 2$, significantly lower than at $\alpha \geq 2$. This reflects the small trapping volume of a single vacancy.

An over-pressurized $\text{He}_m\text{V}_\alpha$ cluster—one whose He content exceeds the equilibrium loading—can relieve its excess energy by two physically distinct mechanisms. These must be distinguished in the rate equations because they produce different products and obey different kinetics.

The results are shown in Fig. B.6 for the binding energy as a function of both m and α . Contours of constant binding energy in bubbles containing m helium atoms and α vacancies are shown in Fig. B.8. The line $E_v^f = E_b^{(v)}(m, \alpha)$ defines the combination of m and α for bubbles in mechanical equilibrium. Those above the line (over-pressurized) tend to grow by absorbing thermal vacancies, while those below the line tend to shrink.

Appendix B.2.2. Dislocation Loops

For dislocation loops, the vacancy exchange chemical potential should be obtained from the loop free energy.

$$\mu = \frac{\partial G_{\text{loop}}}{\partial n}, \quad (\text{B.28})$$

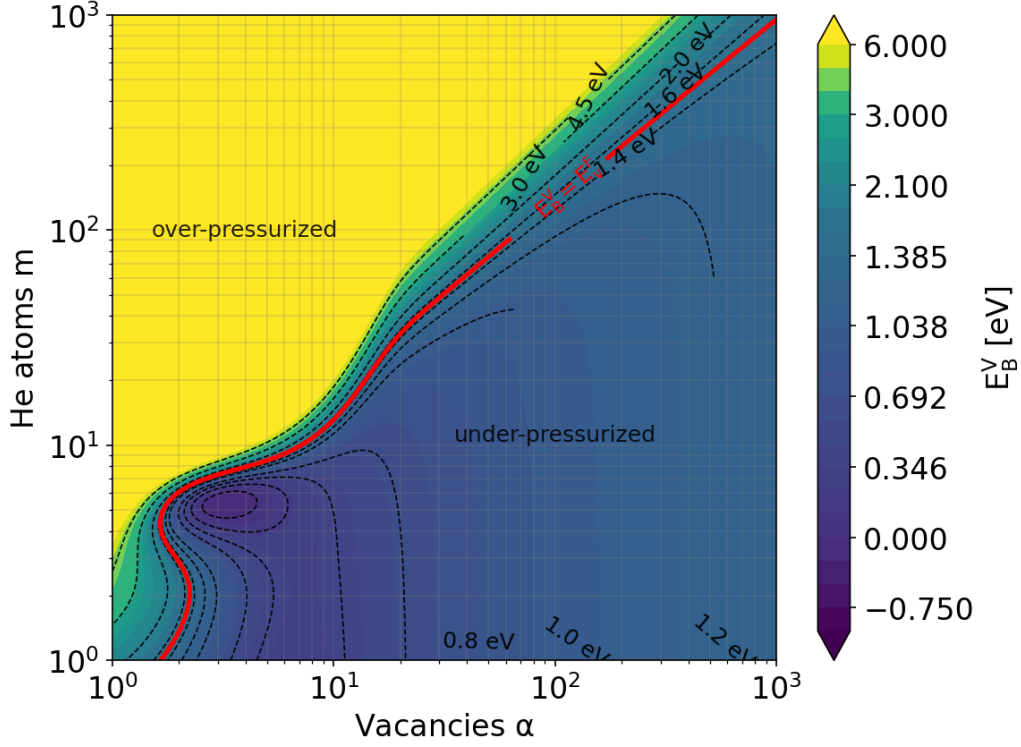


Figure B.8: Vacancy binding energy contours to bubbles containing m helium atoms and α vacancies

where n is the number of point defects contained in the loop. For a faulted circular loop, G_{loop} generally contains both line-energy and fault-energy contributions. For a circular prismatic loop containing n point defects,

$$n\Omega = \pi R^2 b, \quad (\text{B.29})$$

For a loop in an external stress field σ_{ij}^{ext} , the loop free energy should include the mechanical work term,

$$G_{\text{loop}}^{\text{tot}} = G_{\text{self}} + G_{\text{fault}} + G_{\sigma}, \quad (\text{B.30})$$

with

$$G_{\sigma} = -\sigma_{ij}^{\text{ext}} \Omega_{ij}^{\text{loop}}. \quad (\text{B.31})$$

For a circular prismatic loop of radius R , plane normal n_i , and Burgers vector $b_i = b n_i$, the loop relaxation-volume tensor is

$$\Omega_{ij}^{\text{loop}} = s n \Omega n_i n_j = s \pi R^2 b n_i n_j, \quad (\text{B.32})$$

where

$$s = \begin{cases} -1, & \text{vacancy loop,} \\ +1, & \text{interstitial loop.} \end{cases} \quad (\text{B.33})$$

Hence

$$G_\sigma = -s n \Omega \sigma_{ij}^{\text{ext}} n_i n_j = -s \pi R^2 b \sigma_{ij}^{\text{ext}} n_i n_j. \quad (\text{B.34})$$

The point-defect exchange chemical potential is therefore

$$\mu = \frac{\partial G_{\text{loop}}^{\text{tot}}}{\partial n} = \frac{\partial G_{\text{self}}}{\partial n} + \frac{\partial G_{\text{fault}}}{\partial n} - s \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (\text{B.35})$$

For a faulted circular vacancy loop ($s = -1$),

$$\mu_{\text{loop}}^{(v)} = \frac{Gb\Omega}{4\pi(1-\nu)R} \left[\ln\left(\frac{R}{r_c}\right) + 1 \right] + \frac{\gamma_{sf}\Omega}{b} + \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (\text{B.36})$$

For a faulted circular interstitial loop ($s = +1$),

$$\mu_{\text{loop}}^{(i)} = \frac{Gb\Omega}{4\pi(1-\nu)R} \left[\ln\left(\frac{R}{r_c}\right) + 1 \right] + \frac{\gamma_{sf}\Omega}{b} - \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (\text{B.37})$$

r_c is the core radius of a dislocation. Utilizing Eqs. B.5 and B.4, we obtain the binding energy of a vacancy to an interstitial or to a vacancy loop, respectively.

Vacancy Loops

Osetsky et al. [69] studied the stability of vacancy loops in iron using atomistic simulations. Applying the continuum equations B.5 and B.4, together with Eq. B.36 for the chemical potential of a vacancy loop without external stress, we compare the results of atomistic simulations to the continuum equations with $r_c = b/3$ and $\gamma_{sf} = 0.6 \text{ J/m}^2$ in Fig. B.9. A transition correction anchors the continuum to the atomistic divacancy binding energy $E_b(2) = 0.22 \text{ eV}$ via an exponential decay:

$$E_b(\alpha) = E_b^{\text{cont}}(\alpha) + [E_b(2) - E_b^{\text{cont}}(2)] e^{-\lambda(\alpha-2)} \quad (\text{B.38})$$

$$\lambda = \frac{\ln 100}{8} \approx 0.575 \text{ vac}^{-1} \quad (\text{B.39})$$

Interstitial Loops Small- n Power-Law Fit ($n \leq 25$)

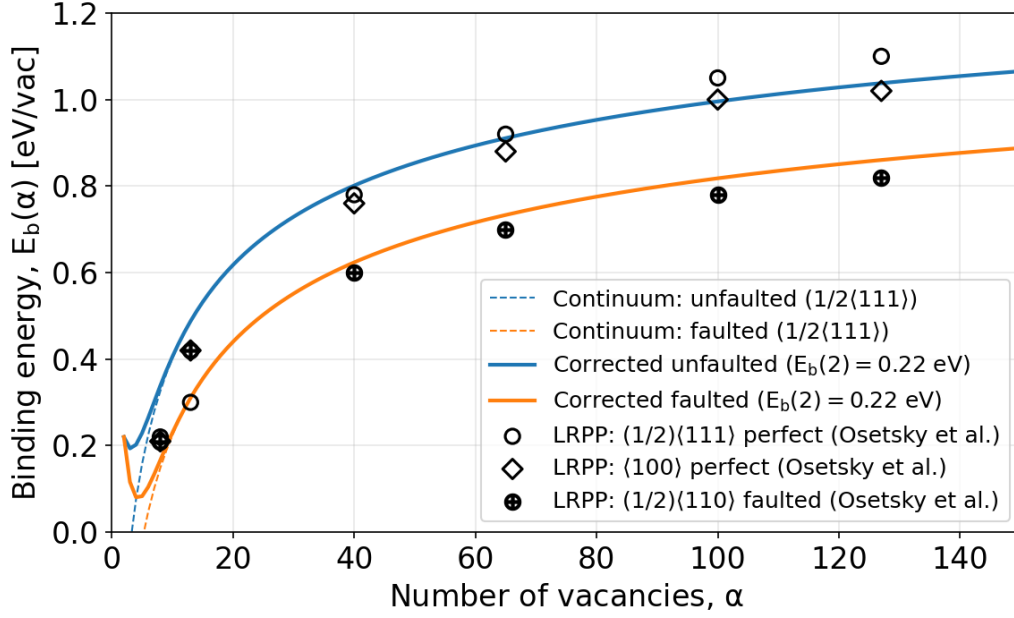


Figure B.9: Binding energy of a vacancy to a vacancy loop in iron using continuum equations (solid lines) and atomistic data from Osetsky et al. [69]

The small-cluster binding energy is fitted by least squares to the atomistic data for $n = 2$ (DFT) and $n = 7, 19$ (Osetsky MD, hexagonal and square loop configurations). The fit is written as

$$E_b^{i,\text{fit}}(n) = An^B. \quad (\text{B.40})$$

The fitted parameters are listed in Table B.17.

Table B.17: Power-law fit parameters for small interstitial loops.

Loop type	A (eV)	B
$\frac{1}{2}\langle 111 \rangle$	0.7501	0.3873
$\langle 100 \rangle$	0.7160	0.3581

The atomistic training data used in the fit is summarized in Table B.18.

Table B.18: Atomistic training data used for the small- n binding-energy fit.

n	$E_b^i - \frac{1}{2}\langle 111 \rangle$ (eV)	$E_b^i - \langle 100 \rangle$ (eV)	Source
2	0.90	0.90	DFT (Domain & Becquart [56]; Marinica et al.[70])
7 (hex)	1.65	1.30	Osetsky LRPP MD
7 (sq)	1.65	1.60	Osetsky LRPP MD
19 (hex)	2.35	2.05	Osetsky LRPP MD
19 (sq)	2.30	2.05	Osetsky LRPP MD

Blended Model with tanh Transition

To connect the small- n fit to the continuum expression, we use a smooth transition function

$$w(n) = \frac{1}{2} \left[1 + \tanh \left(\frac{n_{\text{tr}} - n}{\sigma_{\text{tr}}} \right) \right]. \quad (\text{B.41})$$

The blended binding energy is then written as:

$$E_b^i(n) = w(n) E_b^{i,\text{fit}}(n) + [1 - w(n)] E_b^{i,\text{cont}}(n). \quad (\text{B.42})$$

The transition parameters are listed in Table B.19.

Table B.19: Parameters of the smooth blending function.

Parameter	Value
n_{tr} (transition center)	25 SIAs
σ_{tr} (transition half-width)	5 SIAs

Selected values of the weight function are given in Table B.20.

The binding energy of a single self-interstitial atom to the two competing loop habits is compared in Fig. B.10. Both the $\frac{1}{2}\langle 111 \rangle$ and the $\langle 100 \rangle$ loops show a binding energy that rises steeply for small clusters and then flattens as the loop grows, asymptotically approaching the SIA emission threshold $E_i^f = 3.6$ eV from below. The atomistic data of Osetsky et al. (hexagonal and square configurations) are reproduced by the small- n power-law fit for $n \leq n_{\text{tr}} = 25$ and by the continuum expression at large sizes, blended smoothly through the tanh transition. The di-SIA anchor $E_b^i(2) = 0.90$ eV fixes the low- n end of both curves. Across the entire size range the $\frac{1}{2}\langle 111 \rangle$

Table B.20: Weight function $w(n)$ at selected cluster sizes.

n	$w(n)$
15	0.982
20	0.881
25	0.500
30	0.119
35	0.018

loop consistently binds an SIA more strongly than the $\langle 100 \rangle$ loop, with the gap being largest for small clusters ($n \lesssim 40$) and narrowing gradually as both curves converge toward the emission threshold at large n . The higher per-SIA binding energy of the $\frac{1}{2}\langle 111 \rangle$ habit indicates that it is the more stable, lower-energy configuration and therefore the thermodynamically favored loop, especially in the small-loop regime that controls nucleation; the $\langle 100 \rangle$ loop, being more weakly bound, is comparatively easier to dissolve by interstitial emission. This energetic ordering underlies the observed dominance of $\frac{1}{2}\langle 111 \rangle$ loops and the requirement of a kinetic pathway to generate the $\langle 100 \rangle$ population.

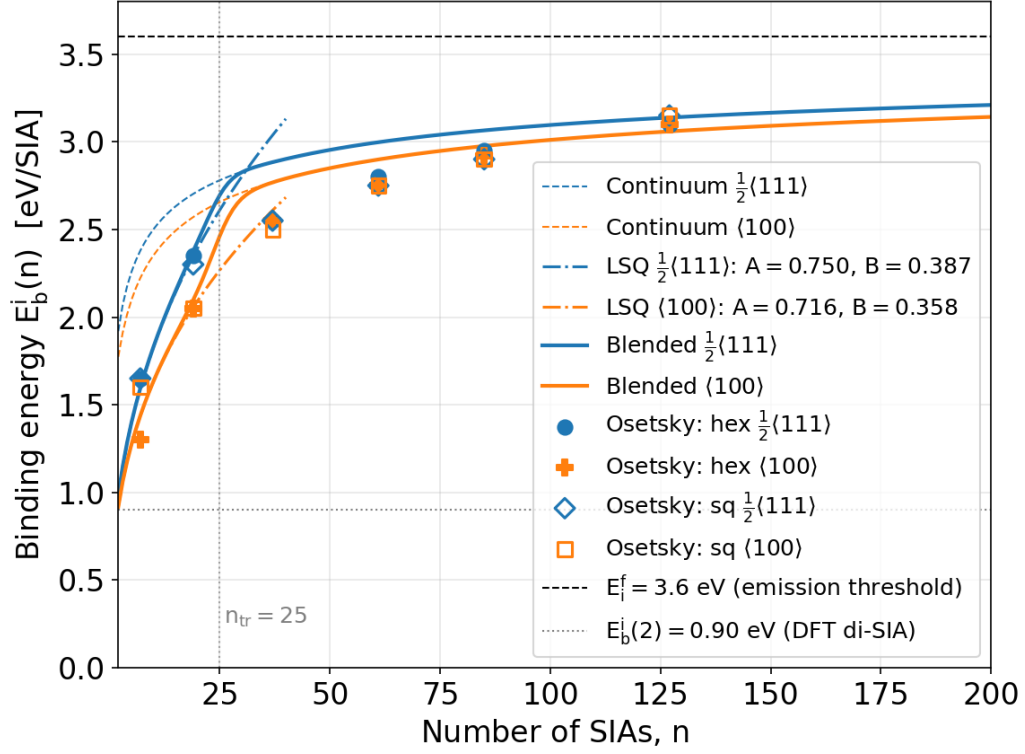


Figure B.10: Binding energy of a self-interstitial atom to a dislocation loop in iron as a function of the number of SIAs, n , for the two loop habits $\frac{1}{2}\langle 111 \rangle$ (blue) and $\langle 100 \rangle$ (orange). The upper horizontal dashed line marks the SIA emission threshold $E_i^f = 3.6$ eV and the lower dotted line the DFT di-SIA anchor $E_b^i(2) = 0.90$ eV. The $\frac{1}{2}\langle 111 \rangle$ loop is more strongly bound than the $\langle 100 \rangle$ loop at every size, confirming its greater relative stability.

Appendix C. Diffusion

The diffusion coefficient for a random walk in 3D is:

$$D = \frac{1}{6} z \lambda^2 \nu_0 \exp(-E_m/k_B T) \quad (\text{C.1})$$

where z is the number of equivalent jump directions, λ is the jump distance, and the factor $\frac{1}{6}$ comes from the 3D random walk. For a bcc lattice with $\langle 111 \rangle$ nearest-neighbor jumps, $z = 8$, and $\lambda = a\sqrt{3}/2$. If the lattice constant is (a), then the diffusion coefficients of interstitials, vacancies, and helium can be written as $D = a^2 \nu \exp(-E_m/k_B T)$. We introduce here 3 basic frequencies in the defect clustering system: ω_i is the frequency of the self-interstitial reaction, ω_h is the frequency of the helium gas reaction, and ω_v is the frequency of the vacancy reaction. For SIAs (and similarly for vacancies and helium atoms), this is given by:

$$\omega_i = \nu_i \exp(-E_m^i/k_B T) = D_i/a^2, \text{ s}^{-1}, \quad (\text{C.2})$$

In bcc iron, vacancy migration proceeds by nearest-neighbor jumps along $\langle 111 \rangle$ directions, with $z = 8$ equivalent sites and the saddle point midway between adjacent lattice sites, so the vacancy executes a 3D random walk at all temperatures of interest. Using the migration energy of Table E.25, the activation energy for self-diffusion follows as $Q_{SD} = E_f^v + E_m^v$, consistent with tracer diffusion experiments.

As summarized in the Energetics section (Table E.25), the most stable SIA in ferromagnetic bcc iron is the $\langle 110 \rangle$ dumbbell, which migrates by a low-barrier translation-rotation mechanism. Because the dumbbell can reorient into any of its six equivalent orientations upon each jump, single-SIA migration is inherently three-dimensional—in contrast to the quasi-1D athermal crowdion glide found in non-magnetic bcc metals [71, 72].

The elementary $\langle 110 \rangle$ dumbbell mechanism carries over to the ferritic matrix of EUROFER97, but the effective SIA mobility is reduced by interactions with the alloy's solute population. EUROFER is a reduced-activation 9Cr ferritic/martensitic steel whose principal alloying additions include Cr, W, Mn, V, Ta, C, and N [3, 73]. After heat treatment, part of the C, N, V, and Ta partitions into MX carbonitrides and $M_{23}C_6$ carbides, so the relevant quantity is the *dissolved* matrix solute content rather than the nominal composition. Two classes of solute–SIA interactions are important. (1) Substitutional solutes (Cr, Mn, W, V, Ta). These form mixed solute–Fe

dumbbells and bound interstitial complexes that create a fluctuating migration landscape of local binding, detrapping, and solute-biased reorientation events [74–77]. (2) Interstitial solutes (C, N). Carbon and nitrogen occupy octahedral sites and bind strongly to SIA and small SIA clusters, acting as efficient traps that shorten the free-flight length and raise the apparent migration energy [78–81]. Both effects are captured by writing the effective SIA diffusivity in the trap-limited form:

$$D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s K_s(T) c_s} \quad (\text{C.3})$$

where D_i^{Fe} is the pure-iron value, c_s is the atomic fraction of solute or trap class s , and $K_s(T)$ is an effective trapping coefficient related to the SIA–solute binding free energy. The dominant trap classes in EUROFER are dissolved C and N, Cr-rich local environments, and carbide/carbonitride interfaces. No single Arrhenius pair (D_0 , E_m) adequately represents SIA diffusion in EUROFER across all conditions. The physically sound approach is to use the pure-Fe mobility as the reference and to introduce alloy-specific corrections through Eq. (C.3).

The migration mechanism of interstitial clusters in bcc iron evolves with cluster size, transitioning from 3D to 1D as the cluster grows:

For $n \geq 4$, clusters adopt the form of prismatic dislocation loops with a Burgers vector $\mathbf{b} = \frac{1}{2}\langle 111 \rangle$ and migrate by thermally activated 1D glide along their Burgers vector direction [82–84]. The 1D migration energy is essentially independent of cluster size for $n \gtrsim 4$ [82], with $E_m^{1D} \lesssim 0.05$ eV, so that $E_m^{1D} \leq k_B T$ is above room temperature and the Arrhenius factor is close to unity. The di-interstitial ($n = 2$), consisting of two parallel $\langle 110 \rangle$ dumbbells, is *less* mobile than the single SIA ($E_m = 0.42$ vs. 0.34 eV) [52], while clusters with $n = 3$ –4 lie in the crossover regime [85].

The 1D diffusion coefficient for a cluster of n SIAs gliding along $\langle 111 \rangle$ is:

$$D_n^{1D} = \frac{1}{2} \lambda_{\langle 111 \rangle}^2 \nu_n \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad (\text{C.4})$$

where $\lambda_{\langle 111 \rangle} = a\sqrt{3}$ is the jump distance and the factor $\frac{1}{2}$ reflects the two-directional 1D random walk. MD simulations show that the attempt frequency scales inversely with size [82, 86]:

$$\nu_n = \frac{\nu_0}{n^s}, \quad \nu_0 \sim 6 \times 10^{12} \text{ s}^{-1}, \quad s \approx 0.5\text{--}1.0, \quad (\text{C.5})$$

giving the expression:

$$D_n^{1D} = \frac{3a^2\nu_0}{2n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right) \quad \text{m}^2 \text{s}^{-1} \quad (\text{C.6})$$

Gliding SIA clusters periodically change their Burgers-vector direction—by thermal activation, by interaction with solutes, or by encounters with other defects [87]. The mean one-dimensional diffusion length L between direction changes controls the effective dimensionality of the transport. If L is finite, the transport may be represented phenomenologically by an effective 3D diffusion coefficient [88, 89]

$$D_n^{3D,\text{eff}} = \frac{D_n^{1D}}{3} \frac{L}{\lambda_{\langle 111 \rangle}}, \quad (\text{C.7})$$

where the factor $1/3$ accounts for the orientational projection of one-dimensional motion onto three spatial dimensions, and $\lambda_{\langle 111 \rangle}$ is the characteristic distance associated with a Burgers-vector reorientation event. Equivalently, if ν_R is the Burgers-vector rotation frequency, then the mean time between rotations is $\tau_R = 1/\nu_R$, and the mean one-dimensional diffusion length between rotations is

$$L = \sqrt{2D_n^{1D}\tau_R} = \sqrt{\frac{2D_n^{1D}}{\nu_R}}. \quad (\text{C.8})$$

Substitution into Eq. (C.7) gives

$$D_n^{3D,\text{eff}} = \frac{D_n^{1D}}{3\lambda_{\langle 111 \rangle}} \sqrt{\frac{2D_n^{1D}}{\nu_R}}. \quad (\text{C.9})$$

The effective 3D reaction rate with spherical sinks of radius R and density C_s is

$$k_n^{3D} = 4\pi R D_n^{3D,\text{eff}} C_s. \quad (\text{C.10})$$

In the pure 1D limit, the 1D sink-strength formalism gives [88, 89]

$$k_n^{1D} = 2(\pi R^2 C_s)^2 D_n^{1D}. \quad (\text{C.11})$$

The critical physical distinction is that the 1D absorption rate scales as $R^4 C_s^2$, i.e., as the square of the geometric cross-section and second order in sink density, whereas the 3D absorption rate scales as $R C_s$, i.e., first order in

both capture radius and sink density. This asymmetry between SIA-cluster and point-defect absorption by sinks is the physical origin of the production bias that drives void swelling under cascade-producing irradiation.

In EUROFER, the elementary 1D glide barrier remains small, but solute interactions modify cluster transport in two coupled ways:

1. **Reduced 1D mobility.** Dissolved C, N, and their vacancy complexes trap and pin glissile $\frac{1}{2}\langle 111 \rangle$ loops, lowering the effective 1D diffusivity [90, 91]:

$$D_{n,\text{eff}}^{1D} = \frac{D_n^{1D,\text{Fe}}}{1 + \sum_s K_{n,s}(T) c_s}, \quad (\text{C.12})$$

where $K_{n,s}(T)$ is the trapping factor for cluster size n with solute class s .

2. **Shortened mean free path.** Substitutional solutes (Cr, W, Mn, V, Ta) promote Burgers-vector changes through local pinning and drag, increasing ν_R and thereby reducing L :

$$L_{\text{eff}} = \sqrt{\frac{2D_{n,\text{eff}}^{1D}}{\nu_{R,\text{eff}}}}. \quad (\text{C.13})$$

Solute effects, therefore, compound: they lower D_n^{1D} and shorten L , pushing cluster transport further from the ideal long-range 1D limit toward 3D-like behavior. The effective 3D coefficient in the alloy is obtained by substituting the alloy quantities into the mixed 1D/3D expression:

$$D_n^{3D,\text{eff},\text{alloy}} = \frac{D_{n,\text{eff}}^{1D}}{3} \cdot \frac{L_{\text{eff}}}{\lambda_{\langle 111 \rangle}}. \quad (\text{C.14})$$

The nominal composition of EUROFER97 is approximately 8.9 Cr–1.1 W–0.47 Mn–0.20 V–0.14 Ta–0.11 C–0.03 N (wt%) [3, 73]. After the standard heat treatment (980°C/30 min normalization + 760°C/90 min tempering), a substantial fraction of C, N, V, and Ta is consumed by $M_{23}C_6$ and MX-type precipitates. Atom-probe and TEM studies indicate that the dissolved matrix concentrations relevant for defect trapping are approximately shown in Table C.21 [92, 93]: The dissolved C and N values are rough estimates; most of the nominal carbon is tied up in precipitates. These concentrations

Table C.21: Dissolved matrix concentrations of main alloying elements in EUROFER.

Species	Nominal (wt%)	Est. dissolved (at. fraction)
Cr (substitutional)	8.9	$c_{\text{Cr}} \approx 0.094$
W (substitutional)	1.1	$c_{\text{W}} \approx 0.0033$
Mn (substitutional)	0.47	$c_{\text{Mn}} \approx 0.0047$
C (interstitial, dissolved)	$\lesssim 0.02$	$c_{\text{C}} \approx 5 \times 10^{-4}$
N (interstitial, dissolved)	$\lesssim 0.01$	$c_{\text{N}} \approx 2 \times 10^{-4}$

are used in the trapping factors below. Table C.22 summarizes the DFT-calculated binding energies of solute atoms to single SIAs ($\langle 110 \rangle$ dumbbells) and to small $\frac{1}{2}\langle 111 \rangle$ interstitial loops in bcc Fe. These govern the trapping factors $K_s(T)$ in Eqs. (C.3) and (C.12).

C and N are the strongest traps for both single SIAs and small loops. The C–SIA binding (~ 0.45 eV) is comparable to the SIA migration energy itself (0.34 eV), meaning that C trapping qualitatively alters the effective migration barrier. Cr binds weakly via the mixed-dumbbell mechanism. At 9 at.% Cr, the cumulative effect is significant despite the small per-atom binding. Mn binds more strongly than Cr to the dumbbell and dominates dumbbell-mediated solute transport [77]. W is repulsive with SIAs and does not contribute to trapping. The trapping coefficient $K_s(T)$ in the effective diffusivity (Eq. C.3) is related to the binding energy by [22]:

$$K_s(T) = z_s \exp\left(\frac{E_b^s}{k_B T}\right), \quad (\text{C.15})$$

where z_s is a coordination factor (number of trapping configurations per solute site; $z_s \approx 4$ –8 for substitutional solutes in bcc, $z_s \approx 2$ –4 for interstitial solutes). With this, the effective SIA diffusivity becomes:

$$D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s z_s c_s \exp(E_b^s/k_B T)}. \quad (\text{C.16})$$

At low temperatures ($T \lesssim 300$ °C), the C and N trapping terms dominate the denominator, and the effective SIA migration energy becomes $E_m^{i,\text{eff}} \approx$

$E_m^i + E_b^C \approx 0.34 + 0.45 = 0.79 \text{ eV}$. At high temperatures ($T \gtrsim 500^\circ\text{C}$), thermal detrapping reduces the effect, and D_i^{eff} approaches the pure-Fe value.

The coordination numbers z_s and binding energies E_b^s used above are rounded values chosen to represent the central range of DFT data from Table C.22. The W contribution is omitted (repulsive binding; $K_W \approx 0$). V and Ta are present at low dissolved concentrations and are omitted for simplicity.

For SIA clusters in the mixed 1D/3D framework (Eq. C.14), the effective rotation frequency and mean free path are:

$$\nu_{R,\text{eff}} = \nu_R^{\text{therm}} + \sum_s \sigma_s^{\text{rot}} c_s \omega_n^{1D}, \quad (\text{C.17})$$

$$L_{\text{eff}} = \sqrt{2D_{n,\text{eff}}^{1D} / \nu_{R,\text{eff}}}, \quad (\text{C.18})$$

where ν_R^{therm} is the thermal rotation frequency in pure Fe (small; often negligible below 500°C for large loops) and σ_s^{rot} is the effective rotation cross-section per solute atom. For Cr at 9 at.%, atomistic estimates give $L_{\text{eff}} \sim 10\text{--}100 a$ ($\sim 3\text{--}30 \text{ nm}$), which is much shorter than the typical sink spacing in irradiated EUROFER ($\sim 100\text{--}1000 \text{ nm}$). This places EUROFER cluster transport firmly in the mixed 1D/3D regime rather than the pure 1D limit applicable to high-purity iron.

Vacancy clusters in bcc iron migrate by a qualitatively different mechanism from interstitial clusters. Whereas SIA clusters glide collectively as dislocation loops, vacancy clusters move only by sequential single-vacancy jumps within or at the surface of the cluster. The motion is always three-dimensional, and the effective migration energy generally increases with cluster size because larger clusters require increasingly coordinated sequences of atomic rearrangements to translate the center of mass by one lattice spacing [66, 94, 95].

For clusters larger than $\sim 5\text{--}6$ vacancies, the dominant transport mechanism transitions from bulk-like nearest-neighbor vacancy hopping to surface (ledge) diffusion of adatom vacancies around the void facets. This mechanism underlies the classical Gruber model for void migration, in which the void diffusion coefficient scales as a high inverse power of the void radius [96, 97].

The anomalously mobile trivacancy. The compact trivacancy (V_3) in the bcc lattice—with all three vacancies at mutual second-nearest-neighbor positions forming an equilateral triangle on a $\{110\}$ plane—has a migration energy

(0.35–0.50 eV) that is *lower* than that of either the monovacancy or the divacancy. This was first reported from DFT-parameterized kinetic Monte Carlo calculations and has been confirmed by k-ART [94] and by independent first-principles studies of vacancy–solute cluster mobility [98]. The mechanism is a concerted three-atom ring rotation in which three Fe atoms adjacent to the cluster simultaneously exchange positions with the three vacancies, translating the cluster center of mass by one lattice vector. Because of this low barrier, V_3 clusters are the most mobile vacancy-type species in α -Fe after isolated monovacancies. [98] further showed that V_3X complexes (with $X = \text{C, N, or O}$) retain much of this mobility—with global migration energies of 0.59 eV (C), 0.86 eV (N), and 0.58 eV (O)—which means that vacancy-mediated interstitial-solute transport via trivacancy dragging is an important mechanism in irradiated ferritic steels.

Divacancy pathway multiplicity. The divacancy in bcc Fe has its ground state at the second-nearest-neighbor (2NN) separation [56]. Its migration involves a two-step process: first, one vacancy jumps to a first-nearest-neighbor position (metastable), then the second vacancy follows to re-establish the 2NN ground state at a translated position. The effective barrier depends on which of several pathways is taken, giving a range of 0.62–0.75 eV [94, 95]. k-ART simulations reveal additional complex pathways involving transient higher-order configurations, with effective barriers as high as 0.84 eV for some routes through the energy landscape [94].

For larger voids containing m vacancies ($m \gtrsim 10$), vacancy cluster migration proceeds by surface self-diffusion of adatom vacancies on the void wall. The classical result due to [96] and [97] gives the void diffusion coefficient as:

$$D_{\text{void}}(m) = \frac{D_s \Omega^{4/3}}{r_m^4} = D_s \Omega^{4/3} \left(\frac{4\pi}{3m\Omega} \right)^{4/3} \propto \frac{D_s}{m^{4/3}}, \quad (\text{C.19})$$

where D_s is the surface self-diffusion coefficient on the void wall, $\Omega = a^3/2$ is the atomic volume, and $r_m = (3m\Omega/4\pi)^{1/3}$ is the void radius. The strong inverse scaling $D_{\text{void}} \propto m^{-4/3}$ means that void mobility drops extremely rapidly with size. Numerical estimates give $D_{\text{void}} \sim 10^{-30} \text{ m}^2/\text{s}$ or less for voids of $m \geq 50$ at temperatures below 500°C, which is effectively zero on any experimentally accessible timescale.

Alternative derivations based on volume diffusion, rather than surface diffusion, give an even steeper size dependence ($D_{\text{void}} \propto m^{-5/3}$), but surface

diffusion is generally accepted as the dominant mechanism for faceted voids in bcc metals [97].

The general size-dependent diffusion coefficient for vacancy clusters used in the rate equations is:

$$D_{(-m,0)} = D_v \cdot m^{-s_v}, \quad s_v \approx 1.0 \quad (m \leq v_{\text{mobile}}), \quad (\text{C.20})$$

where v_{mobile} is a mobility cutoff (typically 3–5). Clusters with $m > v_{\text{mobile}}$ are set to $D = 0$.

For He-containing vacancy clusters (bubbles), the diffusion coefficient is further reduced:

$$D_{(-m,\ell)} = D_{(-m,0)} \cdot f(\ell/m), \quad (\text{C.21})$$

where $f(\ell/m)$ is a reduction factor reflecting the additional pinning by trapped helium. A simple choice is $f = \exp(-E_b^{\text{He}}\ell/k_B T)$, but for practical purposes, He-containing clusters with $m \geq 2$ are treated as immobile.

Vacancy migration in pure α -Fe proceeds by nearest-neighbor jumps along $\langle 111 \rangle$ with a barrier of $E_m^v = 0.67$ eV. In the EUROFER matrix, dissolved solutes modify this mobility through two mechanisms: (i) vacancy-solute binding, which traps vacancies near solute sites and raises the effective migration energy; and (ii) modification of the local saddle-point energy, which can either raise or lower the elementary jump barrier depending on the neighboring chemical environment. Both effects are governed by the vacancy-solute binding energy E_b^{v-s} , which has been computed by DFT for all the principal solutes in EUROFER.

Vacancy-Solute Binding Energies. Table C.22 collects DFT-calculated vacancy-solute binding energies at 1NN and 2NN separations in bcc Fe. Positive values denote attraction (trapping). C and N are the dominant vacancy traps, with binding energies (~ 0.4 – 0.5 eV at 1NN) comparable to the vacancy migration energy itself. Carbon-vacancy (VC) and nitrogen-vacancy (VN) complexes are therefore strongly bound and act as deep traps that immobilize vacancies at low temperatures. This is the primary mechanism by which interstitial solutes suppress vacancy-mediated processes (self-diffusion, void nucleation) in ferritic steels. W is a moderate vacancy trap ($E_b \sim 0.25$ – 0.35 eV at 1NN). At the 0.33 at.% level present in EUROFER, W makes a non-negligible contribution to vacancy trapping, particularly in the intermediate temperature range (300–500°C) where the C/N trapping terms are starting to deactivate. Cr binds very weakly to vacancies ($E_b \sim 0.05$ eV at 1NN) and

Table C.22: RADCLUSTER solute-trapping input parameters for EUROFER. Trapping coefficients are $K_s = z_s \exp(E_b^s/k_B T)$; positive E_b denotes attraction. Adopted values are the rounded central estimates used in the code; DFT ranges and dissolved (after-tempering) concentrations c_s are listed for reference.

Solute	Trap type	z_s	E_b (adopted, eV)	DFT range (eV)	c_s (at/at)	Source
C	SIA dumbbell	4	0.45	0.40–0.50	5×10^{-4}	[74, 99]
N	SIA dumbbell	4	0.40	0.35–0.45	2×10^{-4}	[74, 99]
Cr	SIA dumbbell	8	0.10	0.09–0.12	0.094	[74, 76]
Mn	SIA dumbbell	6	0.20	0.15–0.25	0.0047	[75, 77]
W	SIA dumbbell	—	repulsive	−0.1 to −0.3	0.0033	[99]
C	vacancy	3	0.45	0.41–0.47	5×10^{-4}	[78, 79]
N	vacancy	3	0.40	0.35–0.43	2×10^{-4}	[78, 79]
W	vacancy	8	0.27	0.20–0.35	0.0033	[75, 100]
Mn	vacancy	8	0.10	0.06–0.12	0.0047	[101]
Cr	vacancy	8	0.05	0.03–0.06	0.094	[75]
C	SIA loop	2	0.50	0.42–0.86	5×10^{-4}	[78, 90, 91]
N	SIA loop	2	0.40	~ 0.3 –0.7	2×10^{-4}	[78]
Cr	SIA loop	4	0.10	0.05–0.15	0.094	[74, 99]

is repulsive at 2NN. Despite its high concentration (9.4 at.%), the cumulative trapping effect of Cr on vacancy mobility is modest because $\exp(0.05/k_B T)$ is close to unity at all relevant temperatures. Mn binds slightly more strongly than Cr but still weakly (~ 0.10 eV). Its effect is secondary to C, N, and W.

Effective Vacancy Diffusivity in EUROFER. By analogy with the SIA trap-limited formulation (Eq. C.3), the effective vacancy diffusivity in EUROFER is:

$$D_v^{\text{eff}} = \frac{D_v^{\text{Fe}}}{1 + \sum_s z_s^v c_s \exp(E_b^{v-s}/k_B T)} \quad (\text{C.22})$$

where D_v^{Fe} is the pure-iron value, c_s is the dissolved atomic fraction of solute s , and z_s^v is the coordination number for vacancy trapping configurations near a solute of type s . The dominant trapping species, in decreasing order of importance, are dissolved C, dissolved N, W, Mn, and Cr.

Effect on Vacancy Cluster Mobility. Small vacancy clusters ($m = 2\text{--}5$) are affected by solutes through two mechanisms:

1. **Enhanced binding/pinning:** C and N atoms bind even more strongly to divacancies and trivacancies than to monovacancies, because the larger open volume of the cluster provides a more favorable environment for the interstitial solute [78, 98]. This deepens the trap and raises the effective cluster migration barrier.
2. **Solute decoration:** In EUROFER, small vacancy clusters formed under irradiation rapidly acquire C and N decorations from the dissolved matrix pool, converting bare V_m clusters into $V_m C_p N_q$ complexes. These decorated clusters have substantially reduced mobility relative to bare clusters. The compact trivacancy—anomalously mobile in pure Fe—is particularly affected: $V_3\text{C}$ has a migration energy of 0.59 eV and $V_3\text{N}$ of 0.86 eV [98], compared to 0.35–0.50 eV for bare V_3 .

In RADCLUSTER modeling, the effective vacancy cluster diffusivity in EUROFER is written as:

$$D_{(-m,0)}^{\text{eff}} = \frac{D_{(-m,0)}^{\text{Fe}}}{1 + \sum_s K_{m,s}^v(T) c_s}, \quad (\text{C.23})$$

where $K_{m,s}^v(T) = z_{m,s}^v \exp(E_b^{v_{m-s}}/k_B T)$ and the binding energy $E_b^{v_{m-s}}$ may depend on cluster size.

The solute-modified vacancy diffusivity has three important consequences for cluster dynamics modeling of EUROFER: (1) suppressed void nucleation at low T , (2) delayed thermal emission, and (3) asymmetric SIA–vacancy trapping.

Interstitial helium occupies the tetrahedral site in bcc iron and migrates between nearest-neighbor tetrahedral sites over the very low barrier. This barrier is so small that helium is essentially mobile at all temperatures of interest—it is by far the fastest diffusing species in the defect system. The helium diffusion coefficient and reaction frequency in pure α -Fe are:

$$D_h = a^2 \nu_h \exp\left(-\frac{E_m^h}{k_B T}\right) = 2.5 \times 10^{-8} \exp\left(-\frac{0.06 \text{ eV}}{k_B T}\right) \quad \text{m}^2 \text{ s}^{-1}, \quad (\text{C.24})$$

$$\omega_h = \frac{D_h}{a^2} = 3.0 \times 10^{12} \exp\left(-\frac{0.06 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}, \quad (\text{C.25})$$

with $\nu_h \approx 3.0 \times 10^{12} \text{ s}^{-1}$ and $a = 2.87 \text{ \AA}$ [66, 102]. Because $E_m^h \ll k_B T$ at all reactor-relevant temperatures, the Arrhenius factor is close to unity, and the helium mobility is nearly temperature-independent. Spin-lattice dynamics simulations confirm strongly non-Arrhenius behavior at elevated temperatures, where phonon-drag effects make the diffusion coefficient increase roughly linearly with T rather than exponentially [103].

In the EUROFER matrix, interstitial He can interact with dissolved solutes before reaching a vacancy trap. The relevant interactions, from DFT calculations, are:

Substitutional solutes (Cr, Mn, W). He–substitutional-solute interactions in bcc Fe are generally weak. [66] and [65] find that He interacts only marginally with substitutional Cr and other transition metal solutes, with binding energies in the range 0.00–0.05 eV at 1NN—insufficient to constitute meaningful traps at reactor temperatures. At 9 at.% Cr, the cumulative effect on He mobility is therefore negligible.

Interstitial solutes (C, N). Carbon and nitrogen occupy octahedral sites and interact with interstitial He at tetrahedral sites. The He–C binding energy has been calculated to be ~ 0.05 – 0.10 eV [66, 104], which is comparable to the He migration barrier itself. This means C can cause transient trapping of He

at low temperatures ($T \lesssim 200^\circ\text{C}$), but at higher temperatures, the trapping is thermally overcome and has little effect on long-range He transport.

Vacancy–solute complexes as He traps. The most important solute effect on He transport in EUROFER is *indirect*: C and N form stable vacancy–solute complexes (VC , VN , VC_2) that compete with bare vacancies as He trapping sites. The He binding energy to a VC complex is ~ 1.5 – 1.8 eV—lower than that to a bare vacancy (~ 2.3 eV) but still a deep trap. Since a fraction of the vacancies in EUROFER are decorated by C and N, the effective trap density and the He–trap binding landscape differ from those of pure Fe.

Effective Helium Diffusivity. Because the He–solute binding energies are much smaller than the He–vacancy binding energy, the dominant control on free He transport is the vacancy trap density, not the solute field. The effective He diffusivity in EUROFER can therefore be written as:

$$D_h^{\text{eff}} = \frac{D_h^{\text{Fe}}}{1 + z_V^h c_V^{\text{free}} \exp(E_b^{h-V}/k_B T) + \sum_s z_s^h c_s \exp(E_b^{h-s}/k_B T)} \quad (\text{C.26})$$

where the first trapping term is due to free vacancies (concentration c_V^{free} , binding energy $E_b^{h-V} \approx 2.3$ eV, coordination $z_V^h = 6$) and the sum is over solute traps. In practice, at all temperatures where vacancies exist in meaningful concentrations, the vacancy trapping term completely dominates the denominator, and the solute terms can be neglected. The helium diffusivity reduction in EUROFER relative to pure Fe is therefore controlled primarily by the irradiation-produced vacancy concentration, not by the alloy composition.

Appendix D. Defect Reactions

Appendix D.1. Diffusion-Limited Reactions:

This section derives the rate constants for all reactions between defect clusters, expressed in atomic-fraction units (s^{-1}) and in terms of the reaction frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The cluster notation (m, ℓ) and concentration units (at/at) follow the conventions established in Section Appendix C.

For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant is $\mathcal{K}^{\text{vol}} = 4\pi r_m (D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m) a$, and $\Omega = a^3/2$:

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_\alpha \quad [\text{s}^{-1}]. \quad (\text{D.1})$$

When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}_{AB}^{3D} = 8\pi \xi_{\text{rec}} (\omega_A + \omega_B), \quad (\text{D.2})$$

with $\xi_{\text{rec}} = \sqrt{3}/2 \approx 0.866$ for nearest-neighbor capture in bcc.

Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 4\sqrt{3}\pi (\omega_i + \omega_v) \approx 21.8\omega_i, \quad (\text{D.3})$$

since $\omega_i \gg \omega_v$ at all relevant temperatures. Point defect absorption by a spherical cluster (void or bubble) of m vacancies:

$$\mathcal{K}_{\alpha,m}^{\text{void}} = A_{\text{sph}} m^{1/3} \omega_\alpha, \quad A_{\text{sph}} = (48\pi^2)^{1/3} \approx 7.818. \quad (\text{D.4})$$

Point defect absorption by a dislocation loop of n SIAs requires species-specific rate constants because only SIAs experience a non-trivial elastic bias ($Z_i^{\text{loop}} > 1$), while vacancies are unbiased ($Z_v^{\text{loop}} = 1$):

$$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i, \quad (\text{D.5})$$

$$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v, \quad (\text{D.6})$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{\text{loop}} \approx 1.02\text{--}1.10$ (Table E.23). SIA absorption grows the loop by one; vacancy absorption shrinks it by one.

Point defect absorption by the dislocation network (first-order sink):

$$\left. \frac{dc_\alpha}{dt} \right|_{\text{disl}} = -Z_\alpha \rho_d a^2 \omega_\alpha c_\alpha, \quad (\text{D.7})$$

where ρ_d (m^{-2}) is the dislocation line density and no n_{at} conversion is needed because this term is first-order in c_α .

Diffusion rates of clusters depend on their size, sink strength, and solute interactions as follows:

1. 1D Diffusion-Limited Reactions (SIA Clusters):

SIA clusters of size $n \geq 4$ migrate by 1D glide along $\langle 111 \rangle$ with a reaction frequency (from Section Appendix C):

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad s \approx 0.5-1.0, \quad E_m^{1D} \lesssim 0.05 \text{ eV}. \quad (\text{D.8})$$

2. Mixed 1D/3D regime (finite mean free path):

The effective 3D reaction frequency after Burgers vector rotation is:

$$\omega_n^{3D,\text{eff}} = \frac{\omega_n^{1D} L}{3\sqrt{3} a}. \quad (\text{D.9})$$

The rate constant for an n -SIA cluster reacting with an immobile void of m vacancies is then:

$$\mathcal{K}_{n,m}^{\text{mixed}} = A_{\text{sph}} m^{1/3} \omega_n^{3D,\text{eff}} = \frac{A_{\text{sph}}}{3\sqrt{3}} m^{1/3} \hat{L} \omega_n^{1D} \approx 1.505 m^{1/3} \hat{L} \omega_n^{1D}, \quad (\text{D.10})$$

where $\hat{L} = L/a$ is the dimensionless mean free path.

For a 1D-migrating cluster reacting with randomly distributed spherical sinks of size m and atomic fraction $c_{m,0}$, the loss rate is:

$$\left. \frac{dc_n}{dt} \right|_{\text{sink}} = -\mathcal{K}_{n,m}^{1D} \bar{c}_m^2 c_n, \quad (\text{D.11})$$

where:

$$\mathcal{K}_{n,m}^{1D} = A_{1D} m^{4/3} \omega_n^{1D}, \quad A_{1D} = \frac{9}{8\pi^{2/3}} \approx 2.632. \quad (\text{D.12})$$

The critical difference from 3D is that the rate constant scales as $m^{4/3}$ (not $m^{1/3}$) and is second-order in sink concentration (not first-order). This stronger sensitivity to sink geometry is the physical origin of the production bias.

For intermediate L , the Trinkaus–Singh–Golubov interpolation gives:

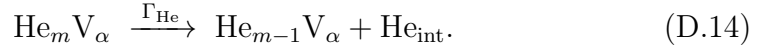
$$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, \quad B_{\text{rot}} = \frac{4}{\pi} \left(\frac{8\pi}{3} \right)^{1/3} \approx 2.627. \quad (\text{D.13})$$

In the limits: $\hat{L} \rightarrow 0$ recovers the 3D rate constant $A_{\text{sph}} m^{1/3} \omega_n^{1D}$; $\hat{L} \rightarrow \infty$ gives a rate proportional to $m^{2/3} \hat{L}^{-2}$, scaling as r^2 as expected for 1D kinetics.

Appendix D.2. Cluster Dissociation Reactions

Appendix D.2.1. Helium thermal emission

The cluster releases a He atom into an interstitial (tetrahedral) site:



The vacancy content is unchanged; one He atom is lost. The emission rate is governed by the helium binding energy:

$$\Gamma_{\text{He}}(\alpha, m) = \mathcal{K}_{h,\alpha} \exp \left[-\frac{E_B^{\text{He}}(\alpha, m)}{k_B T} \right], \quad (\text{D.15})$$

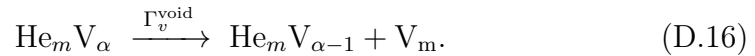
where $\mathcal{K}_{h,\alpha}$ is the He capture rate constant of the daughter cluster $\text{He}_{m-1} \text{V}_\alpha$. This channel dominates when the He binding energy is smaller than the energy cost of creating a Frenkel pair, i.e., when

$$E_B^{\text{He}} < E_f^i - \Omega P_g,$$

which is typically the case at high temperatures ($T \gtrsim 800$ K) and moderate He/V ratios.

Appendix D.2.2. Vacancy emission from a void

A void containing α vacancies may thermally emit one vacancy into the matrix:



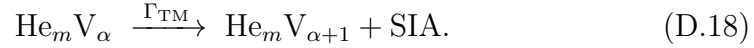
Here V_m denotes a mobile matrix vacancy. The helium content is unchanged, while the void loses one vacancy and therefore shrinks by one atomic volume. The thermal vacancy-emission rate is written in detailed balance form as

$$\Gamma_v^{\text{void}}(\alpha, m) = \mathcal{K}_{v, \alpha-1} \exp \left[-\frac{E_B^{v, \text{void}}(\alpha, m)}{k_B T} \right], \quad (\text{D.17})$$

where $\mathcal{K}_{v, \alpha-1}$ is the vacancy capture rate constant of the daughter void $\text{He}_m V_{\alpha-1}$, and $E_B^{v, \text{void}}(\alpha, m)$ is the vacancy binding energy to the parent void.

Appendix D.2.3. Trap mutation (Frenkel pair emission)

The cluster emits a self-interstitial atom from its periphery, simultaneously creating an additional vacancy within its own volume.



The He content is unchanged; the cluster gains one vacancy and injects one SIA into the lattice. This mechanism was first proposed by [105] as “plastic deformation around an over-pressurized bubble”, and direct experimental evidence for loop punching by helium bubbles was reported by [106] in molybdenum. The atomistic process—sometimes called *self-trapping* when it occurs from a pure interstitial He cluster ($\alpha = 0$)—was characterized in detail by [107] using molecular statics and has been studied by DFT and MD in both bcc Fe [59, 104, 108] and bcc W [109].

The trap-mutation rate can be expressed in Arrhenius form:

$$\Gamma_{\text{TM}}(\alpha, m) = \nu_0 \exp \left(-\frac{E_{\text{TM}}(\alpha, m)}{k_B T} \right), \quad (\text{D.19})$$

where $\nu_0 \sim 10^{12}\text{--}10^{13} \text{ s}^{-1}$ is an attempt frequency and E_{TM} is the activation barrier for Frenkel pair emission. DFT calculations by [109] for W and MD simulations by [59] and [104] for Fe provide the following picture:

- For **pure interstitial He clusters** ($\alpha = 0$): self-trapping (spontaneous Frenkel pair creation) occurs at $m \approx 4\text{--}6$ in bcc Fe at $T > 300 \text{ K}$, with a barrier $E_{\text{TM}} \sim 0\text{--}0.5 \text{ eV}$ that decreases with increasing cluster size [104].
- For **$\text{He}_m V_1$ clusters**: trap mutation occurs at $m \approx 5\text{--}7$, with barriers of $\sim 0.5\text{--}1.5 \text{ eV}$ that decrease steeply with m [59, 108].

- For **larger clusters** ($\alpha \geq 2$): the threshold He content for trap mutation increases roughly as $\alpha^{2/3}$ (reflecting the increased cavity volume), and the emitted SIA is strongly bound to the cluster surface (binding energy $\sim 2\text{--}3$ eV) [109].

For a given $\text{He}_m\text{V}_\alpha$ cluster, the dominant overpressure relief channel is the one with the lower effective barrier. The competition can be summarized as:

$$\text{If } E_B^{\text{He}}(\alpha, m) < E_{\text{TM}}(\alpha, m) : \quad \text{He emission dominates,} \quad (\text{D.20})$$

$$\text{If } E_{\text{TM}}(\alpha, m) < E_B^{\text{He}}(\alpha, m) : \quad \text{trap mutation dominates.} \quad (\text{D.21})$$

In bcc Fe, DFT data indicate that for small clusters ($\alpha \leq 4$) at He/V ratios above $\sim 1.3\text{--}1.7$, the trap-mutation barrier drops below the He binding energy, making Frenkel pair emission the preferred pathway [59, 67]. This is consistent with the observation that the equilibrium He/V ratio for small clusters is $\sim 1.0\text{--}1.3$ (DFT), while for larger bubbles ($R > 1$ nm) the mechanical equilibrium condition $P_g = 2\gamma/R$ gives He/V ~ 0.5 at moderate temperatures [110].

For practical implementation in cluster dynamics, the maximum He loading per vacancy cluster is:

$$\ell_{\text{max}}(\alpha) = \lfloor \alpha_{\text{He}} \alpha^{2/3} \rfloor, \quad \alpha_{\text{He}} \approx 1.5\text{--}2.0. \quad (\text{D.22})$$

This formula should be understood as the *trap-mutation limit*: beyond ℓ_{max} , the cluster spontaneously emits a Frenkel pair rather than accommodating an additional He atom. The He emission limit (where $E_B^{\text{He}} = 0$) occurs at a somewhat higher He/V ratio, but the trap-mutation channel intervenes first and prevents the cluster from reaching that state.

Appendix D.2.4. Vacancy emission from dislocation loops

An interstitial loop may also emit a vacancy into the matrix. This process is equivalent to adding one self-interstitial atom to the loop, since the emission of a vacancy increases the net interstitial content of the defect:

$$I_n \xrightarrow{\Gamma_v^{I\text{-loop}}} I_{n+1} + \text{V}_m. \quad (\text{D.23})$$

Here I_n denotes an interstitial loop containing n self-interstitial atoms. The vacancy-emission rate is again written using detailed balance:

$$\Gamma_v^{I\text{-loop}}(n) = \mathcal{K}_{v,n+1}^{I\text{-loop}} \exp\left[-\frac{E_B^{v,I\text{-loop}}(n)}{k_B T}\right], \quad (\text{D.24})$$

where $\mathcal{K}_{v,n+1}^{I\text{-loop}}$ is the vacancy capture rate constant of the daughter loop I_{n+1} , and $E_B^{v,I\text{-loop}}(n)$ is the effective binding energy associated with removing a vacancy from the matrix and incorporating the corresponding interstitial content into the loop.

Equations (D.17) and (D.24) emphasize the different physical origins of vacancy emission from the two defect classes. For voids, vacancy emission is controlled by surface energy and internal gas pressure. For interstitial loops, vacancy emission is controlled by the incremental line energy required to increase the loop size. In both cases, the emission rate is exponentially sensitive to the appropriate vacancy binding energy and therefore becomes important only at sufficiently high temperatures or for weakly bound small clusters.

Appendix E. Input Parameters

Table E.23 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions. Every rate constant is built from one of the four precomputed geometric constants (A_{sph} , A_{loop} , A_{1D} , B_{rot}), the cluster size raised to a geometry-dependent power, and the appropriate effective reaction frequency $\omega_{\alpha}^{\text{eff}}$ (Eq. D.9). The binding energies $E_b^{\alpha}(m, \ell)$ for voids and loops come from Section Appendix B.2; the bubble binding energies $E_B^v(m, \ell)$ and $E_B^{\text{He}}(m, \ell)$ come from Tables B.15 and B.16. He emission dominates when $E_B^{\text{He}} < E_{\text{TM}}$; trap mutation dominates when $E_{\text{TM}} < E_B^{\text{He}}$, typically at $\text{He}/\text{V} \gtrsim 1.3\text{--}1.7$ for small clusters in bcc Fe.

Table E.24 collects the effective migration barriers for small vacancy clusters in α -Fe from several computational methods. Table E.25 collects the point-defect diffusion and energetics parameters used in RADCLUSTER simulations, including the helium and general diffusion parameters; Pure-Fe baseline values are given first, and EUROFER solute-trapping corrections follow. The effective diffusivity is $D_{\alpha}^{\text{eff}} = D_{\alpha}^{\text{Fe}}/(1 + \sum_s K_s c_s)$ with $K_s = z_s \exp(E_b^s/k_B T)$.

All reaction-rate geometry is captured by four numerical constants:

$$\begin{aligned}
 A_{\text{sph}} &= (48\pi^2)^{1/3} && \approx 7.818 \quad (\text{spherical clusters}), \\
 A_{\text{loop}} &= 8\sqrt{\pi/\sqrt{3}} && \approx 10.78 \quad (\text{dislocation loops}), \\
 A_{1D} &= 9/(8\pi^{2/3}) && \approx 2.632 \quad (\text{pure 1D sinks}), \\
 B_{\text{rot}} &= (4/\pi)(8\pi/3)^{1/3} && \approx 2.627 \quad (1\text{D}/3\text{D interpolation}).
 \end{aligned} \tag{E.1}$$

Table E.26 lists the dissociation and binding energy RADCLUSTER input parameters for EUROFER. The parameters used to compute helium cluster dissociation are given in Table E.27, and those used in RADCLUSTER for trap mutation are listed in Table E.28. For clusters at or above $\ell_{\text{max}}(\alpha)$, trap mutation can be enforced as a hard boundary: any He absorption event that would push $m > \ell_{\text{max}}$ is immediately redirected into a Frenkel pair emission step. The ℓ_{max} formula represents the trap-mutation limit, not the He emission limit. The He emission limit (where $E_B^{\text{He}} = 0$) lies at a somewhat higher He/V ratio, but trap mutation intervenes first.

Table E.23: Reaction rate input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
<i>Geometric prefactors (precomputed constants):</i>			
Spherical cluster	$A_{\text{sph}} = (48\pi^2)^{1/3}$	7.818	—
Dislocation loop	$A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}}$	10.78	—
Pure 1D sink	$A_{1D} = 9/(8\pi^{2/3})$	2.632	—
1D/3D interpolation	$B_{\text{rot}} = (4/\pi)(8\pi/3)^{1/3}$	2.627	—
V-SIA recombination	$4\sqrt{3}\pi$	21.77	—
<i>Dislocation sink:</i>			
Network dislocation density	ρ_d	10^{13} – 10^{14}	m^{-2}
Interstitial bias factor	Z_i^d	1.02–1.15	—
Vacancy bias factor	Z_v^d	1.00	—
<i>Grain-boundary sink:</i>			
Grain diameter	d_g	5–20	μm
Bias factors	$Z_i^{gb} = Z_v^{gb}$	1.00	—
<i>Precipitate sink:</i>			
Mean precipitate radius	r_p	10–50	nm
Precipitate number density	N_p	10^{20} – 10^{21}	m^{-3}
Bias factors	$Z_i^p = Z_v^p$	1.00	—
<i>Mobility cutoffs:</i>			
Max mobile SIA cluster	i_{mobile}	100	SIAs
Max mobile vacancy cluster	v_{mobile}	5	vacancies
<i>He state-space parameters:</i>			
Max He loading parameter	α_{He}	1.5–2.0	—
$\ell_{\text{max}}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$			(trap-mutation limit)
Re-solution parameter (fission)	b_0	10^{-2}	$\text{dpa}^{-1}/\text{He}$
Re-solution parameter (fusion)	b_0	10^{-1}	$\text{dpa}^{-1}/\text{He}$
<i>Size grid (user-defined):</i>			
Max SIA cluster tracked	N	user input	—
Max vacancy cluster tracked	M	user input	—
Max He per cluster	L	user input	—
Grouping transition size	n_{group}	~50–200	—

Table E.24: RADCLUSTER cluster-mobility input parameters. Single-defect values appear in Table E.25; this table covers size-dependent cluster diffusion. SIA clusters glide in 1D as $\frac{1}{2}\langle 111 \rangle$ loops; vacancy clusters migrate in 3D with size-increasing barriers.

Quantity	Symbol	Adopted	Units	Literature range / source
<i>SIA-cluster 1D glide ($n \geq 4$, $D_n^{1D} = \frac{3a^2\nu_0}{2n^{s_i}} \exp(-E_m^{1D}/k_B T)$):</i>				
Glide migration energy	E_m^{1D}	0.03	eV	$\lesssim 0.05$ [61]
Attempt frequency	ν_0^{1D}	6×10^{12}	s ⁻¹	[61]
Size exponent	s_i	0.7	—	0.5–1.0
Mobility cutoff	i_{mobile}	100	SIAs	—
<i>Small-cluster SIA migration (size-resolved, for reference):</i>				
$n = 1$ ($\langle 110 \rangle$ dumbbell), 3D	E_m	0.34	eV	[52]
$n = 2$ (parallel $\langle 110 \rangle$), 3D	E_m	0.42	eV	[52]
$n = 3$ (3D/1D transition)	E_m	0.15–0.20	eV	—
$n \geq 4$ ($\frac{1}{2}\langle 111 \rangle$ loop, 1D)	E_m^{1D}	$\lesssim 0.05$	eV	[61]
<i>Vacancy-cluster diffusion ($D_{(-m,0)} = D_v m^{-s_v}$, $m \leq v_{\text{mobile}}$):</i>				
Size exponent	s_v	1.0	—	—
Mobility cutoff	v_{mobile}	5	vacancies	—
$m = 1$ (monovacancy), 3D	E_m	0.67	eV	0.65–0.70 [52, 56, 111]
$m = 2$ (divacancy), 3D	E_m	0.70	eV	0.62–0.75 [94, 95]
$m = 3$ (trivacancy, fast), 3D	E_m	0.42	eV	0.35–0.50 [94, 98]
$m = 4$ –5 (weakly mobile)	E_m	0.7–1.1	eV	[94]
$m \geq 6$ (immobile sink)	—	$D = 0$	—	[94]
<i>Mixed 1D/3D transport (EUROFER):</i>				
Mean free path	$\hat{L} = L/a$	30–100	—	—
1D/3D interpolation constant	B_{rot}	2.627	—	—

Table E.25: RADCLUSTER point-defect diffusion and energetics input parameters for bcc Fe / EUROFER. The Adopted column gives the single value used in the code; the Literature range / source column records the supporting data. Effective EUROFER diffusivities follow $D_{\alpha}^{\text{eff}} = D_{\alpha}^{\text{Fe}} / (1 + \sum_s K_s c_s)$ with $K_s = z_s \exp(E_b^s / k_B T)$ (solute trapping in Table C.22).

Quantity	Symbol	Adopted	Units	Literature range / source
<i>Lattice and elastic constants:</i>				
Lattice constant	a	0.2867	nm	[49]
Atomic volume	$\Omega = a^3/2$	1.18×10^{-29}	m ³	—
Burgers vector ($\frac{1}{2}\langle 111 \rangle$)	b_{111}	0.2482	nm	—
Burgers vector ($\langle 100 \rangle$)	b_{100}	0.2867	nm	—
Shear modulus (RT)	μ	82	GPa	—
Poisson ratio	ν	0.29	—	—
Surface energy	γ_s	2.0	J/m ²	—
<i>Vacancy:</i>				
Formation energy	E_f^v	2.0	eV	—
Migration energy	E_m^v	0.67	eV	0.55–0.70 [52, 56, 95]
Attempt frequency	ν_v	1.0×10^{13}	s ⁻¹	6×10^{12} – 6×10^{13} [53, 112]
Pure-Fe diffusion prefactor	D_0^v	8.2×10^{-8}	m ² /s	—
Divacancy binding (2NN)	E_B^{2v}	0.22	eV	—
Divacancy migration	E_m^{2v}	0.70	eV	0.62–0.75 [94, 95]
<i>Self-interstitial ($\langle 110 \rangle$ dumbbell):</i>				
Formation energy	E_f^i	3.64	eV	—
Migration energy (T-R)	E_m^i	0.34	eV	[52, 57]
Attempt frequency	ν_i	1.0×10^{13}	s ⁻¹	$\sim 4 \times 10^{12}$ [52, 57, 71]
Pure-Fe diffusion prefactor	D_0^i	8.2×10^{-8}	m ² /s	—
Di-SIA binding	E_B^{2i}	0.80	eV	—
Di-SIA migration	E_m^{2i}	0.42	eV	[52]
<i>Interstitial helium (tetrahedral):</i>				
Migration energy	E_m^h	0.06	eV	0.06–0.09 [64]
Attempt frequency	ν_h	3.0×10^{12}	s ⁻¹	10^{12} – 3×10^{12} [65]
Pure-Fe diffusion prefactor	$D_0^h = a^2 \nu_h$	2.5×10^{-8}	m ² /s	—
He–vacancy binding (1st He)	E_b^{h-V}	2.30	eV	[64, 65]
He–vacancy binding (2nd He)	$E_b^{h-V_2}$	2.00	eV	[64]
He–divacancy binding	E_b^{h-V}	2.5–2.8	eV	[102]
He trapping coordination	z_V^h	6	—	—
<i>He equation of state (virial coefficients, 600–1000 K):</i>				
Second virial coefficient	B_2	1.67×10^{-29}	m ³ /atom	—
Third virial coefficient	B_3	1.84×10^{-58}	m ⁶ /atom ²	—

Table E.26: Dissociation / binding energy input parameters for EUROFER cluster dynamics: vacancy binding to voids and bubbles, and point-defect binding to dislocation loops.

Parameter	Symbol	Value	Units
<i>Void (vacancy cluster) binding — vacancy emission:</i>			
Vacancy formation energy	E_f^v	2.0	eV
Surface energy	γ_s	2.0	J/m ²
Divacancy binding (DFT)	$E_b^v(2)$	0.22	eV
Atomistic–continuum decay constant	λ	0.5756	vac ^{−1}
Atomistic–continuum anchor size	$\alpha_{1/2}$	10	vacancies
<i>Bubble — vacancy binding (Eq. B.18):</i>			
He–vacancy binding (1st He in V_1)	$E_B^V(1, 1)$	2.30	eV
Fitting amplitude $A(m = 0)$	$A(0)$	1.2353	eV
Fitting amplitude $A(m = 1)$	$A(1)$	2.9064	eV
Fitting amplitude $A(m = 2)$	$A(2)$	3.4147	eV
Fitting amplitude $A(m = 3)$	$A(3)$	2.1504	eV
Fitting amplitude $A(m = 4)$	$A(4)$	−0.1590	eV
<i>Vacancy loop binding:</i>			
Stacking fault energy	γ_{sf}	0.6	J/m ²
Core radius	r_c	$b/3$	—
Divacancy binding anchor	$E_b(2)$	0.22	eV
Decay constant	λ	0.575	vac ^{−1}
<i>Interstitial loop binding (power-law fit, $n \leq 25$):</i>			
$\frac{1}{2}\langle 111 \rangle$ loop: A	A_{111}	0.7501	eV
$\frac{1}{2}\langle 111 \rangle$ loop: B	B_{111}	0.3873	—
$\langle 100 \rangle$ loop: A	A_{100}	0.7160	eV
$\langle 100 \rangle$ loop: B	B_{100}	0.3581	—
Blending transition center	n_{tr}	25	SIAs
Blending transition width	σ_{tr}	5	SIAs

Table E.27: Helium binding energy model parameters for bubbles (Eq. B.25). The continuum term uses the He solution energy, the thermal de Broglie wavelength, and the virial coefficients from Table E.26. The atomistic correction decays exponentially with m .

Parameter	Symbol	Value	Units
<i>Continuum He binding (Eq. B.24):</i>			
He interstitial solution energy	E_s^{He}	2.35	eV
He atomic mass	m_{He}	4.0026	amu
He virial B_2	B_2	1.67×10^{-29}	m^3/atom
He virial B_3	B_3	1.84×10^{-58}	m^6/atom^2
<i>Atomistic-continuum blending (Eq. B.25):</i>			
Decay constant	μ	0.658	He^{-1}
Fitting amplitude $A^{\text{He}}(\alpha = 2)$	$A^{\text{He}}(2)$	0.55	eV
Fitting amplitude $A^{\text{He}}(\alpha = 3)$	$A^{\text{He}}(3)$	0.40	eV
Fitting amplitude $A^{\text{He}}(\alpha = 4)$	$A^{\text{He}}(4)$	0.75	eV

Table E.28: Trap mutation (Frenkel pair emission) parameters for EUROFER cluster dynamics. Trap mutation is the reaction $\text{He}_m\text{V}_\alpha \rightarrow \text{He}_m\text{V}_{\alpha+1} + \text{SIA}$, with rate $\Gamma_{\text{TM}} = \nu_0 \exp(-E_{\text{TM}}/k_B T)$. Self-trapping refers to the $\alpha = 0$ case (pure interstitial He cluster). Values from [59], [104], [67], and [109].

Parameter	Symbol	Value	Units
<i>Trap mutation rate: $\Gamma_{\text{TM}} = \nu_0 \exp(-E_{\text{TM}}/k_B T)$</i>			
Attempt frequency	ν_0	1.0×10^{12}	s^{-1}
<i>Activation barriers $E_{\text{TM}}(\alpha, m)$ for $\text{He}_m\text{V}_\alpha$ in bcc Fe:</i>			
He_5V_1	$E_{\text{TM}}(1, 5)$	~ 1.0	eV
He_6V_1	$E_{\text{TM}}(1, 6)$	~ 0.5	eV
He_7V_1	$E_{\text{TM}}(1, 7)$	~ 0 (spontaneous)	eV
<i>Self-trapping barriers ($\alpha = 0$, pure He clusters):</i>			
He_4	$E_{\text{TM}}(0, 4)$	~ 0.3	eV
He_5	$E_{\text{TM}}(0, 5)$	~ 0.1	eV
He_6	$E_{\text{TM}}(0, 6)$	~ 0 (spontaneous)	eV
<i>Maximum He loading (trap-mutation limit):</i>			
He/V ratio parameter	α_{He}	1.5–2.0	—
$\ell_{\text{max}}(\alpha) = \lfloor \alpha_{\text{He}} \alpha^{2/3} \rfloor$			
<i>Competition criterion:</i>			
He emission dominates if $E_B^{\text{He}}(\alpha, m) < E_{\text{TM}}(\alpha, m)$			
Trap mutation dominates if $E_{\text{TM}}(\alpha, m) < E_B^{\text{He}}(\alpha, m)$			
In bcc Fe: trap mutation wins at $\text{He}/\text{V} \gtrsim 1.3\text{--}1.7$ for $\alpha \leq 4$			

Appendix F. Parameter Identification Algorithm

The two-fidelity emulator of stage (ii) is constructed and queried as follows. The procedure is written so that each step maps directly onto a callable routine in the calibration code; symbols follow the notation of the proposed workflow above.

Inputs.. Parameter bounds $\{[\theta_i^{\min}, \theta_i^{\max}]\}_{i=1}^p$ for the $p \approx 10$ –12 screened parameters; condition grid $\{(T_k, d_k)\}$ of temperatures and target doses; low-fidelity dose cap $d_{\text{LF}} \approx 0.1$ dpa and coarse bin-moment resolution n_{LF} ; high-fidelity dose cap $d_{\text{HF}} \approx 100$ dpa and full bin resolution n_{HF} ; low- and high-fidelity design sizes $N_{\text{LF}} \gg N_{\text{HF}}$; verified integrator tolerances (rtol, atol) held fixed at both fidelities.

1. **Low-fidelity design.** Draw N_{LF} points $\{\boldsymbol{\theta}^{(j)}\}$ from a space-filling Sobol sequence over the bounded parameter box and rescale each coordinate to its physical range.
2. **Low-fidelity runs.** For every design point, integrate RADCLUSTER with a resolution n_{LF} to dose d_{LF} under tolerances (rtol, atol), and record the early-time identifiable observables $\mathbf{y}_{\text{LF}}^{(j)}$ (incipient loop number density, He-stabilized bubble density, and mean size). Discard and flag any run that fails the tolerance check rather than relaxing it.
3. **High-fidelity subset.** Select N_{HF} design points as a maximin subset of the low-fidelity design so that the expensive runs remain space-filling and nested within the cheaper ones.
4. **High-fidelity runs.** For each selected point, integrate at full resolution n_{HF} to d_{HF} , capturing the cavity bifurcation and supercritical void swelling, and record the full observable vector $\mathbf{y}_{\text{HF}}^{(j)}$ and size distributions.
5. **Transform.** Map all density observables to log space and standardize every output channel to zero mean and unit variance; store the transform parameters for later inversion.
6. **Fit low-fidelity GP.** Train a Gaussian process $\widehat{\mathcal{M}}_{\text{LF}}(\boldsymbol{\theta})$ on $\{(\boldsymbol{\theta}^{(j)}, \mathbf{y}_{\text{LF}}^{(j)})\}$ with an anisotropic Matérn-5/2 kernel; optimize length scales and noise by maximizing the marginal likelihood.

7. **Fit discrepancy GP.** At the high-fidelity points, evaluate $\mu_{\text{LF}}^{(j)} = \widehat{\mathcal{M}}_{\text{LF}}(\boldsymbol{\theta}^{(j)})$ and form the autoregressive residual $r^{(j)} = \mathbf{y}_{\text{HF}}^{(j)} - \rho \mu_{\text{LF}}^{(j)}$. Train a second GP $\delta(\boldsymbol{\theta})$ on $\{(\boldsymbol{\theta}^{(j)}, r^{(j)})\}$ and estimate the scaling ρ jointly with the discrepancy hyperparameters by marginal likelihood.

8. **Assemble predictor.** Define the fused high-fidelity emulator

$$\widehat{\mathcal{M}}_{\text{HF}}(\boldsymbol{\theta}) = \rho \widehat{\mathcal{M}}_{\text{LF}}(\boldsymbol{\theta}) + \delta(\boldsymbol{\theta}), \quad \sigma_{\text{HF}}^2(\boldsymbol{\theta}) = \rho^2 \sigma_{\text{LF}}^2(\boldsymbol{\theta}) + \sigma_{\delta}^2(\boldsymbol{\theta}),$$

returning both the mean prediction and its predictive variance in the standardized log space.

9. **Validate.** On held-out high-fidelity runs, check that the standardized prediction errors have unit-variance and that nominal coverage of the predictive intervals is achieved; if coverage fails, increase N_{HF} and return to step 4.

10. **Serve to calibration.** Expose a function that accepts $\boldsymbol{\theta}$, returns $(\widehat{\mathcal{M}}_{\text{HF}}, \sigma_{\text{HF}}^2)$ in physical units (inverting the transform of step 5), and adds σ_{HF}^2 to the measurement and discrepancy variances in the likelihood of Eq. (55).

11. **Active refinement.** After each calibration pass, place the next high-fidelity run at the $\boldsymbol{\theta}$ that maximizes expected posterior-variance reduction per unit predicted CPU cost, append it to the high-fidelity set, and refit from step 7.

The output is the calibrated posterior, together with a propagated predictive uncertainty in which the dose-extrapolation error is carried by the high-fidelity discrepancy term $\delta(\boldsymbol{\theta})$ rather than by an uncontrolled polynomial fit.

Appendix G. Experimental Database

We compile here and analyze all experimentally reported size-distribution data for dislocation loops and voids in F/M steels and pure iron, covering the primary alloys of interest—EUROFER97, F82H, T91, HT9, Fe-9Cr model alloys, and pure α -Fe—irradiated by neutrons, self-ions, and dual-beam ions over a wide range of doses (0.1–650 dpa) and temperatures (60–600 °C). Table G.29 documents the data sources.

Appendix G.1. Temperature Dependence of Loops at Fixed Dose:

The most dramatic influence on both loop density and mean size is irradiation temperature. Figure G.11 shows the temperature dependence of the dislocation loop number density and mean size. The data establish the following trends across all F/M steels studied:

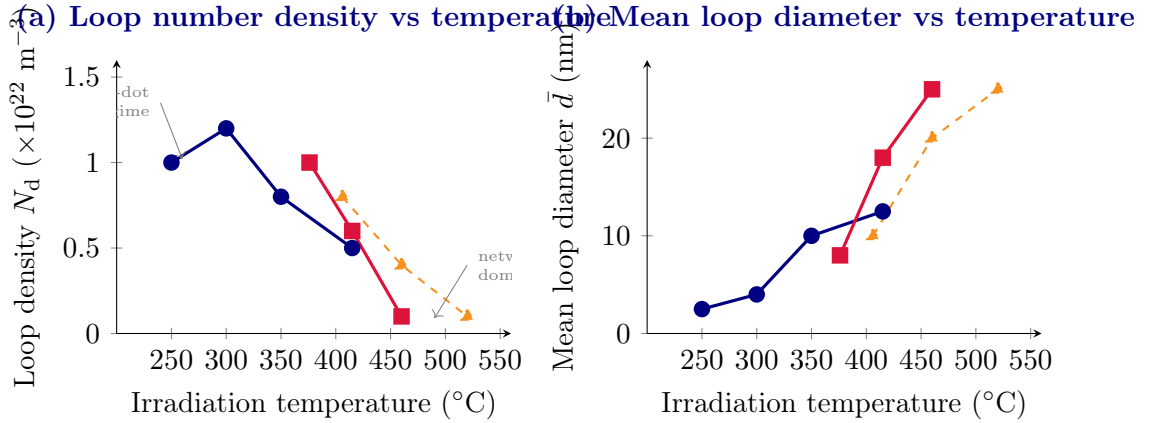


Figure G.11: Temperature dependence of (a) loop number density N_d and (b) mean loop diameter $\bar{d}$ at fixed dose. In both panels, blue circles denote EUROFER97 at 16.3 dpa [5], red squares denote T91 at 15.4 dpa under BOR-60 neutron irradiation [6], and orange dashed triangles denote T91 under dual-ion irradiation at 35 dpa [14]. Both quantities show a clear monotonic trend: loop density decreases and mean size increases with increasing temperature, consistent with enhanced thermal diffusion accelerating coarsening and loop dissolution into the network dislocation structure. The peak loop density in EUROFER97 occurs at approximately 300 °C, above which coarsening dominates. The ion-irradiated T91 data are shifted by approximately +60–70 °C relative to the neutron data, as expected from the higher dose rate [14].

1. Loop density reaches a maximum in the range 280–320 °C. Below this temperature, the visible-loop density is artificially suppressed because

most defects remain as sub-resolution “black dots” ($d < 3$ nm). Above this temperature, thermal diffusion accelerates coarsening, and large loops are absorbed into the pre-existing network dislocation structure. EUROFER97 at 16.3 dpa shows $N_d = 1.2 \times 10^{22} \text{ m}^{-3}$ at 300 °C, dropping to $0.5 \times 10^{22} \text{ m}^{-3}$ at 415 °C [5].

2. In T91 at 15.4 dpa in BOR-60, N_d decreases from $1.0 \times 10^{22} \text{ m}^{-3}$ at 376 °C to $\sim 0.1 \times 10^{22} \text{ m}^{-3}$ at 460 °C, where network dislocations dominate [6].
3. In EUROFER97, $\bar{d}$ rises from 2.5 nm at 250 °C to 12.5 nm at 415 °C [5]. In T91, $\bar{d}$ grows from ~ 8 nm at 376 °C to > 25 nm at 460 °C [6].
4. Below ~ 300 °C in EUROFER97, $\frac{1}{2}\langle 111 \rangle$ loops dominate. At and above 350 °C, $\langle 100 \rangle$ loops become dominant, consistent with their lower migration energy and higher thermal stability [4, 5]. In T91 and HT9 under BOR-60 irradiation at 376–460 °C, $a\langle 100 \rangle$ loops dominate at all doses [6, 7].

Appendix G.2. Dose Dependence of Loops at Fixed Temperature:

Figure G.12 shows the loop density and mean size as functions of dose at representative fixed temperatures.

The dose dependence separates into two regimes:

1. Accumulation regime ($< \sim 5$ dpa): Both density and mean size increase with dose. The growth of N_d follows a sub-linear power law $N_d \propto \phi^n$ with $n < 1$, reflecting simultaneous nucleation and clustering. The mean size grows as $\bar{d} \propto \phi^m$ with $m \approx 0.3$ – 0.5 . HT9 in-situ Kr data show the density peak at ~ 4 dpa [39].
2. Saturation regime ($\gtrsim 15$ – 20 dpa): Density either saturates (EUROFER97 at 330 °C) or decreases as large loops absorb smaller ones and convert to network dislocations. Mean size continues to grow. T91 and HT9 (BOR-60) show density decreasing from ~ 1.0 to $\sim 0.7 \times 10^{22} \text{ m}^{-3}$ and mean size growing from ~ 8 to ~ 14 nm between 15 and 35 dpa.

Self-ion irradiation of Fe-9Cr at 2 dpa/300 °C produces a loop density of $6.8 \times 10^{22} \text{ m}^{-3}$ and a mean size of 6.5 nm — a *homogeneous* field comparable to the near-dislocation values under neutron irradiation but 2–3 orders of magnitude higher than the matrix value under neutrons [13]. The temperature must be raised by +60–70 °C for ion irradiation to match the

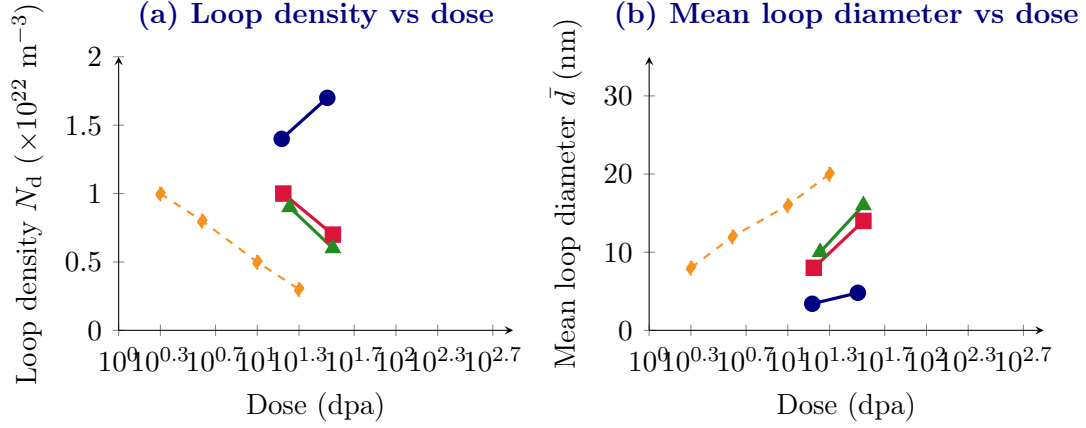


Figure G.12: Dose dependence of (a) loop number density and (b) mean loop diameter at fixed temperature. In both panels, blue circles denote EUROFER97 at approximately 330 °C under BOR-60 neutron irradiation [4], red squares denote T91 at 376 °C [6], green triangles denote HT9 at 377 °C [7], orange dashed diamonds denote in situ Kr-ion irradiation of HT9 at 420 °C [39], and the purple dotted pentagon in panel (a) marks the Fe-9Cr self-ion point at 2 dpa and 300 °C from Lambrecht et al. [13], which lies above the plotted vertical range. EUROFER97 at ~ 330 °C shows near-saturation behavior: the density increases by only about 20% from 15 to 32 dpa, while $\bar{d}$ increases by about 1.4 nm. T91 and HT9 at 376–377 °C show coarsening: the density decreases while $\bar{d}$ grows by about 6 nm over the same dose range. The in situ Kr-ion data for HT9 at 420 °C reveal a density peak near 4 dpa followed by a monotonic decrease as loops convert to network dislocations.

neutron-irradiated microstructure [14, 15]. The $\langle 100 \rangle$ -loop fraction is higher under ions ($\sim 40\%$) than under neutrons ($\sim 20\text{--}30\%$) at the same nominal conditions.

Appendix G.3. Temperature Dependence of Voids at Fixed Dose:

Unlike dislocation loops, void/cavity microstructures in F/M steels exhibit a qualitative change in character with increasing dose: a *bimodal size distribution* develops once the dose exceeds ~ 35 dpa at 376–420 °C. This bimodal structure consists of a high-density population of He-stabilized subcritical bubbles ($d < r_c \approx 2$ nm) coexisting with a lower-density population of supercritical growing voids ($d > 2\text{--}4$ nm). Table G.30 summarizes the key measured density and size values that delineate the transition.

The temperature dependence of void density and swelling (which is proportional to $N_v \bar{d}_v^3$) follows the well-known bell-shaped swelling curve. The

key quantitative observations on the temperature dependence of void density and mean size are:

1. Unimodal regime below $\sim 415^\circ\text{C}$: In T91 and HT9 (BOR-60, 15–17dpa), only sub-2 nm bubbles are present with densities $\sim 10^{21} \text{ m}^{-3}$ and swelling $< 0.01\%$. EUROFER97 remains in this regime even at 32dpa/335 $^\circ\text{C}$, with bubble density $0.11\text{--}0.14 \times 10^{22} \text{ m}^{-3}$ and mean size 1.6–2.6 nm [4].
2. Void density peaks in the bimodal regime: The total cavity density (bubbles + voids) is highest near the peak-swelling temperature. At 460 $^\circ\text{C}$ (dual-ion T91), both bubble and void populations are most developed; swelling peaks here [9].
3. Void mean size increases with temperature in the bimodal regime: In dual-ion T91, the void-population mean diameter grows from ~ 5.5 nm at 406 $^\circ\text{C}$ to ~ 7 nm at 460 $^\circ\text{C}$ before the distribution collapses at higher temperatures [14]. In HT9 at 35.1dpa (BOR-60), the void-population mean is ~ 7 nm at 377 $^\circ\text{C}$ [7].
4. He/dpa ratio elevates bubble density and nucleates bimodal earlier: Under mixed-spectrum neutron irradiation (HFR, Petten), EUROFER97 develops a higher void density than under fast-spectrum BOR-60 at the same dose and temperature, attributed to the higher helium production rate [10]. In dual-ion T91, higher He co-injection lowers the critical cavity radius and nucleates bimodal distributions at lower doses [9].

Appendix G.4. Dose Dependence of Voids at Fixed Temperature:

The dose dependence of void density and mean size differs fundamentally from that of dislocation loops. Three stages are identified:

- Incubation ($\lesssim 20\text{dpa}$, $T \approx 380^\circ\text{C}$): Only sub-critical bubbles exist. Bubble density is high ($\sim 10^{21} \text{ m}^{-3}$) and mean size small (< 2 nm). Swelling is $< 0.02\%$. The microstructure is effectively in the unimodal regime.
- Transition and bimodal development ($\sim 20\text{--}50\text{dpa}$): A fraction of bubbles escape the subcritical bubble class and form supercritical voids. The total cavity density develops a bimodal character; swelling increases rapidly (T91: 0.01% at 15dpa $\rightarrow 0.02\%$ at 35dpa; HT9: 0.02% at 17dpa $\rightarrow 0.07\%$ at 35dpa [6, 7]).

- Steady-state swelling ($\gtrsim 100\text{dpa}$): Under self-ion irradiation, HT9 reaches a steady-state swelling rate of $\sim 0.2\%/ \text{dpa}$ at 460°C beyond $\sim 200\text{dpa}$ [8]. At 600dpa , the F/M HT9 reaches 2.6% maximum swelling (at 334 local dpa) [132]. Void mean diameter in this regime reaches $20\text{--}30\text{ nm}$ and continues to grow.

Table G.31 summarizes the swelling–dose relationship for the best-studied alloys.

Pure α -iron provides a reference system free of alloying effects. At 60°C (neutron, 0.79dpa), cavity densities exceed 10^{24} m^{-3} with sub- 1 nm mean size—an extremely high density of athermal vacancy clusters [115]. At 400°C (neutron, $\sim 1\text{dpa}$), voids develop a truncated octahedral morphology with $\{111\}$ facets and $\{100\}$ truncations; the mean size reaches $\sim 5\text{ nm}$ and swelling peaks at 0.07% [116]. Above 400°C , thermal vacancy emission suppresses void growth.

Appendix G.5. Summary of Quantitative Trends

Table G.32 condenses the principal quantitative trends. The following quantitative anchors summarize the data:

- Loop density maximum: $\sim 1.2 \times 10^{22}\text{ m}^{-3}$ near 300°C ; decreases to $< 0.1 \times 10^{22}\text{ m}^{-3}$ at 460°C (T91).
- Loop mean size range: 2.5 nm (EUROFER97, 250°C) to $> 25\text{ nm}$ (T91, 460°C).
- Void density (unimodal regime): $\sim 10^{21}\text{ m}^{-3}$, mean $1.2\text{--}2.6\text{ nm}$.
- Void density (bimodal regime): bubble population $\sim 2\text{--}6 \times 10^{21}\text{ m}^{-3}$, mean $1.2\text{--}1.4\text{ nm}$; void population $\sim 5 \times 10^{20}\text{ m}^{-3}$, mean $5\text{--}7\text{ nm}$.
- Swelling: 0.02% at $15\text{--}17\text{dpa}$, 0.07% at 35dpa ; $\sim 2.6\%$ at 600dpa (HT9).

Appendix G.6. Dislocation Loop Size Distributions

Dislocation loops in bcc iron-based alloys form with two predominant Burgers vectors: $\mathbf{b} = \frac{1}{2}\langle 111 \rangle$ (glissile, $a/2$ type) and $\mathbf{b} = \langle 100 \rangle$ (sessile, a type). The relative proportion shifts with temperature and alloy composition: $\frac{1}{2}\langle 111 \rangle$ loops dominate below $\sim 300^\circ\text{C}$ in EUROFER97, while $\langle 100 \rangle$ loops become dominant at and above 350°C [4, 5]. In T91 and HT9 under BOR-60

irradiation at 376–420 °C, $a\langle 100 \rangle$ loops are dominant at all doses studied [6, 7].

The most dramatic variable controlling the loop size distribution is irradiation temperature. Figure G.13 illustrates the progressive evolution schematically.

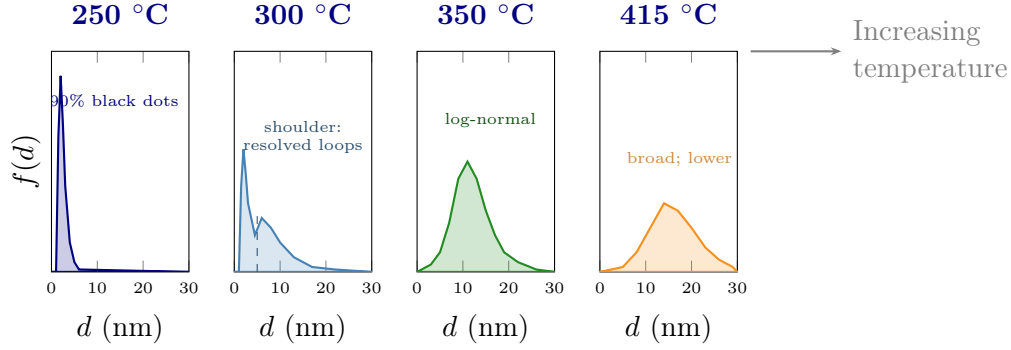


Figure G.13: Schematic representation of the temperature-dependent evolution of dislocation loop size distributions in EUROFER97 (16.3 dpa, BOR-60 neutron irradiation). At 250 °C, the distribution is dominated by a narrow spike of sub-3 nm black-dot defects ($\sim 90\%$). At 300 °C, a resolved-loop shoulder develops at 5–15 nm while the black-dot peak persists. Above 350 °C, the distribution becomes a well-developed log-normal centred at 10–13 nm. At 415 °C, the distribution broadens further and the total density decreases as large loops begin transforming into network dislocations.

For a fixed irradiation temperature, increasing the dose shifts the distribution rightward and narrows the relative spread. Figure G.14 illustrates this for EUROFER97 (BOR-60) and HT9 (BOR-60).

Key experimental observations on the dose dependence are:

- In EUROFER97 at 330–335 °C, the mean diameter increases only from 3.4 nm (15 dpa) to 4.8 nm (32 dpa), while the number density increases from $1.4 \times 10^{22} \text{ m}^{-3}$ to $1.7 \times 10^{22} \text{ m}^{-3}$ — an almost saturated state after 15 dpa [4].
- In T91 at 376 °C, the mean diameter grows from ~ 8 nm (15.4 dpa) to ~ 14 nm (35.1 dpa), while density *decreases*, indicating coarsening by the absorption of smaller loops into larger ones and into network dislocations [6].
- In HT9 at 377 °C, the mean diameter grows from ~ 10 nm (17.1 dpa) to ~ 16 nm (35.1 dpa) with a commensurate density decrease [7].

- At very high doses under self-ion irradiation (>100 dpa), loop density first peaks around 4 dpa and then decreases monotonically as loops are absorbed into the network dislocation structure [39].

Loop Size Distribution Evolution with Dose — HT9 at 377°C (BOR-60)

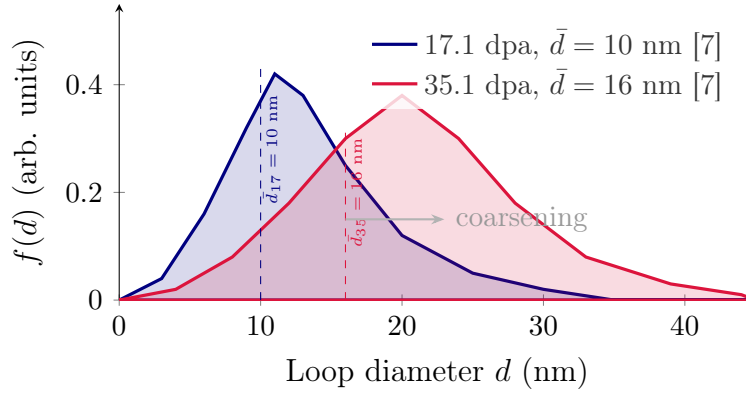


Figure G.14: Schematic dose evolution of the dislocation loop size distribution in HT9 at 377 °C (BOR-60 neutron irradiation). The distribution shifts rightward and the peak height decreases, indicative of coarsening: mean diameter increases from ~ 10 nm at 17.1 dpa to ~ 16 nm at 35.1 dpa while total loop number density decreases [7].

Self-ion irradiation at the same nominal dose and temperature as neutron irradiation produces dislocation loop distributions that differ in two important respects:

1. The distribution is *more symmetric* and *more homogeneous spatially* —neutron irradiation produces a strongly heterogeneous distribution with high loop densities near pre-existing dislocation lines ($\sim 10^{23} \text{ m}^{-3}$) and sparse populations in the matrix ($\sim 10^{20} \text{ m}^{-3}$), while self-ion irradiation produces a uniform loop field at intermediate density [13].
2. The fraction of sessile $\langle 100 \rangle$ loops is larger under self-ion irradiation ($\sim 40\%$) than under neutron irradiation for the same Fe-9Cr alloy at 300 °C [13].

The temperature must be raised by $+60\text{--}70$ °C for ion irradiation to reproduce the same microstructure as neutron irradiation, due to the ~ 3 orders of magnitude higher dose rate [14, 15].

Appendix G.7. Void and Cavity Size Distributions

At doses below ~ 17 – 20 dpa in T91 and HT9 (BOR-60, 376–377 °C), and at all doses in EUROFER97 at 330–340 °C, cavity distributions are *unimodal* and confined to sub-2 nm diameters. These cavities are helium-stabilized bubbles that remain below the critical radius for spontaneous void growth. The distributions are narrow and right-skewed, with mean diameters of 1.0–2.6 nm and number densities of 10^{20} – 10^{21} m $^{-3}$ (Figure G.15).

The most distinctive feature of cavity microstructure evolution in F/M steels is the development of a bimodal size distribution once cavities exceed the critical radius for void growth. This transition is observed at ~ 35 dpa in both T91 and HT9 under BOR-60 neutron irradiation at 376–377 °C [6, 7], and is reproduced across a wide range of parameters under dual-ion irradiation of T91 [9].

The bimodal distribution consists of:

- A high-density small-bubble population ($d < 2$ nm, density $\sim 10^{21}$ m $^{-3}$) of He-stabilized, sub-critical cavities.
- A lower-density growing void population ($d > 2$ nm, density $\sim 10^{19}$ – 10^{20} m $^{-3}$) of super-critical cavities that have escaped He stabilization and grow by net vacancy absorption.

The two populations are separated by a valley near the critical radius $r_c \approx 1$ nm. Despite its lower density, the void population dominates swelling because swelling scales as $\Delta V/V \propto \sum N_i d_i^3$.

Figure G.15 illustrates the transition from unimodal to bimodal and the temperature evolution of the bimodal distribution.

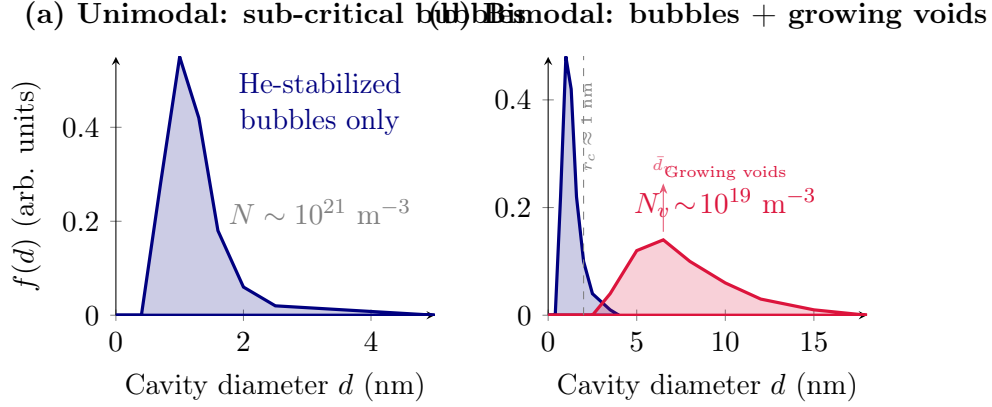


Figure G.15: Schematic comparison of (a) the unimodal cavity size distribution characteristic of low-dose or low-temperature conditions in F/M steels, where only He-stabilized sub-critical bubbles exist, and (b) the *bimodal* distribution that develops in T91 and HT9 beyond ~ 35 dpa at $376\text{--}377^\circ\text{C}$, where a high-density population of small bubbles coexists with a lower-density population of growing super-critical voids. The critical radius $r_c \approx 1$ nm separates the two populations.

Appendix G.8. Comparison: Neutron vs. Ion Irradiation

Table G.33 summarizes the key differences and agreements between loop and void size distributions measured under neutron and ion irradiation.

The principal conclusions from the experimental size-distribution database are:

- Loop distributions are right-skewed log-normals that narrow and shift to larger mean diameter with increasing temperature ($250\text{--}415^\circ\text{C}$) and dose (coarsening regime, >15 dpa in BOR-60 conditions).
- At low temperatures ($<300^\circ\text{C}$), the distribution is dominated by a sharp spike of sub-3 nm black-dot defects; resolved loops form a secondary shoulder.
- Void distributions are *unimodal and sub-2 nm* at low doses or temperatures, transitioning to a *robust bimodal form* (small He-bubbles + growing voids) at ~ 35 dpa and $376\text{--}460^\circ\text{C}$ in T91 and HT9.
- The bimodal distribution is a fundamental feature of F/M steels in the peak-swelling temperature window and is reproduced across a wide range of dual-ion irradiation parameters.

- Ion irradiation produces narrower, more homogeneous loop distributions and requires a +60–70 °C temperature shift to match neutron-irradiated microstructures.

Acknowledgments

The author acknowledges support from the U.S. Department of Energy, Office of Fusion Energy Sciences, under Award No. DE-SC0018410:0008 to UCLA.

Declarations

- **Ethics Declaration:** The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Competing interests:** The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Author contributions:** Nasr M. Ghoniem conceived the work, performed the analysis, developed the code, and wrote the manuscript.
- **Declaration of generative AI-assisted technologies :** During the preparation of this work, the author(s) used Anthropic Claude in order to research published literature, assist in coding, and prepare figures. After using this service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Table G.29: Primary sources of size-distribution data compiled in this study. Abbreviations: N = neutron, SI = self-ion, DI = dual-ion, WBDF = weak-beam dark-field TEM, BF/DF = bright/dark-field TEM.

Material	Irrad.	Facility	Dose (dpa)	T (°C)	Method	Reference
EUROFER97	N fast	BOR-60	15, 32	330–335	WBDF	[4]
EUROFER97	N fast	BOR-60	16.3	250–415	BF/DF	[5]
T91	N fast	BOR-60	15.4, 35.1	376–524	STEM	[6]
HT9	N fast	BOR-60	17.1, 35.1	377	STEM	[7]
HT9	N fast	ATR	4.02	300–600	STEM	[113]
Fe-9Cr	N fission	BR2	2.0	300	WBDF	[13]
Pure Fe	N fission	Multiple	0.19	300	BF/DF	[114]
Pure Fe	N fission	Multiple	0.79	60	BF/DF	[115]
Pure Fe	N fission	Reactors	1.0	400	BF/DF	[116]
HT9	SI Kr	In-situ TEM	4, 20	420–470	In-situ	[39]
Fe-9Cr	SI Fe ⁺	Ion facility	2.0	300	WBDF	[13]
Pure Fe	SI Fe ³⁺	Ion fac.	30	300	BF/DF	[117]
T91	DI Fe+He	Dual-ion fac.	14.6–35	406–570	TEM/STEM	[14]
T91	DI Fe+He	Dual-ion fac.	14.6–35	406–570	TEM	[9]
HT9	SI Fe ²⁺	Self-ion	140–650	440–460	STEM	[8]
F82H	DI Fe+He	Dual-ion fac.	20	470	TEM	[118]
F82H	DI Fe+He	Dual-ion fac.	≤ 82	500	TEM	[119]
Fe-9Cr	SI Fe	Ion fac.	3	450	TEM	[120]
Fe-9Cr	SI Fe	Ion fac.	3	450	TEM	[121]
F82H	SI Fe ³⁺	Ion fac.	30	300	TEM	[122]
HT9	SI Fe	Self-ion	140	440	STEM	[123]
T91	DI Fe+He	Dual-ion fac.	16.6–72	369–520	TEM/STEM	[124]
T91	DI Fe+He	Dual-ion fac.	47	376	TEM/STEM	[125]
EK-181	N fast	Reactor	6	250–400	APT/TEM	[126]
CLAM	DI Fe+He	Dual-ion fac.	11	350–550	TEM	[127]
F82H	Spall.	SINQ	10.7–19.6	~250	TEM	[128]
T91	Spall.	SINQ	11.8	~250	TEM	[129]
Fe-12Cr	SI He ⁺	Ion fac.	<1	250–400	TEM	[130]
EM12/F17	N fast	Reactor	—	<500	TEM	[131]

Table G.30: Key void/cavity density and mean-size data illustrating the unimodal-to-bimodal transition in T91 and HT9 at 376–377 °C under BOR-60 neutron irradiation.

Material	Dose (dpa)	Regime	Bubble population		Void population	
			N_b ($\times 10^{21} \text{ m}^{-3}$)	$\bar{d}_b$ (nm)	N_v ($\times 10^{20} \text{ m}^{-3}$)	$\bar{d}_v$ (nm)
T91	15.4	Unimodal	~ 10	1.2	—	—
T91	35.1	Bimodal	~ 6	1.3	~ 5	5.0
HT9	17.1	Unimodal	~ 3	1.4	—	—
HT9	35.1	Bimodal	~ 2	1.2	~ 5	7.0

References: T91 [6]; HT9 [7].

Table G.31: Summary of swelling and void size data at key dose levels in T91 and HT9.

Material	Source	Dose	T	Regime	$\bar{d}_v$ (nm)	Swelling (%)
T91	BOR-60 neutron	15.4dpa	376 °C	Unimodal	—	~ 0.01
T91	BOR-60 neutron	35.1dpa	376 °C	Bimodal	5.0	~ 0.02
HT9	BOR-60 neutron	17.1dpa	377 °C	Unimodal	—	0.02
HT9	BOR-60 neutron	35.1dpa	377 °C	Bimodal	7.0	0.07
HT9	EBR-II/FFTF	200dpa	420 °C	Steady-state	—	0.09–1.02
HT9	Self-ion (Fe^{2+})	650dpa	460 °C	Steady-state	~ 22	$\sim 0.2\%/ \text{dpa}$
HT9 (F/M)	Self-ion	600dpa	450 °C	Steady-state	~ 30	2.6 (max)

References: BOR-60 [6, 7]; EBR-II/FFTF [133]; self-ion [8, 132].

Table G.32: Summary of quantitative trends in loop and void density and mean size as functions of temperature and dose in ferritic–martensitic steels. The symbols $\uparrow$ and $\downarrow$ denote increase and decrease, respectively.

Quantity	With T at fixed dose		With dose ϕ at fixed T	
	$T < 300\text{ }^{\circ}\text{C}$	$T > 300\text{ }^{\circ}\text{C}$	$\phi < 5\text{ dpa}$	$\phi > 15\text{ dpa}$
Loop density N_{d}	$\uparrow$; black-dot regime	$\downarrow$	$\uparrow$	$\downarrow$ or saturation
Loop mean size $\bar{d}_{\ell}$	$\uparrow$; small loops	$\uparrow\uparrow$	$\uparrow$	$\uparrow$; coarsening
Void density N_{v}	Low; unimodal bubbles	$\uparrow$ then $\downarrow$	Low	$\uparrow$; bimodal regime
Void mean size $\bar{d}_{\text{v}}$	Very small, $< 2\text{ nm}$	$\uparrow\uparrow$	$< 2\text{ nm}$	$\uparrow\uparrow$
Swelling S	≈ 0	Bell-shaped; $\uparrow$ then $\downarrow$	≈ 0	$\uparrow$

Table G.33: Comparison of defect size distributions under neutron and ion irradiation in ferritic–martensitic (F/M) steels.

Observable	Neutron irradiation	Ion irradiation
Loop mean diameter	5–20 nm depending on temperature; heterogeneous spatial distribution	6–7 nm at similar temperature under self-ion irradiation; more uniform spatial distribution
Loop distribution shape	Right-skewed log-normal distribution; black-dot spike at low temperature	More symmetric and narrower distribution; fewer black dots
$\langle 100 \rangle$ loop fraction	$\sim 20\text{--}30\%$ at 300 °C	$\sim 40\%$ at 300 °C
Spatial heterogeneity	Strong: decoration of pre-existing dislocation lines dominates; density varies by three orders of magnitude	Weak: nearly homogeneous field; no significant dislocation-decoration effect
Void/cavity regime	Bimodal at >35 dpa and 376–420 °C	Bimodal across the investigated parameters at $T \approx 460$ °C
Temperature shift	—	+60–70 °C required to match neutron microstructure
G-phase / α'	Well developed under neutron irradiation at equivalent conditions	Suppressed under ion irradiation: less segregation and fewer Ni/Si clusters

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